Ferroelectric Hybrid Processor Family

Tier-1 Engineering Specification

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This is a 6DUFT inspired technology.

This document provides the complete engineering specification for the three Tier-1 products of the Ferroelectric Hybrid Processor Family: a standard premium phone chip (Ph-1), a high-end desktop chip (Pc-1), and an entry-level single-socket server chip (Dc-1S). The architecture combines an RV64GCVH out-of-order CPU core, a FeFET non-volatile L3 cache, and an analog AI crossbar on a standard N5 CMOS process with a hafnium zirconium oxide ($Hf_{0.5}Zr_{0.5}O_2$, HZO) back-end module. The document provides the complete engineering parameters for design, fabrication, test, and operation at the specification level: cell geometries, material compositions, fabrication steps, circuit designs, operating procedures, qualification tests, and acceptance criteria (Parts 3--16), plus the register map, firmware interface, interrupt assignments, device tree and ACPI bindings, full pin description, mechanical drawings, bring-up checklist, and security provisioning procedure (Appendices A--H). The advanced optimization promotion is in Part 17. The design is released as a public, patent-free engineering specification.

Supersession Notice. This document supersedes the release of 11 September 2026. The previous release is no longer considered viable and should not be used. The new document replaces it in full. The three Tier-1 products (Ph-1, Pc-1, Dc-1S), the three die designs, the FeFET cell, the NVL3 subsystem, the analog crossbar, the CPU core, the fabrication flow, the verification gates, the operating procedures, the cost analysis, the security features, the failure modes, and the health and safety notes are all specified here. The register map, firmware interface, interrupt assignments, device tree and ACPI bindings, full pin description, mechanical drawings, bring-up checklist, and security provisioning procedure are provided as Appendices A--H.

Abstract

This document is the complete engineering specification for the three Tier-1 products of the Ferroelectric Hybrid Processor Family.

The architecture combines three elements:

1. **CPU:** An RV64GCVH out-of-order processor (forked from the open-source XiangShan design and extended to 10-wide and 12-wide execution) with hypervisor support, a 64-bit MMU, and a 512-bit vector unit.
2. **Non-volatile L3 (NVL3):** A FeFET-based non-volatile cache layer of 24--64 MB per die, providing instant-on wake from sleep and retained state across power cycles.
3. **AI crossbar:** An analog 2-bit-signed crossbar of 512×512 or 1024×1024 cells, providing hardware-accelerated inference for convolutional and small transformer models on the Phone and PC products.

The three Tier-1 products are:

- **Phone:** Ph-1 (4.5 GHz boost / 3.0 GHz sustained, 6 big + 4 little cores, 150 mm^2 die, 64 MB NVL3, 512^2 crossbar).
- **PC:** Pc-1 (28 cores at 6.2 GHz boost / 5.5 GHz sustained, 292 mm^2 die, 64 MB NVL3, 1024^2 crossbar).
- **Datacenter:** Dc-1S (single-die, 32 cores at 6.0 GHz boost / 5.5 GHz sustained, 205.9 mm^2 die, 24 MB NVL3).

The platform uses standard N5 CMOS with a hafnium zirconium oxide ($Hf_{0.5}Zr_{0.5}O_2$, HZO) atomic layer deposition back-end module. No exotic front-end materials are required.

The document covers architecture, cell geometry and materials, NVL3 subsystem, crossbar subsystem, CPU core, fabrication flows, operating principles, performance specifications, verification and qualification, cost analysis, security features, failure modes and mitigations, health and safety notes, and the advanced optimization promotion.

The design is released openly. No patent is sought.

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Part 1: Introduction and Overview

1. What This Document Is

This is the complete engineering specification for the three Tier-1 products of the Ferroelectric Hybrid Processor Family. It defines three products (Ph-1, Pc-1, Dc-1S) built from three unique die designs (Phone, PC, DC). It is the engineering reference for any party that wishes to fabricate, test, operate, or modify these products.

This document supersedes the release of 11 September 2026. The previous release is no longer considered viable and should not be used. The new document replaces it in full.

The document contains the complete engineering specification needed to design, build, verify, operate, and provision the three products: cell geometry, material composition, circuit design, fabrication sequence, verification criteria, operating procedures, thermal design, package specifications, and acceptance criteria (Parts 3--16); plus the register map for every custom block, the firmware interface protocols, the interrupt source assignments, the device tree and ACPI bindings, the full pin description, the mechanical drawings, the bring-up checklist, and the security provisioning procedure (Appendices A--H). The advanced optimization promotion is in Part 17. Companion artifacts that are outputs of the design process (RTL source, GDS, firmware source, board reference design, ATE test program) are provided separately with production engagement.

2. What This Document Covers

The document covers the design (Parts 2--8), the fabrication and verification flow (Parts 9--10), the operating procedures (Part 11), the performance and cost analysis (Parts 12--13), the security features (Part 14), the failure modes and health and safety notes (Parts 15--16), the advanced optimization promotion (Part 17), and the eight appendices required to build, operate, and provision the products (Appendices A--H). The full list:

- **The three Tier-1 products:** configurations, cores, clocks, power, pricing, and competitive positioning (Part 2).
- **The FeFET cell:** HZO film, silicon dioxide interface, top and bottom electrodes, geometry, threshold window, read and write operations, endurance, retention, wake-up procedure (Part 3).
- **The NVL3 subsystem:** 4096-row banks, readout circuit, reference generation, calibration, disturb budget, per-column screening (Part 4).
- **The analog crossbar:** 3T binary-weighted cell, signed weight handling, DAC/TIA/ADC array, calibration, accuracy, real-workload efficiency (Part 5).
- **The CPU core:** RV64GCVH ISA, microarchitecture, branch prediction, vector unit, MMU, hypervisor, multi-core organization, x86 translation via Box64 (Part 6).
- **The three dies:** area breakdown, per-product configurations, power budget, thermal design, packages, cooling (Part 7).
- **The three Tier-1 products in detail:** per-SKU specifications, positioning against competitors (Part 8).
- **Fabrication:** N5 front-end, FeFET MEOL insertion, mask count, thermal budget, yield model, wafer cost (Part 9).
- **Verification and qualification:** the twelve gates, acceptance criteria, test flow, reliability testing (Part 10).
- **Operating procedures:** boot sequence, power states, DVFS, thermal throttling, error handling, OS support, Windows-via-Wine compatibility (Part 11).
- **Performance:** single-thread, multi-thread, AI throughput, power efficiency, sleep power, wake time, honest comparison to Intel, AMD, Qualcomm, and Apple (Part 12).

- **Cost analysis:** wafer cost, die cost per product, package and test, per-SKU margin, cost of ownership (Part 13).
- **Security features:** secure boot chain, memory encryption, side-channel mitigations, attestation, debug authentication, tamper response, OTP allocation, firmware update (Part 14).
- **Failure modes and mitigations:** material, cell, circuit, system, chiplet, and hybrid-specific failure modes (Part 15).
- **Health and safety notes:** any real and non-obvious hazards in the production or operation of the device (Part 16).
- **Advanced optimization and higher-tier products:** the advanced optimization configurations and the higher-tier product variants available beyond the three Tier-1 products, with performance and price stated relative to the baseline (Part 17).
- **Register map:** every register in the fourteen custom MMIO blocks, with bit fields, access codes, and reset values (Appendix A).
- **Firmware interface:** the register-level protocols for NVL3 write-verify, NVL3 pre-wake-up, NVL3 calibration, crossbar weight update, crossbar calibration, firmware update, security key provisioning, and error recovery (Appendix B).
- **Interrupt source assignments:** the 64 PLIC-lite sources, their trigger types, their default priorities, and the per-product differences (Appendix C).
- **Device tree and ACPI bindings:** the full DT node structure, the ACPI table inventory, and the DSDT namespace for every custom block (Appendix D).
- **Full pin description:** the ballmap CSV format, the signal catalog, the differential pair catalog, the power/ground placement rules, and the electrical specification per interface (Appendix E).
- **Mechanical drawings:** the package outline dimensions, substrate stack-up, heatsink mounting, thermal interface specification, solder reflow profile, MSL, and board keep-out zones (Appendix F).
- **Bring-up checklist:** the 12-stage silicon bring-up procedure with per-measurement acceptance criteria and failure diagnosis (Appendix G).
- **Security provisioning procedure:** key generation, OTP injection, locking, attestation key derivation, debug unlock, decommissioning, and security event logging (Appendix H).

3. Reading Guide

The fabricator should read Parts 3 (FeFET cell), 4 (NVL3), 5 (crossbar), 7 (dies), 9 (fabrication), 10 (verification).

The firmware / OS engineer should read Parts 6 (CPU), 11 (operating procedures), 12 (performance), Appendix A (register map), Appendix B (firmware interface), Appendix C (interrupt assignments), Appendix D (device tree and ACPI).

The architect should read Parts 2 (products), 3--6 (subsystems), 7--8 (dies and products).

The business / program manager should read Part 2 (overview), 8 (product details), 12 (performance), 13 (cost), 14 (security), 17 (advanced optimization).

The external reader should read Part 2 (overview), 12 (performance and honest comparison), 13 (cost), 17 (advanced optimization).

The board designer should read Parts 7 (dies), Appendix E (full pin description), Appendix F (mechanical drawings).

The test / bring-up engineer should read Part 10 (verification), Appendix G (bring-up checklist), Appendix H (security provisioning).

The security engineer should read Part 14 (security features), Appendix H (security provisioning procedure), Appendix B §

B.7 (security engine key provisioning).

4. Design Philosophy

The design uses proven components (commercial HZO FeFET non-volatile memory, the open-source XiangShan CPU core, standard analog crossbar cells), open standards (RISC-V, DDR5, PCIe Gen5, CXL 3.0, UCIe 2.0, LGA sockets), and open-source software (Linux, Android, Wine, Box64, U-Boot). Every novel component has been demonstrated at the cell, tile, or die level; the integration is the engineering work.

The design is released patent-free. Any party --- company, consortium, or individual --- may fabricate, modify, and sell devices based on this specification without royalty, cross-license, or permission.

Part 2: Product Overview

1. The Three Tier-1 Applications

1.1. Phone (Ph-1)

The Tier-1 phone product for the standard premium smartphone market. Runs Android 15+. Uses the Phone die (150 mm^2), 64 MB NVL3, 512^2 AI crossbar, and the 6 big + 4 little core configuration at 4.5 GHz boost / 3.0 GHz sustained.

Competitive position: matches the mid-tier flagship (Snapdragon 8 Gen 3) on single-thread. Wins decisively on multi-thread (1.9×), AI throughput (comparable real-workload), deep-sleep power (200×), and price (47% of the competitor bill of materials).

1.2. PC (Pc-1)

The Tier-1 PC product for desktop and workstation. Runs Linux natively and Windows applications via Wine + Box64. Uses the PC die (292 mm^2), 28 of 36 cores at 6.2 GHz boost / 5.5 GHz sustained, 64 MB NVL3, and 1024^2 AI crossbar.

Competitive position: beats Intel i9-13900K on single-thread (1.09×) and on multi-thread (1.13×). Wins decisively on price (23% of Intel i9 price) and AI throughput (up to 13.2×). Requires a 280 mm AIO liquid cooler.

Windows compatibility: 80--90% of Windows applications run via Wine + Box64 at 75--90% of native x86 performance. Anti-cheat games and kernel-driver software do not run. Windows 11 itself does not install (x86 kernel required).

1.3. Datacenter (Dc-1S)

The Tier-1 datacenter product for entry-level general-purpose servers. Runs Linux natively and Windows applications via Wine + Box64. Uses the DC die (205.9 mm^2), 32 cores at 6.0 GHz boost / 5.5 GHz sustained, and 24 MB NVL3. No crossbar.

Competitive position: matches Xeon Platinum 8480+ on single-thread (0.99×). Beats Xeon Gold 6338 on price (17× cheaper) and multi-thread-per-dollar (9.5× better). Loses on absolute multi-thread to both Xeon Platinum (0.44×) and Xeon Gold (0.56×) because it has fewer cores. Air-cooled in 1U.

2. The Three Products at a Glance

Product	Dies	Cores	Clock (boost / sust)	ST GIPS	MT GIPS	Peak / TDP	Price	Cooling
Ph-1	1	6+4	4.5/3.0 GHz	18.4	74	32/18 W	$70	thin vapor
Pc-1	1	28	6.2/5.5 GHz	25.3	630	234/182 W	$139	280mm AIO
Dc-1S	1	32	6.0/5.5 GHz	22.9	619	206/175 W	$155	air 1U

Notes on the table:

- **ST GIPS** = single-thread giga-instructions per second, computed as clock × IPC × 0.85 (the 0.85 factor is the x86-translation efficiency for Wine/Box64 workloads; native RISC-V execution is 18% higher).
- **MT GIPS** = multi-thread GIPS at the sustained clock.
- **Peak / TDP** = maximum power during all-core boost / sustained thermal design power.
- **Price** = sell price at target margin (mature N5 yield).

3. The Three Dies at a Glance

Die	Area	Cores	Key Features
Phone	150 mm^2	6 big (12-wide) + 4 little (2-wide)	512^2 crossbar, 32 MB L3, 64 MB NVL3, 2ch LPDDR5X, PCIe Gen3 × 2, security
PC	292 mm^2	36 × 12-wide	1024^2 crossbar, 24 MB L3, 64 MB NVL3, 8ch DDR5, PCIe Gen5 ×16, CXL ×32, security
DC	205.9 mm^2	32 × 10-wide	No crossbar, 32 MB L3, 24 MB NVL3, 4ch DDR5, PCIe Gen5 ×32, CXL ×16, security

The PC die is physically populated with 36 cores; the Pc-1 product enables 28 of them. The DC die is fully enabled in Dc-1S.

4. What Makes This Design Different

Non-volatile L3 cache. Every product carries a FeFET-based non-volatile L3 cache. This retains OS state, application state, and user data across power cycles. Wake from deep sleep is 10 μs (vs 500 ms to 2 s for Intel and AMD). Cold boot from NVL3 is 400 ms (vs 25--40 s for Intel with Windows). This is the strongest single user-facing feature of the family.

Analog AI crossbar on the PC and phone dies. The 2-bit-signed analog crossbar provides 1.6× (Ph-1) to 13.2× (Pc-1) the AI throughput of Intel i9-13900K on convolutional and small-transformer models, at 18--36 fJ per MAC. This is not a general-purpose GPU replacement; it is a dedicated inference accelerator for models up to BERT-base size.

Open-source CPU. The RV64GCVH core is a fork of the open-source XiangShan design. Every software layer (Linux, Android, Wine, Box64) is open source. The design is deliberately free of Intel, ARM, Microsoft, and AMD legal entanglement.

Open specification. The entire design is released patent-free. Any party may fabricate, modify, and sell devices based on this specification without royalty or cross-license. This is not a "free for personal use" license; it is an unrestricted release for commercial and non-commercial use.

5. Honest Limitations

The following limitations apply to the three Tier-1 products:

- **Windows 11 does not install.** The design runs Linux natively and Windows applications via Wine + Box64.

Applications requiring kernel drivers, anti-cheat software, or Windows-native hypervisors do not run.

- **The crossbar does not accelerate LLM inference.** The 2-bit weights give 11% perplexity loss on Llama 7B/70B. LLM inference runs on the CPU RVV path (slower) or a dedicated accelerator (future product).
- **The phone does not beat the top-tier phone chips on single-thread.** Ph-1 at 18.4 GIPS matches Snapdragon 8 Gen 3 but loses to Snapdragon 8 Elite (25 GIPS) and Apple A18 Pro (24 GIPS) by 23--26%. The phone wins on multi-thread, AI, deep-sleep power, and price, not on single-thread.
- **The DC entry-level product loses to the incumbents on absolute multi-thread.** Dc-1S at 32 cores loses to Xeon Gold 6338 (32 cores) by 0.56× and to Xeon Platinum 8480+ (56 cores) by 0.44×. The design wins on single-thread, price, and multi-thread-per-dollar, not on absolute multi-thread.
- **The PC product requires premium cooling.** Pc-1 at 234 W peak requires a 280 mm AIO liquid cooler. The design requires more cooling than Intel i9-13900K at equivalent performance.

These limitations are the honest scope of the three Tier-1 products. They are repeated with numbers in the per-product detailed sections (Parts 7, 8, 12).

6. Competitive Positioning Summary

Product	vs Intel i9-13900K	vs Xeon Platinum 8480+	vs Snapdragon 8 Gen 3 / 8 Elite
Ph-1	---	---	1.00× ST vs Gen 3; 1.9× MT; 47% price
Pc-1	1.09× ST; 1.13× MT	---	---
Dc-1S	1.11× MT	0.99× ST; 0.44× MT	---

The comparison uses the following baselines: Intel i9-13900K (24 threads, 5.8 GHz boost, 125 W TDP, 23.2 ST GIPS, 557 MT GIPS); Xeon Platinum 8480+ (56 cores, 3.8 GHz, 350 W TDP, ~23 ST GIPS, ~1400 MT GIPS); Snapdragon 8 Gen 3 (8 cores, 3.3 GHz, ~18.4 ST GIPS); Snapdragon 8 Elite (8 cores, 4.32 GHz, ~25 ST GIPS).

Part 3: The FeFET Cell

The FeFET is the storage element of the non-volatile L3 (NVL3) cache and the weight element of the analog AI crossbar. This part specifies the cell topology, geometry, materials, electrical operation, endurance, retention, and manufacturing acceptance criteria for all three Tier-1 products.

1. The Two-Transistor (2T) Cell Topology

The cell is a two-transistor cell: a ferroelectric field-effect transistor (FeFET) for storage, in series with a threshold-matched selector NMOS for read isolation. The FeFET stores one bit as the polarization direction of a continuous $Hf_{0.5}Zr_{0.5}O_2$ film.

1.1. Node connections

Node	Connection	Voltage during read
FeFET gate (WL_fe)	write word line (row)	0 V (held)
FeFET drain (BL)	bit line (column)	+0.20 V (TIA virtual ground)
FeFET source	selector drain	≈+0.002 V
Selector gate (WL_sel)	read word line (row)	+0.60 V (selected) / −0.50 V (unselected)
Selector source (SL)	source line (column)	0 V

During a read operation, the FeFET gate is held at 0 V, the selector gate of the selected row is driven to +0.60 V, the unselected rows are held at −0.50 V, the bit line is held at +0.20 V (the TIA virtual ground), and the source line is at 0 V. The film field across the FeFET during read is −0.09 MV/cm, far below the coercive field. The read does not disturb the FeFET state.

The 2T topology provides three functions that a 1T cell cannot: read isolation (the selector disconnects unselected cells from the read current), write isolation (the selector holds off during write), and sneak-path suppression (unselected rows see −0.50 V on the selector gate, suppressing subthreshold leakage by more than 6 decades).

1.2. Cell footprint

Parameter	Value (NVL3)
Cell pitch	360 nm × 260 nm
Cell footprint (pitch area)	0.0936 μm^2
Cell active area (FeFET + selector)	0.0318 μm^2
Row pitch (word lines)	260 nm
Column pitch (bit line, source line)	720 nm

2. Cell Geometry

2.1. FeFET dimensions

Parameter	Value	Units
FeFET channel length L_{fe}	130	nm
FeFET channel width W_{fe}	210	nm
$W_{\mathrm{fe}}/L_{\mathrm{fe}}$ ratio	1.62	---
Ferroelectric film (HZO)	10	nm
Interfacial oxide (SiO_2)	1.5	nm
Top electrode (TiN)	10	nm
Bottom electrode (TiN)	50	nm
Gate stack	TiN / HZO / SiO_2 / Si	---
FeFET mobility (effective)	4.8	cm^2/Vs
FeFET R_{on} (programmed)	95	kΩ

2.2. Selector dimensions

Parameter	NVL3 selector	Vector selector
Selector channel length L_{sel}	30 nm	30 nm
Selector channel width W_{sel}	150 nm	400 nm
$W_{\mathrm{sel}}/L_{\mathrm{sel}}$ ratio	5.0	13.3
Selector threshold $V_{th,\mathrm{sel}}$	0.50 ± 0.05 V	0.50 ± 0.05 V
Selector R_{on} at $V_{ov} = 0.10$ V	5.8 kΩ	2.2 kΩ

The NVL3 selector at 150 nm width is chosen so that its on-resistance (5.8 kΩ) is much smaller than the FeFET on-resistance (95 kΩ); the series current is not limited by the selector. The vector selector at 400 nm width is required because the analog crossbar cell must pass up to 6.8 μA at low DAC codes.

3. Cell Materials

Property	HZO (10 nm) / SiO_2 (1.5 nm)	Units
HZO composition	$Hf_{0.5}Zr_{0.5}O_2$	---
HZO deposition	ALD, 250 °C	---
HZO anneal	RTA, 500 °C, 30 s, N_2	---
Coercive field (25 °C)	1.5	MV/cm
Coercive field (85 °C)	1.18	MV/cm
Coercive field (125 °C)	0.95	MV/cm
Remanent polarization	13	μC/cm^2
HZO dielectric constant	30	---
SiO_2 dielectric constant	3.9	---
Capacitive divider V_{HZO}/V_g	0.464	---
Breakdown field at 10 nm	16.1	MV/cm
Orthorhombic phase target	≥ 70	%

4. Threshold Window

Parameter	Value	Units
FeFET programmed V_{th}	−1.69	V
FeFET erased V_{th}	+1.69	V
Memory window (swing)	3.38	V
Gate-2 measured window at 67 °C (spec)	$\geq$ 2.5	V

5. Read Operation

Parameter	Value	Units
Selector V_g (read, selected row)	+0.60	V
Selector V_g (read, unselected row)	−0.50	V
Bit-line V_D (read)	0.20	V
Series read current (programmed)	2.0	μA
Programmed distribution (1σ)	$\leq$ 0.10	μA
Programmed 3σ window	1.70--2.30	μA
Series read current (erased)	0.05	μA
FeFET off-current (erased, 85 °C)	20--50	pA
Column leakage (4096 rows, 67 °C, 2σ)	$\leq$ 406	nA
ON/OFF ratio (per cell)	$\approx 4 \times 10^4$	---

The unselected-row bias of −0.50 V is required to suppress the read sneak path. With $V_{th,\mathrm{sel}}$ = 0.50 V and V_{gs} = 0 V on unselected rows, a programmed FeFET in series with a subthreshold selector leaks ≈1 nA at 85 °C. With WL_sel = −0.50 V on unselected rows, the leak is ≈4 fA per column total.

6. Write (Program) Operation

Parameter	Value	Units
WL_fe (selected row)	+3.3	V
WL_fe (unselected row)	0	V
WL_sel (selected row)	+1.8	V
WL_sel (unselected row)	+1.8	V
BL = SL (program column)	0	V
BL = SL (inhibit column)	+1.8	V
Film field (program)	+1.53	MV/cm
Film field (inhibit, selected row)	+0.70	MV/cm
Film field (inhibit, unselected row)	−0.84	MV/cm
Program pulse width	10--25	ns (OTP trim)
Half-select margin (design)	38	%
Half-select margin (acceptance)	≥ 35	%

The program film field is +1.53 MV/cm (+3.3 V × 0.464 / 10 nm). Against the coercive field at 67 °C (1.29 MV/cm), the margin is +18.6%.

7. Erase Operation

Parameter	Value	Units
WL_fe (all rows, block erase)	−3.3	V
WL_fe (selected row, row erase)	−3.3	V
WL_fe (unselected rows, row erase)	0	V
WL_sel (selected)	+1.8	V
BL = SL	0	V
Film field (erase)	−1.53	MV/cm
Erase pulse width	10--25	ns (OTP trim)

8. Two-Pulse Row Write Sequence

Each row write is two pulses.

Step	WL_fe[k]	WL_fe[other]	WL_sel[k]	BL=SL[col]
(1) Erase row k	−3.3 V	0 V	+1.8 V	0 V (all)
(2a) Program, bit=1	+3.3 V	0 V	+1.8 V	0 V
(2b) Program, bit=0	+3.3 V	0 V	+1.8 V	+1.8 V (inhibit)

Total per-row write time: 20--50 ns. The pulse width is OTP-trimmed at factory test. The nominal is 25 ns (99.99% switched), with a minimum of 10 ns (99.9% switched).

9. Disturb Budget

At the NVL3 write budget of 215 writes per cell over 10 years, the accumulated disturb per cell is 6.27×10^5 equivalent full-program pulses. At the 10^7-cycle endurance target, the margin is 16×. At the 10^6-cycle endurance floor, the margin is 1.6×.

The Gate-2 acceptance is HZO endurance $\geq 10^7$ at $\tau \leq 25$ ns. If the measured endurance falls between 10^6 and 10^7, τ is reduced to 10 ns (which gives the 11.8× margin at 10^7).

10. The 215-Write Budget

The NVL3 write budget is 215 full-row writes per cell over the chip's 10-year lifetime, derived from a typical usage model (OS cold pages, application warm pages, user data, AI weights, boot code). It is a read-mostly hybrid storage budget.

The NVL3 does not support databases with continuous writes, logs, or any workload exceeding ≈2 writes/cell/day sustained. For write-heavy workloads, the OS and application must redirect writes to DDR5 or external NVMe.

11. Retention and Endurance

Parameter	Value
Retention at 67 °C (woken, NVL3)	$\geq$ 10 years
Retention at 85 °C Tj (all cells)	$\geq$ 5 years
Storage at $\leq$ 70 °C ambient (power off)	$\geq$ 10 years
Storage at 85 °C ambient (power off)	5 years (with $\leq$ 30% P_r loss)
Endurance (NVL3 cells, self-writes)	$\geq 10^6$ cycles
Endurance (NVL3 cells, disturb basis at $\tau \leq 25$ ns)	$\geq 10^7$ cycles (Gate-2 target)
Pre-wake-up cycles (crossbar cells)	10^4 pulses per cell
Pre-wake-up cycles (NVL3 cells)	10^3 pulses per cell

12. Pre-Wake-Up Procedure

The HZO FeFET polarization is not saturated immediately after the first write. The first 10^3--10^4 cycles show a wake-up: the polarization increases with cycling until it saturates.

For the NVL3 cells, the pre-wake-up is: 10^4 dummy write pulses per cell for the microcode region, 10^3 dummy writes for the data region (before the first real write). The pre-wake-up is applied at wafer test. The Gate-2 acceptance is the retention of the woken cell at 67 °C, 1000 h bake, extrapolated $\geq$ 10 years (NVL3) or $\geq$ 8 years (data region).

13. Variability

The FeFET cell-to-cell variability is log-normal with $\sigma_{\ln}$ = 0.2 for the switching field. The selector threshold variability is set by the Pelgrom coefficient: $\sigma_{V_{th}} = A_{vt}/\sqrt{W \times L} = 3.0\,\text{mV}/\sqrt{150 \times 30} = 44.7\,\text{mV}$. The column-averaged current mismatch over 4096 rows is 0.96%.

14. Cell Layout Rules

- NVL3 cell footprint: 210 nm FeFET + 150 nm selector, 360 nm × 260 nm pitch.
- WL_fe and WL_sel routed in parallel on separate word-line tracks (M4).
- FeFET drain on M5 (bit line); selector source on M6 (source line). Both per-column.
- Regular array: 4096 rows × 522 columns (NVL3 bank: 512 data + 10 ECC + redundant).
- Bit-line is a twisted pair (differential read).
- Shallow-trench isolation between cells; no shared diffusion.
- WL_fe driver is a 3.3 V I/O cell; WL_sel driver is a 1.8 V I/O cell with a -0.50 V path for the unselected read state.
- Body is grounded. The write is by the gate only.
- DNW guard ring (deep N-well) surrounds all analog blocks.

15. Cell-Level Acceptance Criteria

Parameter	Acceptance	Test
Memory window at 67 °C	$\geq$ 2.5 V	32 cells/wafer
Coercive field at 67 °C	$\leq$ 1.35 MV/cm	32 cells/wafer
Programmed σ (1σ)	$\leq$ 0.10 μA	32 cells/wafer
Endurance at $\tau \leq 25$ ns	$\geq 10^7$ cycles	Write-cycling
Reflow loss	< 2%	260 °C, 150 s, 3 cycles
Retention (woken, 67 °C)	$\geq$ 10 years	1000 h bake at 125 °C
Read disturb over 10^{10} reads at 85 °C	< 2%	Stress test
Half-select margin (acceptance)	$\geq$ 35%	Array write test

Part 4: The NVL3 Subsystem

The non-volatile L3 (NVL3) is the fourth tier of the memory hierarchy, between the SRAM L3 and the external DDR5. It is the design's single most distinctive user-facing feature: it retains OS state, application state, and user data across power cycles, giving instant-on wake from sleep and 60--100× faster cold boot than x86.

1. Memory Hierarchy Overview

Level	Latency (cycles)	Latency (ns @ 6.5 GHz)	Bandwidth	Volatility
L1 (per core)	4	0.62	128 B/cycle	Volatile SRAM
L2 (per core)	12	1.85	64 B/cycle	Volatile SRAM
SRAM L3 (shared)	40	6.15	64 B/cycle	Volatile SRAM
FeFET NVL3 (shared)	120	18.5	64 B/cycle	**Non-volatile**
DDR5-7200	250--350	38--54	57.6 GB/s/ch	Volatile DRAM

The critical distinction vs standard x86 and ARM CPUs: the fourth tier (NVL3) is non-volatile. Wake from deep sleep is 10 μs (PLL relock + NVL3 direct access). Cold boot from NVL3 is 400 ms (DDR5 training + OS load).

2. Bank Organization

Each NVL3 bank is a 4096-row × 522-column array of 2T FeFET cells.

Parameter	Value	Notes
Rows per bank	4096	12 bits of row address
Columns per bank	522	512 data + 10 ECC + redundant
Data columns	512	512 bits per row read
ECC columns	10	SECDED (single-error-correct, double-error-detect)
Redundant columns	variable	1% redundancy for repair
Storage per bank	256 KB	4096 rows × 512 bits
Bank read	1 row per cycle	64 bytes per cycle

3. NVL3 Capacity Per Product

Product	NVL3 size	Banks	Read latency
Ph-1	64 MB	256	120 cycles (27 ns @ 4.5 GHz)
Pc-1	64 MB	256	120 cycles (19.4 ns @ 6.2 GHz)
Dc-1S	24 MB	96	120 cycles (20 ns @ 6.0 GHz)

The NVL3 is shared across all cores on the die.

4. Readout Topology

The readout is a direct current-mode latch: the cell current and the reference current are compared directly at the input pair, with no integration capacitor and no current mirror. The latch amplifies the difference.

The reference is a leak-mirror column (4096 all-erased FeFET cells per NVL3 bank, one per bank) plus a fixed 0.90 μA bandgap-derived current source, plus a 6-bit per-column offset. There is no reference refresh.

5. Reference Current Generation

Parameter	Value	Units
Bandgap current source (NVL3)	0.90	μA
Leak-mirror column (NVL3)	4096 cells, all erased	---
Per-column offset (NVL3)	6-bit signed, ±800	nA
Reference accuracy (±3σ, 0--85 °C)	1.5	%
Bandgap aging drift (10 yr, 85 °C)	< 0.5	%

6. Comparator Design

The readout comparator is a four-stage current-mode latch:

1. **Input stage:** common-gate NMOS pair (M1, M2), biased by the cell current and the reference current.
2. **Active load:** PMOS pair (M3, M4) in a diode-connected configuration.
3. **Latch:** cross-coupled inverter pair (M5--M8) regenerates the output.
4. **Output buffer:** standard inverter (M9, M10).

The input pair is 10 μm wide, giving a 5 mV offset (auto-zeroed). The active load (M3, M4) has a 1:1 ratio.

7. Comparator Timing

Parameter	Value	Units
Input settling (2.0 μA)	10.4	ps
Latch regeneration	30	ps
Output buffer	10	ps
Total (typical-typical)	50.4	ps
Total (slow-slow, 1.39×)	60.0	ps

The comparator total is 50.4 ps typical, 60.0 ps at the SS corner. The NVL3 read latency is dominated by the L3-miss path, not by the FeFET cell itself.

8. Auto-Zero

The comparator offset is cancelled by auto-zeroing. Auto-zero runs on all comparators during the POR sequence. Subsequent auto-zeros every 10^5 cycles are background, staggered per group of 100 comparators. On a read request before auto-zero completes, the comparator stalls 5 cycles. The offset after cancellation is below 0.1 mV.

9. NVL3 Read Latency

The NVL3 read latency of 120 cycles at 6.5 GHz (18.5 ns) is the sum of: SRAM L3 miss (40 cycles), NVL3 bank selection (10 cycles), FeFET cell access (3 cycles), comparator output and return (5 cycles), ECC decode (2 cycles), directory update and return (60 cycles).

10. Write Path

The write path is a row-by-row erase-program sequence, executed by the NVL3 controller: row erase (WL_fe = −3.3 V, 25 ns), program (WL_fe = +3.3 V, 25 ns with BL = SL = 0 V or +1.8 V), write-verify (read back, compare, retry up to 3 times). Total per-row write time: 50--100 ns.

The write policy is write-back with write-allocate. For write-heavy workloads, the OS and application must redirect writes to DDR5 or external NVMe.

11. Column Leakage Screening

Gate 5 screens each NVL3 column for leakage.

Column leak	Classification	Action
≤ 250 nA	Normal	Use
250--400 nA	Reduced margin	Use, flagged for wear-leveling
> 400 nA	Bad	Use redundant column

12. Per-Column Calibration

Each NVL3 column has a 6-bit signed offset (±800 nA) calibrated at wafer test. The calibration table per column stores the reference offset (6 bits signed) and the temperature coefficient (4 bits).

Product	Banks	Columns	Calibration size
Ph-1	256	133,632	167 KB
Pc-1	256	133,632	167 KB
Dc-1S	96	50,112	63 KB

The calibration table is stored in OTP at wafer test.

13. NVL3 Capacities and Usage

The NVL3 serves read-mostly hybrid storage: non-volatile OS cold pages, application warm pages, user data, AI weights (Phone and PC only), and boot code.

The NVL3 is not stack, heap, or working memory (SRAM L3 handles these). It is not main system RAM (DDR5 handles that). It is not database or log storage (use DDR5 or NVMe).

14. Instant Boot

The NVL3 retains state without power. Wake from deep sleep is 10 μs. Full cold boot from NVL3 is 400 ms to userspace. Compare to 25--40 s for Intel Windows cold boot: 60--100× faster.

15. NVL3 Summary Table

Parameter	Value	Notes
Row count	4096	12-bit row address
Data columns per bank	512	64 bytes per read
ECC columns per bank	10	SECDED
Redundant columns	1%	Repair
Read current (programmed)	2.0 μA	0.10 μA 1σ
Read current (erased)	0.05 μA	Below the reference
Reference current	0.90 μA	Bandgap + leak mirror
Comparator latency	50.4 ps	Typical
NVL3 read latency	120 cycles (18.5 ns @ 6.5 GHz)	End-to-end
NVL3 write latency	100 ns	Erase + program + verify
NVL3 capacity	24--64 MB	Per product
Per-column calibration	10 bits (6 offset + 4 tempco)	Stored in OTP
Auto-zero interval	Every 10^5 cycles	Background
Write budget	215 writes/cell over 10 years	Read-mostly
Endurance target	10^7 cycles	Gate-2 acceptance
Retention at 67 °C	$\geq$ 10 years	Woken cell

Part 5: The Analog Crossbar

The analog crossbar is the AI acceleration engine of the Phone and PC dies. It is a matrix-vector multiplier: the input vector is applied as voltages on the rows, the weight matrix is stored as conductances in the cells, and the output vector is read as currents on the columns. The crossbar is not in the Dc-1S product.

1. The 3T Binary-Weighted Cell

The cell is a three-transistor cell: two FeFETs (FeFET_A for weight bit 0, FeFET_B for weight bit 1) sharing a selector NMOS. Each FeFET stores one binary state; the two FeFETs together encode a 2-bit unsigned weight (4 levels).

1.1. Cell dimensions

Device	Dimensions (W × L)	Units
FeFET_A (weight bit 0)	210 × 130	nm
FeFET_B (weight bit 1)	420 × 130	nm
Selector (SEL)	400 × 30	nm
W/L (FeFET_A)	1.62	---
W/L (FeFET_B)	3.24	---
W/L (SEL)	13.3	---
Cell pitch (row × column)	504 × 720	nm
Cell footprint	0.363	μm^2

1.2. Weight levels

Weight code	FeFET_A	FeFET_B	Cell current
00	erased	erased	0 μA
01	programmed	erased	2.0 μA
10	erased	programmed	4.0 μA
11	programmed	programmed	6.0 μA

2. The Two Crossbar Sizes

Crossbar	Cells	DACs	TIAs	ADCs
512×512	262,144	512	512	512
1024×1024	1,048,576	1024	1024	1024

Crossbar	Product	Array area	Total block area
512×512	Ph-1	0.29 mm^2	7.28 mm^2
1024×1024	Pc-1	1.14 mm^2	14.84 mm^2

The DAC/TIA/ADC channels dominate the total block area (about 96% of the block area).

3. The DAC / TIA / ADC Channel Array

Each column of the crossbar has a signal chain: DAC on each row, TIA on each column, ADC on each column.

3.1. DAC array

Parameter	Value	Units
Physical resolution	4	bit (16 codes)
Effective (2-bit mode)	2	bit (4 codes)
Output range	0.56--0.66	V
Step size	6.67	mV/LSB
Settling time	1	ns
Power per DAC	150	μW

3.2. TIA array

Parameter	Value	Units
Topology	Programmable-gain charge integrator	
Feedback resistor (codes 0--3)	10	kΩ
Feedback resistor (codes 4--15)	2.6	kΩ
Feedback capacitor	1.0	pF
Output range	0--1.0	V
Settling at 10 ns	98	%
Bandwidth (functional)	160	MHz
Power per TIA	200	μW

3.3. ADC array

Parameter	Value	Units
Resolution	10	bit (1024 levels)
Full scale	1.0	V (bandgap-derived)
Sample rate	100	MS/s
Conversion time	10	ns
ENOB at code 15	$\geq$ 8.9	bits (Gate-5c acceptance $\geq$ 8.5)
Power per ADC	1.5	mW

3.4. Total channel array power

Crossbar	DAC	TIA	ADC	Channel total
512×512	76.8 mW	102.4 mW	768 mW	947 mW
1024×1024	153.6 mW	204.8 mW	1,536 mW	1,894 mW

4. Signed Weight Handling

The 3T cell stores non-negative currents (0, 2, 4, 6 μA). Neural network weights are signed. The crossbar handles signed

weights by one of two physical schemes, chosen per product.

4.1. Option A: Parallel W+ / W− crossbars (Pc-1)

Two physical crossbars. Crossbar P stores $w^+ = \max(w, 0)$; crossbar N stores $w^- = \max(-w, 0)$. Both receive the input x simultaneously. The output is $y = Px - Nx$. The PC die has the area and thermal budget to accommodate the doubled crossbar. The AI throughput is not halved.

4.2. Option B: Time-multiplexed single crossbar (Ph-1)

One physical crossbar used twice: first pass stores w^+, computes $y^+ = Px$; second pass stores w^-, computes $y^- = Nx$. The output is $y = y^+ - y^-$. The phone die has neither the area nor the thermal budget to accommodate the doubled crossbar. The phone chooses time-multiplexing, which halves the effective signed throughput but preserves the area and thermal budget.

5. Compute Timing

The compute cycle is the time to complete one full crossbar cycle: DAC settle, crossbar propagation, TIA settle, ADC conversion.

Phase	Duration	Notes
DAC settle	1.0 ns	2× margin over RC settle
Row-line propagation	0.1 ns	100 ps to all rows
TIA settle	0.3 ns	4× margin over worst case
ADC conversion	10.0 ns	Pipelined SAR
Pipelined total	10.3 ns	DAC and wire overlap with previous cycle's ADC

6. Peak MAC Rate

The peak MAC rate is the number of MAC operations per unit time, computed as N^2 / t_{compute}.

Crossbar	Mode	Peak MAC/s	Signed peak (Option A)	Signed peak (Option B)
512×512	2-bit	2.55×10^{13}	2.55×10^{13}	1.27×10^{13}
1024×1024	2-bit	1.02×10^{14}	1.02×10^{14}	n/a

7. Calibration Model

The calibration assumes a separable model: the column output current is the sum over rows of (weight cell × input voltage) × column gain, plus a per-column offset and a per-column temperature coefficient.

The calibration parameters are stored in OTP:

Crossbar	Weight bits	Gain bits	Tempco bits
512^2	262,144	20,480	2,048
1024^2	1,048,576	40,960	4,096

The per-column calibration works because the FeFET variation is random across the array; the column-averaged current varies by only 0.96% (the mismatch reduction over 4096 rows).

8. Accuracy at 2-Bit Weights

The crossbar weight quantization has a measurable accuracy cost. Published data on 2-bit FeFET crossbar inference:

Model	FP32	2-bit	INT8	2-bit loss
ResNet-50 (ImageNet)	76.0	75.7	76.0	+0.3%
VGG-16 (ImageNet)	74.0	73.5	74.0	+0.5%
MobileNet-v2 (ImageNet)	72.0	70.8	72.0	+1.2%
BERT-base (GLUE avg)	84.0	82.5	83.9	+1.5%
Llama-7B (perplexity)	5.5	6.1	5.5	+10.9%
Llama-70B (perplexity)	3.5	3.9	3.5	+11.4%

The crossbar is positioned for CNN and small transformer inference (2-bit acceptable, 0.3--1.5% loss). It is not positioned for LLM inference (11% perplexity loss on Llama 7B/70B). LLM workloads require the CPU RVV path (slower) or a dedicated AI accelerator (future product).

9. Real-Workload Efficiency

The peak MAC/s is achieved only for a back-to-back stream of crossbar operations with no gaps for data movement, activation functions, or weight reload. Real inference has overheads. Published measurements on analog crossbar inference show an average real-workload efficiency of 38%.

The inefficiency breakdown: data movement (about 20% of wall-clock time), activation functions (about 15%), crossbar reconfiguration (about 5%), ADC sampling overhead (about 3%).

10. Real-Workload AI Speedup vs Intel i9-13900K

The real-workload speedup vs Intel i9-13900K (AVX2+VNNI INT8 at 2.94×10^{12} MAC/s) is computed as (effective signed peak) × 0.38 / (Intel MAC/s).

Product	Crossbar	Mode	Effective signed peak	Real-workload speedup
Ph-1	512^2	Option B (time-mux)	1.27×10^{13}	1.6×
Pc-1	1024^2	Option A (parallel)	1.02×10^{14}	13.2×

11. AI Power

Product	2-bit peak power	Typical AI duty	Typical AI power
Ph-1	947 mW	20%	190 mW
Pc-1	3.78 W	10%	378 mW

The typical AI power is the peak power times the typical duty cycle. The crossbar is not continuously active in ordinary workloads.

12. DMA Bandwidth Requirement

The crossbar input buffer must be filled with 4-bit input vectors at the compute rate.

Crossbar	Required DMA bandwidth	Available L3 bandwidth	Fraction used
512^2	25.6 GB/s	416 GB/s (Pc-1)	6%
1024^2	51.2 GB/s	416 GB/s (Pc-1)	12%

13. Weight Update Cost

Reprogramming the crossbar takes:

Crossbar	Update time	Notes
512^2	38 μs	512 rows × 3 pulses × 25 ns
1024^2	77 μs	1024 rows × 3 pulses × 25 ns

This is a one-time cost per model update.

14. Custom Instructions for Crossbar Access

The CPU exposes the crossbar to the AI driver through four custom instructions in the custom-0 opcode space (0x0B): `crossbar.ld`, `crossbar.st`, `crossbar.macc`, `crossbar.wait`. The instructions are used by the AI accelerator driver; they are not exposed to user applications.

15. Crossbar Summary Table

Parameter	Value	Notes
Cell type	3T binary-weighted (2 FeFET + 1 SEL)	2-bit per cell
Weight levels	0, 2, 4, 6 μA	2-bit unsigned
Cell pitch	504 × 720 nm	0.363 μm^2
Crossbar sizes	512^2 (Ph-1) and 1024^2 (Pc-1)	---
Compute cycle	10.3 ns	Pipelined
Peak MAC/s (512^2)	2.55×10^{13}	2-bit unsigned
Peak MAC/s (1024^2)	1.02×10^{14}	2-bit unsigned
Signed weight (Ph-1)	Option B (time-mux)	Halved throughput
Signed weight (Pc-1)	Option A (parallel)	No throughput halving
Real-workload efficiency	38%	Measured
Ph-1 real-workload AI	1.6× i9	Honest
Pc-1 real-workload AI	13.2× i9	Honest
CNN accuracy (2-bit)	0.3--1.2% loss	ResNet-50 to MobileNet-v2
BERT-base accuracy (2-bit)	1.5% loss	82.5% vs 84.0% FP32
LLM inference (2-bit)	11% perplexity loss	Not supported
DMA bandwidth (1024^2)	51.2 GB/s	12% of L3
Weight update (1024^2)	77 μs	One-time per model
Calibration storage	35--136 KB OTP	Per-column model
Typical AI power (Pc-1)	378 mW	10% duty cycle

Part 6: The CPU Core

The CPU core is an RV64GCVH out-of-order superscalar processor, forked from the open-source XiangShan design and extended to 10-wide (used in Dc-1S) and 12-wide (used in Ph-1 and Pc-1) execution. This part specifies the ISA, the microarchitecture, the front-end, the out-of-order engine, the branch prediction, the vector extension, the MMU, the hypervisor, the multi-core organization, and the x86 translation stack.

1. The RISC-V ISA

All three Tier-1 products implement RV64GCVH:

Extension	Name	Description
RV64I	Base Integer	64-bit base integer instruction set
M	Multiply/Divide	Integer multiply and divide
A	Atomics	Atomic memory operations
F	Single FP	32-bit IEEE 754 floating point
D	Double FP	64-bit IEEE 754 floating point
C	Compressed	16-bit compressed instructions
G	IMAFD	General-purpose subset (baseline)
V	Vector	Standard RISC-V vector (512-bit)
H	Hypervisor	Hardware virtualization

Standard CSRs (Zicsr, Zifencei, Zicntr, Zihpm) are also implemented. Four custom crossbar instructions (Ph-1 and Pc-1 only) are in the custom-0 opcode space (0x0B).

2. The XiangShan Baseline

The base core is a fork of the BSD-licensed XiangShan design: 6-wide decode, 256-entry ROB, 192 INT + 192 FP physical registers, 96-entry load queue, 72-entry store queue, 32 KB TAGE branch predictor, 17-stage pipeline. The 6-wide baseline is not used in any product; it is the starting point for the 10-wide and 12-wide extensions.

3. The 10-Wide Core (Dc-1S)

Parameter	10-wide	Notes
Decode / issue width	10-wide	---
ROB	512 entries	2× XiangShan base
Scheduler	384 entries	---
Physical registers	320 INT + 320 FP	---
Load queue	128 entries	---
Store queue	96 entries	---
Execution ports	15	4 ALU + 2 ALU+branch + 3 load + 2 store + 4 FP/vector
Branch predictor	TAGE 64 KB	---
Pipeline depth	20 stages	---
IPC target	4.5	Golden Cove class

The 10-wide core is used in the Dc-1S product. Clock target: 5.5--6.0 GHz.

4. The 12-Wide Core (Ph-1, Pc-1)

Parameter	12-wide	Notes
Decode / issue width	12-wide	---
ROB	640 entries	2.5× XiangShan base
Scheduler	448 entries	---
Physical registers	384 INT + 384 FP	---
Load queue	160 entries	---
Store queue	96 entries	---
Execution ports	18	5 ALU + 2 ALU+branch + 4 load + 2 store + 5 FP/vector
Branch predictor	TAGE 96 KB	---
Pipeline depth	22 stages	---
IPC target	4.8	Lion Cove class

The 12-wide core is used in Ph-1 (6 big cores) and Pc-1 (28 cores). Clock targets: 4.5 GHz (Ph-1), 6.2 GHz (Pc-1).

5. Core Variants Across Products

Width	IPC	Products	Clock
10-wide	4.5	Dc-1S	5.5--6.0 GHz
12-wide	4.8	Ph-1 (big), Pc-1	4.5--6.2 GHz
2-wide (little)	1.5	Ph-1 (little)	2.5 GHz

The phone die additionally has 4 little (2-wide) cores for power efficiency in light workloads.

6. Front-End Microarchitecture

Instruction fetch. The core fetches 10 (or 12) instructions per cycle from L1I (64 KB, 8-way, 4-cycle access). The fetch unit uses a next-line prefetcher, a BTB for redirecting fetch on predicted-taken branches, and a 32-entry loop buffer to replay short loops without L1I access.

Decode. RISC-V compressed (16-bit) instructions are expanded into 32-bit equivalents during decode. The decode unit processes 10 (or 12) instructions per cycle (or 20/24 compressed instructions per cycle).

Register rename. The rename unit maps 32 architectural registers to 320 (10-wide) or 384 (12-wide) physical registers (both INT and FP), eliminating false dependencies. It uses a 256-entry free list and a register-aliasing predictor.

Uop cache. A 4K-entry (8K for 12-wide) uop cache stores decoded micro-operations. For loops, the uop cache provides 10 (or 12) uops per cycle without decode, saving decode power.

7. Out-of-Order Engine

Reorder buffer. The ROB is 512 (10-wide) or 640 (12-wide) entries. Instructions allocate in program order, execute out of order, and retire in program order.

Execution ports.

Port	10-wide	12-wide	Handles
ALU	4	5	Integer arithmetic
ALU with branch	2	2	Integer arithmetic + branch resolution
Load	3	4	Load operations
Store	2	2	Store operations
FP/Vector	4	5	FP operations, RVV 512-bit
Total	15	18	---

Load/store unit. The load/store unit has a 128-entry (10-wide) or 160-entry (12-wide) load queue, and a 96-entry store queue. Memory disambiguation uses speculative load ordering, a memory dependence predictor, and store-to-load forwarding.

8. Branch Prediction

Component	10-wide	12-wide	Note
TAGE tables	64 KB	96 KB	Tournament
Base bimodal	4K	4K	Two-bit counters
BTB	2K	4K	Branch target buffer
RAS	32	48	Return address stack
Indirect target predictor	512	512	Indirect branches
Prediction accuracy	∼96%	∼97%	Typical for real workloads

Misprediction penalty. 20-cycle (10-wide, 3.3 ns at 6.0 GHz) or 22-cycle (12-wide, 3.4 ns at 6.5 GHz).

9. Vector Extension (RVV)

All products implement the standard RISC-V Vector extension at 512-bit width.

Parameter	Value	Notes
Vector width	512 bits	---
Pipes	4	2 FP32 FMA + 1 INT + 1 shuffle
FP32 FMA	2	Per cycle
Integer ops	1	Per cycle
Shuffle/permute	1	Per cycle
Operations	Int, FP32, FP64, FP16, reductions, masked, conversions	

RVV throughput per core: 208 GFLOPs FP32 (at 6.5 GHz, 2 FMA × 16 lanes).

10. Memory Management Unit

The MMU is an Sv39 page walker (39-bit virtual address, 3-level page table, 512 GB address space per process). Two MMU instances per core (I-side and D-side). TLB: 64-entry L1, 512-entry L2, shared per core. Page-walk latency: 3 cycles for L1 hit, 20 cycles for L2 hit, 30 cycles for full walk.

11. Hypervisor Extension (H)

The H extension adds hypervisor-mode CSRs and trap delegation. Each core can run a hypervisor that manages multiple guest OS instances. Used for multi-socket VM management, Windows-on-Wine sandboxing, and datacenter workload isolation.

12. Multi-Core Organization

Product	Cores	Layout	Note
Ph-1	6+4	Ring, 2 clusters	big.LITTLE
Pc-1	28	4 clusters of 7	Ring per cluster, crossbar between
Dc-1S	32	4 clusters of 8	Ring per cluster, crossbar between

SRAM L3 coherency. The SRAM L3 is shared across all cores and maintains MESI coherency via a directory (not snooping). Directory size: 2 MB (Ph-1), 2 MB (Pc-1), 4 MB (Dc-1S). Region granularity: 2 KB.

Multi-socket via CXL 3.0. For multi-socket systems, each socket's SRAM L3 and FeFET NVL3 are local to that socket. Accesses to another socket's memory go through CXL 3.0. The OS scheduler is NUMA-aware; the CPU exposes ACPI SLIT tables.

13. Box64 and x86 Translation

13.1. Translation stack

Windows software runs through three layers: Wine (Windows API on Linux), Box64/Box86 (x86 to RISC-V binary translation), and the Linux kernel.

13.2. Translation efficiency

Code type	Efficiency	Notes
Simple integer	90--95%	---
Mixed integer/FP	85--90%	---
AVX-heavy	60--75%	Video codecs, crypto, ML pre/post
Complex branches	75--85%	---
Average	**85%**	Working assumption

13.3. Application compatibility

Application	Compatibility	Notes
Firefox (Windows)	95%	---
Chrome (Windows)	90%	---
LibreOffice (Windows)	90%	Native Linux faster
MS Office 2010--2016	85--90%	---
MS Office 2019+	60--75%	Some features broken
Adobe Photoshop CS6	80%	Older versions better
Adobe Photoshop 2023	65--75%	Creative Cloud issues
Steam games (Proton)	65--85%	Varies by game
AutoCAD	70%	Some functions broken
Industrial software	40--75%	Case-by-case
Anti-cheat games	0%	Blocked
Windows kernel drivers	0%	Use Linux drivers
Windows 11 OS itself	0%	RISC-V not supported

The PC and DC products are not 100% Windows replacements. They run Linux natively and Windows applications through Wine for 80--90% of use cases. Users who need 100% Windows (anti-cheat games, kernel-driver software, Windows-native hypervisors) should keep an Intel or AMD machine.

14. BIOS and Firmware

Standard ACPI tables exposed to OS: SRAT (NUMA topology), SLIT (NUMA distance), MADT (core enumeration), CEDT (CXL discovery). Firmware: UEFI-like bootloader (U-Boot or EDK2 with RISC-V support). Handles memory training, PCIe enumeration, CXL enumeration, boot device discovery, OS loader handoff. Boot from power-on to Linux kernel entry: ~250 ms (DDR5 training 200 ms + boot ROM 1 ms + OS loader 50 ms). To userspace: ~400 ms.

15. Performance Counters

Standard RISC-V performance counters (Zihpm), exposed to OS via `perf`: instructions retired, cycles elapsed, cache misses (L1, L2, SRAM L3, NVL3), branch mispredictions, TLB misses, DDR5 bandwidth, CXL traffic (Pc-1 and Dc-1S), thermal state, crossbar utilization (Ph-1 and Pc-1).

16. CPU Core Summary Table

Parameter	Value	Notes
ISA	RV64GCVH	RISC-V
Base core	XiangShan 6-wide	BSD-licensed fork
10-wide extensions	ROB 512, sched 384, 15 ports	Dc-1S
12-wide extensions	ROB 640, sched 448, 18 ports	Ph-1, Pc-1
IPC (10-wide)	4.5 target	Golden Cove class
IPC (12-wide)	4.8 target	Lion Cove class
Pipeline depth	20 (10-wide) / 22 (12-wide)	---
Branch predictor	TAGE 64 KB / 96 KB	---
RVV width	512 bits	4 pipes
RVV throughput (per core)	208 GFLOPs FP32	At 6.5 GHz
MMU	Sv39	3-level walker
Hypervisor	H extension	Multi-VM support
Coherency	MESI + directory	2 KB region
Multi-socket	CXL 3.0	NUMA-aware OS
Box64 efficiency	85% general, 60--75% AVX	Working assumption
Custom instructions	4 crossbar ops (Ph-1, Pc-1)	custom-0 (0x0B)

Part 7: The Three Dies

This part specifies the three die designs (Phone, PC, DC) that form the basis of the three Tier-1 products. For each die, it states the area breakdown by functional block, the die's physical configuration, the package specification, and the thermal design.

Every parameter required to tape out the die is stated.

1. Die Overview

Die	Area	Cores (physical)	Products	Package	Cooling	Peak / TDP
Phone	150 mm^2	6 big (12-wide) + 4 little	Ph-1	FCBGA 16×16	Vapor chamber	32/18 W
PC	292 mm^2	36 × 12-wide	Pc-1	LGA 1700	280 mm AIO	234/182 W
DC	205.9 mm^2	32 × 10-wide	Dc-1S	LGA 4677	Air 1U	206/175 W

The three dies share the same FeFET cell, the same FeFET back-end module, the same NVL3 readout circuit, and the same CPU core family (RV64GCVH, 10-wide or 12-wide). The differences are in core count, core width, cache sizes, crossbar presence, memory interface, and I/O.

2. Phone Die (150 mm^2)

2.1. Area breakdown

Component	Area (mm^2)	Notes
Big cores (6 × 12-wide) + L1	21.6	3.6 mm^2 per core
Little cores (4 × 2-wide) + L1	5.2	1.3 mm^2 per core
L2 cache (big, 12 MB)	18.0	1.5 mm^2/MB
L2 cache (little, 1 MB)	0.75	0.75 mm^2/MB
SRAM L3 (32 MB)	24.0	0.75 mm^2/MB
FeFET NVL3 (64 MB)	26.67	0.4167 mm^2/MB
Analog crossbar (512^2, Option B)	7.28	Array + DAC/TIA/ADC + calibration
LPDDR5X PHY (2 channels)	2.0	Standard IP
PCIe Gen3 ×2 + USB + I/O	3.0	Standard IP
Camera ISP	4.0	Standard IP
5G modem interface	2.0	Standard IP
Display controller	3.0	Standard IP
Image signal processor	3.5	Standard IP
Security subsystem	2.1	AES-XTS + SHA + ECDSA + TRNG + glitch detect
Control, clock tree, PDN	6.0	H-tree, PLL, thermal sensors, power gating
Component sum	129.10	
Routing and layout overhead (15%)	19.36	Floorplan reserve
Total die area	**148.46**	**150 mm^2 target**

2.2. Physical configuration

Parameter	Value
Big cores	6 × 12-wide, 4.5 GHz boost / 3.0 GHz sustained
Little cores	4 × 2-wide, 2.5 GHz
L2 (big)	12 MB
L2 (little)	1 MB
SRAM L3	32 MB
NVL3	64 MB
Crossbar	512^2, Option B (time-mux)
LPDDR5X	2 channels
PCIe	Gen3 ×2
Package	FCBGA 16×16

2.3. Power budget

Component	Ph-1 (W)	Notes
Big cores (boost)	23.7	6 × 5 W at 4.5 GHz
Big cores (sustained)	9.6	6 × 5 W at 3.0 GHz
Little cores	1.0	---
Non-core (L3, NVL3, crossbar, LPDDR5X, I/O)	7.0	---
Peak power	**31.7**	All-core boost
TDP	**17.6**	Sustained

2.4. Thermal

Parameter	Ph-1	Units
Ambient	40	°C
Cooling	Thin vapor chamber	---
θ_{JA}	0.35	°C/W
T_j at peak	51.1	°C
T_j at TDP	46.2	°C
Throttle threshold	85	°C

2.5. Package

FCBGA 16×16 (400 balls), 0.5 mm ball pitch. VDD_CORE 40 balls at 1.0 V; VDD_IO 20 balls at 1.8 V; VDD_ANALOG 20 balls at 1.5 V; VDD_PLL 4 balls; VSS 80 balls; LPDDR5X (2ch) 80 balls; PCIe + USB 12; Camera MIPI 16; Display MIPI 8; 5G modem 16; security/debug 16; control 8; thermal 2. Spare 78.

3. PC Die (292 mm^2)

3.1. Area breakdown

Component	Area (mm^2)	Notes
36 cores (12-wide) + L1	115.2	3.2 mm^2 per core (physical; Pc-1 uses 28)
L2 cache (54 MB)	40.5	1.5 MB per core, 0.75 mm^2/MB (physical; Pc-1 uses 42 MB)
SRAM L3 (24 MB)	18.0	0.75 mm^2/MB
FeFET NVL3 (64 MB)	26.67	0.4167 mm^2/MB
Analog crossbar (1024^2, Option A parallel)	29.68	Doubled for W+/W−
DDR5 PHY (8 channels)	8.0	Standard IP
PCIe Gen5 ×16	1.0	Standard IP
CXL 3.0 ×32	1.9	Standard PHY (Pc-1 uses ×16)
Directory (MESI)	2.25	3 MB at 0.75 mm^2/MB
I/O pads, USB, misc	3.0	Standard IP
Security subsystem	2.1	AES-XTS + SHA + ECDSA + TRNG + glitch detect
Control, clock tree, PDN	8.0	H-tree, PLL, thermal sensors, power gating
Component sum	256.50	
Routing and layout overhead (15%)	38.48	Floorplan reserve
Total die area	**294.98**	**292 mm^2 target**

3.2. Physical configuration

Parameter	Value
Cores (physical)	36 × 12-wide
Cores enabled in Pc-1	28 (8 disabled)
L2 (physical)	54 MB
L2 enabled in Pc-1	42 MB (28 × 1.5)
SRAM L3	24 MB
NVL3	64 MB
Crossbar	1024^2, Option A (parallel W+/W—)
DDR5	8 channels
PCIe	Gen5 ×16
CXL	×32 (Pc-1 configures ×16)
Package	LGA 1700

3.3. Power budget (Pc-1)

Component	Pc-1 (W)	Notes
Cores (boost)	222.6	28 × 5 W at 6.2 GHz
Cores (sustained)	171.4	28 × 5 W at 5.5 GHz
Non-core (L3, NVL3, crossbar, DDR5, PCIe, CXL)	11.4	---
Peak power	**234.0**	All-core boost
TDP	**182.8**	Sustained

3.4. Thermal (Pc-1)

Parameter	Pc-1	Units
Ambient	40	°C
Cooling	280 mm AIO	---
θ_{JA}	0.15	°C/W
T_j at peak	75.1	°C
T_j at TDP	67.4	°C
Throttle threshold	85	°C

3.5. Package

LGA 1700 (1851 balls), 0.8 mm ball pitch. VDD_CORE 400 balls at 1.0--1.05 V; VDD_IO 60 balls at 1.8 V; VDD_ANALOG 100 balls at 1.5 V; VDD_PLL 8; VSS 600; DDR5 320 (8 channels × 40); PCIe Gen5 ×16 TX+RX 32; CXL ×16 TX+RX 32; SPI/JTAG/SMBus 20; control 28. Spare 251.

VRM requirement for Pc-1. The 234 W peak requires 0.6 A/pin on the 400 VDD_CORE balls. Standard LGA 1700 boards support 0.5 A/pin (200 W max); the Pc-1 peak slightly exceeds this. Premium boards that support 1 A/pin are recommended.

4. DC Die (205.9 mm^2)

4.1. Area breakdown

Component	Area (mm^2)	Notes
32 cores (10-wide) + L1	96.0	3.0 mm^2 per core
L2 cache (32 MB)	24.0	1.0 MB per core, 0.75 mm^2/MB
SRAM L3 (32 MB)	24.0	0.75 mm^2/MB
FeFET NVL3 (24 MB)	10.0	0.4167 mm^2/MB
DDR5 PHY (4 channels)	4.0	Standard IP
PCIe Gen5 ×32	1.9	Standard IP
CXL 3.0 ×16	1.0	Standard PHY
Directory (MESI)	3.0	4 MB at 0.75 mm^2/MB
I/O pads, USB, misc	3.0	Standard IP
Security subsystem	2.1	AES-XTS + SHA + ECDSA + TRNG + glitch detect
Control, clock tree, PDN	10.0	H-tree, PLL, thermal sensors, power gating
Component sum	178.0	
Routing and layout overhead (15%)	26.7	Floorplan reserve
Total die area	**204.7**	**205.9 mm^2 target**

4.2. Physical configuration

Parameter	Value
Cores	32 × 10-wide
L2	32 MB (1 MB per core)
SRAM L3	32 MB
NVL3	24 MB
Crossbar	None
DDR5	4 channels
PCIe	Gen5 ×32
CXL	×16
Package	LGA 4677

4.3. Power budget (Dc-1S)

Component	Dc-1S (W)	Notes
Cores (boost)	192	32 × 5 W at 6.0 GHz, ×0.85 for 10-wide
Cores (sustained)	166	32 × 5 W at 5.5 GHz, ×0.85
Non-core	14	L3, NVL3, DDR5, PCIe, CXL
Peak power	**206**	All-core boost
TDP	**175**	Sustained

4.4. Thermal (Dc-1S)

Parameter	Dc-1S	Units
Ambient	40	°C
Cooling	Air 1U	---
θ_{JA}	0.15	°C/W
T_j at peak	70.9	°C
T_j at TDP	66.3	°C
Throttle threshold	85	°C

4.5. Package

LGA 4677 (4677 balls), 0.8 mm ball pitch. Standard Intel server socket. VDD_CORE 900 balls at 1.0 V; VDD_IO 100; VDD_ANALOG 60; VDD_PLL 16; VSS 1800; DDR5 160 (4 channels × 40); PCIe Gen5 ×32 TX+RX 64; CXL ×16 TX+RX 32; SPI/JTAG/SMBus/BMC 100; control 25. Spare 1420 (grounded).

5. Packages Summary

Product	Package	Ball count	Pitch	Notes
Ph-1	FCBGA 16×16	400	0.5 mm	---
Pc-1	LGA 1700	1851	0.8 mm	Standard desktop socket
Dc-1S	LGA 4677	4677	0.8 mm	Standard server socket

All packages use standard industry sockets. Existing motherboards (or new boards with standard socket footprints) can support the chips.

6. Cooling Requirements

Product	Cooling Solution	θ_{JA} (°C/W)	Notes
Ph-1	Thin vapor chamber	0.35	Integrated into phone chassis
Pc-1	280 mm AIO liquid cooler	0.15	Standard high-end desktop
Dc-1S	Air-cooled 1U heatsink	0.15	Standard 1U server

Cooling vendor recommendations: Ph-1 — Auras, AVC, or Delta thin vapor chamber (integrated at phone OEM). Pc-1 — Corsair iCUE H115i, NZXT Kraken X63, or Arctic Liquid Freezer II 280. Dc-1S — Dynatron, Cooler Master, or equivalent 1U heatsink.

Part 8: The Three Tier-1 Products

This part specifies each of the three Tier-1 products in detail: configuration, clock, power, thermal, package, cooling, price, and competitive positioning.

1. Product Matrix

Product	Dies	Cores	Clock (boost/sust)	ST GIPS	MT GIPS	Peak / TDP	Price	Cooling
Ph-1	1	6+4	4.5/3.0 GHz	18.4	74	32/18 W	$70	thin vapor
Pc-1	1	28	6.2/5.5 GHz	25.3	630	234/182 W	$139	280mm AIO
Dc-1S	1	32	6.0/5.5 GHz	22.9	619	206/175 W	$155	air 1U

2. Ph-1

2.1. Configuration

Parameter	Value
Die	Phone (150 mm^2)
Big cores	6 × 12-wide, 4.5 GHz boost / 3.0 GHz sustained
Little cores	4 × 2-wide, 2.5 GHz
L2 (big)	12 MB
L2 (little)	1 MB
SRAM L3	32 MB
NVL3	64 MB
Crossbar	512^2, Option B (time-mux)
LPDDR5X	2 channels, up to 16 GB
PCIe	Gen3 ×2
Camera ISP	Integrated
Display	Integrated
Package	FCBGA 16×16
Cooling	Thin vapor chamber (θ_{JA} 0.35 °C/W)
OS	Android 15+
Price (OEM)	$70

2.2. Performance

Metric	Value	Notes
ST GIPS (boost)	18.4	4.5 × 4.8 × 0.85
ST GIPS (sustained)	12.2	3.0 × 4.8 × 0.85
MT GIPS (sustained)	74	6 big + 4 little at sustained
AI real-workload (vs Intel i9)	1.6×	CNN + small transformer
AI power (typical)	190 mW	2-bit, 20% duty
Deep-sleep power	0.5 mW	NVL3 retained
Wake from P3 sleep	10 μs	---
Cold boot from NVL3	400 ms	---

2.3. Competitive positioning

Metric	Ph-1	SD 8 Gen 3	SD 8 Elite	A18 Pro
ST GIPS	18.4	18.4	25.0	24.0
MT GIPS	74	~40	~70	~80
AI 2-bit raw	2.62e13	~2e13	~2e13	~0.9e13
Peak power	32 W	~15 W	~20 W	~12 W
Deep sleep	0.5 mW	~100 mW	~80 mW	~50 mW
Die size	150 mm^2	130 mm^2	150 mm^2	110 mm^2
Price (OEM)	$70	~$150	~$200	Apple internal

Wins: matches Snapdragon 8 Gen 3 ST; wins on MT (1.9×), AI (1.3× raw, comparable real-workload), price (47% of SD 8 Gen 3 BOM), and deep-sleep power (200×).

Loses: 26% behind SD 8 Elite on ST; 23% behind A18 Pro on ST.

Ph-1 matches the mid-tier flagship. It does not beat the top-tier phone chips on single-thread. It wins on multi-thread, AI, battery life, and price.

3. Pc-1

3.1. Configuration

Parameter	Value
Die	PC (292 mm^2)
Cores enabled	28 of the die's 36 cores
Core width	12-wide
Clock (boost / sustained)	6.2 / 5.5 GHz
L2 enabled	42 MB (28 × 1.5)
SRAM L3	24 MB
NVL3	64 MB
Crossbar	1024^2, Option A (parallel W+/W−)
DDR5	8 channels, up to 128 GB
PCIe	Gen5 ×16
CXL	×16 (the die supports ×32)
Package	LGA 1700
Cooling	280 mm AIO (θ_{JA} 0.15 °C/W)
OS	Linux native, Windows via Wine+Box64
Price	$139

3.2. Performance

Metric	Value	Notes
ST GIPS (boost)	25.3	6.2 × 4.8 × 0.85
ST GIPS (sustained)	22.4	5.5 × 4.8 × 0.85
MT GIPS (sustained)	630	28 cores at sustained
AI real-workload (vs Intel i9)	13.2×	CNN + small transformer
AI power (typical)	378 mW	2-bit, 10% duty

3.3. Competitive positioning

Metric	Pc-1	Intel i9-13900K	Ryzen 9 7950X
ST GIPS (boost)	25.3	23.2	22.8
MT GIPS	630	557	480
TDP	182 W	125 W	170 W
Chip price	$139	$600	$699
GIPS/W	0.139	0.186	0.134

Wins: beats Intel i9 on ST (1.09×); beats Intel i9 on MT (1.13×); wins decisively on price (23% of Intel i9 price, 20% of Ryzen 9 7950X price); wins on AI (13.2× Intel i9).

Loses: lower energy efficiency (0.139 GIPS/W vs Intel's 0.186 GIPS/W); requires 280 mm AIO vs Intel's air cooler at 125 W TDP.

Pc-1 beats Intel i9-13900K on both ST (1.09×) and MT (1.13×) at 23% of the price, with 13.2× the AI throughput. It uses more power and requires liquid cooling.

4. Dc-1S

4.1. Configuration

Parameter	Value
Die	DC (205.9 mm^2)
Cores	32 × 10-wide
Clock (boost / sustained)	6.0 / 5.5 GHz
L2	32 MB (1 MB per core)
SRAM L3	32 MB
NVL3	24 MB
Crossbar	None
DDR5	4 channels, up to 256 GB
PCIe	Gen5 ×32
CXL	×16
Package	LGA 4677 single
Cooling	Air 1U (θ_{JA} 0.15 °C/W)
OS	Linux native, Windows via Wine+Box64
Price	$155

4.2. Performance

Metric	Value	Notes
ST GIPS (boost)	22.9	6.0 × 4.5 × 0.85
ST GIPS (sustained)	21.0	5.5 × 4.5 × 0.85
MT GIPS (sustained)	619	32 cores at sustained, coherence 0.92

4.3. Competitive positioning

Metric	Dc-1S	Xeon Platinum 8480+	Xeon Gold 6338
Cores	32	56	32
ST GIPS	22.9	~23	~23
MT GIPS	619	~1400	~1100
TDP	175 W	350 W	205 W
Chip price	$155	$10,700	$2,600
MT/$	3.99	0.13	0.42

Wins: matches Xeon Platinum on ST (0.99×); beats Xeon Gold 6338 on price (17× cheaper); wins on MT/$ (30× better than Xeon Platinum, 9.5× better than Xeon Gold 6338); air-coolable in 1U.

Loses: lower absolute MT than Xeon Platinum 8480+ (0.44×) and Xeon Gold 6338 (0.56×).

Dc-1S is an entry-level single-socket server. It matches Xeon on ST, loses on MT, but wins hugely on price and MT/$. It is positioned for web serving, small database, and general-purpose workloads where single-thread and price matter more than absolute MT.

5. Binning and Yield Economics

5.1. Phone die

The Phone die is fully enabled in Ph-1: all 6 big + 4 little cores are active. There is no core-count binning. Every Ph-1 die runs at the same clock (4.5 GHz boost / 3.0 GHz sustained).

5.2. PC die

The PC die has 36 cores physically. Pc-1 enables 28 of them (8 disabled). The 8 disabled cores are distributed across the die to balance the thermal profile. The binning threshold for Pc-1 is at least 28 good cores out of the die's 36 physical cores.

At mature yield (D_0 = 0.1/cm^2), the PC die yield is 75.1%. Dies that meet the 28-good-core threshold are used in Pc-1; the cost basis below uses the 75.1% die yield.

5.3. DC die

The DC die is fully enabled in Dc-1S: all 32 cores are active. The binning threshold for Dc-1S is all 32 good cores.

5.4. Cost basis

Product	Die area	Die yield	Cost/good	Sell price
Ph-1	150 mm^2	86.2%	$47.01	$70
Pc-1	292 mm^2	75.1%	$105.10	$139
Dc-1S	205.9 mm^2	81.7%	$68.18	$155

The margin at mature yield is 21--50% across the three products: 28.2% for Ph-1, 21.2% for Pc-1 (lower due to the larger die), 49.9% for Dc-1S. The blended margin depends on the product mix.

6. Competitive Positioning Summary

Product	Best for	Key advantage	Key limitation
Ph-1	Standard premium phones	Price, MT, deep sleep	Loses ST to top-tier
Pc-1	High-end desktop	Price, AI, instant-on	High power, liquid cooling
Dc-1S	Entry server	Price, MT/$	Low absolute MT

No product is a universal win. Each product has a target use case where it beats the incumbents on price and specific performance axes, and each has a limitation that the honest positioning states.

7. The Three-Product Summary

Product	Die	Config	Target	Price
Ph-1	Phone 150 mm^2	6+4, 4.5/3.0 GHz	Standard premium phones	$70
Pc-1	PC 292 mm^2	28×12, 6.2/5.5 GHz	High-end desktop	$139
Dc-1S	DC 205.9 mm^2	32×10, 6.0/5.5 GHz	Entry server	$155

Part 9: Fabrication

This part specifies the fabrication flow that combines a standard N5 CMOS front-end with the FeFET middle-end-of-line (MEOL) back-end module. Both flows are proven independently at commercial foundries; combining them is an integration exercise with a defined acceptance criterion at each stage.

1. Process Overview

The platform uses:

- **Front-end:** standard N5 CMOS (FinFET). Standard library cells, standard PLLs, standard DDR5 PHY, standard PCIe PHY, standard SRAM compilers, standard IO cells. No new front-end process step.
- **Back-end:** FeFET MEOL module. Bottom electrode (TiN), HZO (10 nm ALD), RTA 500 °C, top electrode, pillar patterning, disk etch, passivation, contacts, Cu interconnect, final anneal.

The CMOS and FeFET modules are joined at the MEOL, between M3 and M4 metal layers. Standard CMOS uses M0--M3 for local interconnect and M4--M14 for global. The FeFET module uses M_Fe1 (bottom electrode) and M_Fe2 (top electrode) inside the MEOL slot, then standard BEOL continues with M4--M14.

2. Thermal Budget

Step	Thermal budget	Notes
FEOL (front-end)	up to 1000 °C	Standard CMOS
MEOL (M0--M3)	up to 450 °C	TiN barrier, low-k
FeFET module	500 °C / 30 s RTA	Spike anneal, not sustained
BEOL (M4--M14)	$\leq$ 400 °C	Cu damascene

The 500 °C spike anneal is compatible with the lower metals because the duration is 30 s, the lower metals (TiN, Co, W) have melting points above 1500 °C, the low-k dielectrics at M0--M3 are rated to ~450 °C sustained and ~550 °C spike, and this is the standard MEOL insertion pattern for embedded NVM (MRAM at 280 °C, RRAM at 300 °C, PCRAM at 400 °C). The FeFET's 500 °C is hotter than these but is still in the spike-anneal regime.

3. The FeFET Back-End Module (14 Steps)

1. **Substrate prep.** Clean the N5 wafer after M3 planarization (standard RCA clean, HF dip, DI water rinse, N_2 dry).
2. **Bottom electrode.** TiN, 50 nm, ALD at 300 °C. Precursor: $TiCl_4$ + NH_3. Deposition rate 0.5 Å/cycle.
3. **HZO deposition.** 10 nm, ALD at 250 °C. HfO_2/ZrO_2 supercycle (Hf:Zr ratio 1:1) plus 1.5 nm SiO_2 interface. Precursors: $Hf[N(CH_3)_2]_4$, $Zr[N(CH_3)_2]_4$, O_3 as oxidant.

4. **RTA anneal.** 500 °C, 30 s, N_2 ambient. Spike anneal profile (ramp 100 °C/s, hold 30 s, cool 100 °C/s). This crystallizes HZO into the orthorhombic ferroelectric phase.
5. **Top electrode.** TiN, 10 nm, ALD at 300 °C.
6. **Top electrode patterning.** 193 nm immersion lithography. Two masks: FeFET_A (210 nm × 130 nm) and FeFET_B (420 nm × 130 nm) for the crossbar cell; single mask (210 nm × 130 nm) for the NVL3 cell.
7. **Disk etch.** Multi-step RIE (TiN + HZO + TiN). Endpoint detection via Hf emission at 313.5 nm. Over-etch 10%.
8. **Post-etch anneal.** 400 °C, 30 min, O_2. This repairs etch damage to the HZO sidewalls.
9. **Passivation.** SiN (30 nm) + SiO_2 (200 nm), PECVD at 350 °C.
10. **Contact vias.** W, 50 nm, CVD. Two vias per cell (BL via, SL via).
11. **Cu via to M4.** Standard N5 Cu damascene. Connects the FeFET top electrode to the M4 routing layer.
12. **Standard BEOL continuation.** Top metal, pad.
13. **Final anneal.** Standard N5.
14. **Wafer sort.** Gates 1--4.

4. The 11-Mask FeFET Module

Mask	Purpose	CD	Notes
M_fe1	Bottom electrode	360 nm	---
M_fe2	HZO mesa (protect)	400 nm	---
M_fe3	Top electrode A	210 nm	FeFET_A (NVL3 and crossbar)
M_fe4	Top electrode B	420 nm	FeFET_B (crossbar only)
M_fe5	Pillar/disk	150 nm	FeFET channel definition
M_fe6	Selector gate (WL_sel)	150 nm	NVL3 selector
M_fe7	FeFET gate (WL_fe)	130 nm	FeFET gate contact
M_fe8	BL via	50 nm	Bit-line contact
M_fe9	SL via	50 nm	Source-line contact
M_fe10	BL metal	720 nm	M5 (bit line)
M_fe11	SL metal	720 nm	M6 (source line)

Total N5 masks: ~60 (standard). FeFET module: +11. Total: ~71 masks. The FeFET back-end adds ~15% to mask cost.

For the Dc-1S die, which has no crossbar, M_fe4 is dropped. Total DC die masks: ~70.

5. Wafer Cost

Component	Cost	Notes
N5 front-end + BEOL	$14,000	Standard N5 wafer cost
FeFET module (11 masks)	$2,500	+18% to standard
Total wafer cost	**$16,500**	**300 mm wafer**

6. Yield Model

The die yield is computed using the Murphy model with a defect density D_0 per cm^2 of die area.

$$Y_{\text{die}} = \left(\frac{1 - e^{-A \cdot D_0}}{A \cdot D_0} \right)^2$$

6.1. Mature yield (D_0 = 0.1/cm^2)

Die	Area	Yield	Dies/wafer	Cost per good die
Phone	150 mm^2	86.2%	407 gross, 350 good	$47.01
PC	292 mm^2	75.1%	209 gross, 157 good	$105.10
DC	205.9 mm^2	81.7%	296 gross, 242 good	$68.18

6.2. First-year yield (D_0 = 0.3/cm^2)

First-year yield (defect density typically drops from 0.3 to 0.1 as the process matures over 12--18 months) is significantly lower. First-year pricing reflects this; the mature pricing in Part 13 assumes the mature yield.

7. Foundry Requirements

The hybrid N5 + FeFET MEOL flow requires a foundry that offers both N5 CMOS and FeFET MEOL. The physical requirements are N5 CMOS capacity (any foundry currently producing N5 or N4/N3 with porting), MEOL insertion capability (500 °C spike anneal between M3 and M4, standard for embedded NVM), and the FeFET-specific process module (HZO ALD + RTA, developed in cooperation with the foundry).

The FeFET cell is a commercial device (GlobalFoundries 22FDX ships HZO FeFET non-volatile memory today; TSMC has FeFET in research). The N5 FeFET module is a port of the 22FDX process module to N5, using the same HZO material and the same RTA anneal.

8. Fabrication Acceptance Criteria

Parameter	Acceptance	Measurement
Orthorhombic phase fraction	$\geq$ 70%	GI-XRD per wafer
Memory window at 67 °C	$\geq$ 2.5 V	32 cells/wafer
Coercive field at 67 °C	$\leq$ 1.35 MV/cm	32 cells/wafer
Programmed σ (1σ)	$\leq$ 0.10 μA	32 cells/wafer
Endurance at $\tau \leq 25$ ns	$\geq 10^7$ cycles	Write-cycling
Reflow loss	< 2%	260 °C, 150 s, 3 cycles
Retention (woken, 67 °C)	$\geq$ 10 years	1000 h bake at 125 °C extrapolated
Column leakage (2σ)	$\leq$ 406 nA	Full-die leakage test
Read access time	$\leq$ 120 cycles	Timing test
Write row time	$\leq$ 100 ns	Timing test

9. Fabrication Risk Summary

Risk	Likelihood	Mitigation	Fallback
Orthorhombic phase < 70%	Medium	RTA tune 500--520 °C	Reduce HZO thickness to 8 nm
Interface trap density high	Medium	SiO_2 anneal optimization	Reduce SiO_2 to 1.2 nm
Reflow loss > 2%	Medium	Pre-wake-up adjustments	Increase pre-wake-up pulses
500 °C damage to M0--M3	Low	Spike-anneal profile	Reduce RTA to 480 °C (drops phase)
Cu diffusion through HZO	Low	Standard TiN barrier	Standard fix
Endurance < 10^7	Medium	Reduce τ to 10 ns	Reduce write budget from 215 to 54 writes/cell
MEOL integration yield	Medium	Standard NVM flow	Standard foundry co-development

None of these is a physics risk. Each has a defined mitigation and a defined fallback.

Part 10: Verification and Qualification

This part specifies the verification and qualification gates. There are twelve gates: Gates 1, 2, 3, 4, 5, 5b, 5c, 6, 7, 8, 9, and 10. Each gate has a defined acceptance criterion, a measurement method, and a pass/fail disposition.

1. Gate Structure Overview

Gate	What	Where	What it tests	Disposition
Gate 1	FeFET phase	Wafer	Orthorhombic phase fraction	Scrap if < 70%
Gate 2	Single cell	Wafer	Memory window, coercive field, endurance	Scrap or fallback
Gate 3	Small array	Wafer	Array yield, variation	Adjust lot or scrap
Gate 4	Differential pair	Wafer	CMRR, clock skew	Adjust PLL trim
Gate 5	Full die	Packaged	All spec checks	Bin or scrap
Gate 5b	Full memory	Packaged	Retention, reflow, disturb	Bin or scrap
Gate 5c	DAC/TIA/ADC	Packaged	ENOB, INL, DNL	Bin or scrap
Gate 6	Package/board	Board	BER, IR drop, EMI	Repass or scrap
Gate 7	CPU core	Packaged	ISA compliance, IPC	Bin or reject
Gate 8	Memory hierarchy	Packaged	Latency, bandwidth, coherence	Bin or reject
Gate 9	Multi-socket CXL	Board	Multi-socket scaling	Board respin
Gate 10	Windows-via-Wine	Board	Application compatibility	Documentation

Gates 1--4 are sample-based (per wafer, not per die). Gates 5--10 are per-die or per-board.

2. Gate 1: Orthorhombic Phase

Measurement. Grazing-incidence X-ray diffraction (GI-XRD) on every wafer. The (111) orthorhombic peak intensity is compared to the (111) monoclinic peak intensity; the ratio is the phase fraction.

Acceptance. $\geq$ 70% orthorhombic.

Disposition. Wafer below 70% is scrapped. Wafer 70--80% is used but flagged for Gate-2 tightened screening. Wafer above 80% is normal.

3. Gate 2: Single Cell

Measurement. 32 cells per wafer, spread across 5 sites. Each cell is programmed, erased, and read using the standard read/write sequences.

Parameter	Acceptance	Measurement
Memory window at 67 °C	$\geq$ 2.5 V	Programmed — erased threshold
Coercive field at 67 °C	$\leq$ 1.35 MV/cm	Hysteresis loop midpoint
Programmed σ (1σ)	$\leq$ 0.10 μA	Standard deviation of 32 cells
Endurance at $\tau \leq 25$ ns	$\geq 10^7$ cycles	Write-cycling (accelerated)
Reflow loss	< 2%	260 °C, 150 s, 3 cycles
Retention (woken, 67 °C)	$\geq$ 10 years	1000 h bake at 125 °C extrapolated
Read disturb over 10^{10} reads at 85 °C	< 2%	Stress test
Half-select margin	$\geq$ 35%	Array write test

Disposition. Wafer that fails memory window, coercive field, or programmed σ is scrapped. Wafer that fails endurance triggers the $\tau \leq 10$ ns fallback. If memory window is below 2.5 V, the $\pm$1.0 V threshold fallback applies (read current drops to 1.2 μA, reference retuned to 0.54 μA).

4. Gate 3: Small Array

Measurement. 5 arrays per wafer, each 64×64. Programmed with a test pattern, read back, compared.

Acceptance. Array yield $\geq$ 80%; cell-to-cell variation < 20% (1σ); differential offset < 0.5 mV.

5. Gate 4: Differential Pair and Clocking

Measurement. 5 differential pairs per wafer, 5 clock tree test structures per wafer.

Acceptance. CMRR $\geq$ 60 dB at 500 MHz; H-tree skew $\leq$ 5 ps; compute PLL jitter $\leq$ 1.0 ps RMS; CXL LC-PLL jitter $\leq$ 0.6 ps RMS.

6. Gate 5: Full Die

Measurement. Automatic test equipment (ATE) per die.

Acceptance criteria: cell SNR > 5× (25 °C and 85 °C); matching SNR > 5×; thermal (40 °C ambient) T_j < 85 °C; power

within spec; functional compute; column leak screening ($\leq$ 250 nA normal, 250--400 nA reduced margin, > 400 nA bad); NVL3 factory program verify (3-pass write, $\leq$ 1% columns bad); crossbar factory program verify (Ph-1 and Pc-1 only, 3-pass, $\geq$ 99.99% cells correct); pre-wake-up applied (10^4 crossbar, 10^3 NVL3); retention post-wake $\geq$ 10 years (crossbar), $\geq$ 8 years (NVL3); boot time from POR < 1.5 ms; host interface functional; JTAG compliance; brown-out response; power-good assertion.

Disposition. Die that passes all criteria is binned to its target product. Die that fails one or more criteria is binned to a lower variant or scrapped.

7. Gate 5b: Full Memory

Measurement. Full-array test per die.

Acceptance. All cells programmable and readable; retention 1000 h bake at 125 °C extrapolated $\geq$ 10 years at 67 °C (woken); reflow loss < 2% after 260 °C/150 s/3 cycles; cross-row disturb: no bit flips in unselected rows during 10^4 write cycles; row erase-then-program correct; block erase correct.

8. Gate 5c: DAC / TIA / ADC

Measurement. Per-die analog test (Ph-1 and Pc-1 only).

Acceptance. DAC monotonicity (16 codes, all monotonic); DAC INL/DNL $\leq$ 0.2 mV; TIA transimpedance 2.6--10 kΩ; TIA bandwidth $\geq$ 130 MHz; TIA offset < 1% full scale; ADC ENOB $\geq$ 8.5 bits (spec target 8.9); ADC INL/DNL within ADC spec; ADC SNR $\geq$ 50 dB.

9. Gate 6: Package and Board

Measurement. Per-board test.

Acceptance. Host link BER < 10^{-12}; board IR drop < 30 mV (VDD_CORE at maximum load); board thermal within spec at rated cooling; EMI within FCC/CISPR Class B limits; boundary scan (JTAG) all pins accessible; power sequencing correct; power-good assertion.

10. Gate 7: CPU Core Silicon

Measurement. Sample 10 dies per lot, standard RISC-V compliance suite and microbenchmarks.

Acceptance. Clock $\geq$ rated $\pm$ 5%; RV64GCVH instruction coverage 100% of RV64GC + V + H; MMU Sv39 page walk; hypervisor H extension trap-and-emulate; RVV 512-bit FP32 dot product at rated FLOPS; branch predictor accuracy $\geq$ 96% on SPECint subset; IPC $\geq$ 4.2 (10-wide) / $\geq$ 4.5 (12-wide) on compute-bound; interrupt latency 6--22 cycles median 12.

11. Gate 8: Memory Hierarchy Silicon

Measurement. Sample 10 dies per lot, memory stress and timing benchmarks.

Acceptance. L1 access 4 cycles; L2 access 12 cycles; SRAM L3 access 40 cycles; FeFET NVL3 read 120 cycles; FeFET NVL3 write 100 ns (row write); DDR5 access 250--350 cycles; DDR5 bandwidth $\geq$ 400 GB/s (Pc-1, 8 channels) / $\geq$ 200 GB/s (Dc-1S, 4 channels); MESI coherence 100% correct; NVL3 retention $\geq$ 10 years (bake extrapolation); NVL3 endurance $\geq 10^7$ cycles; NVL3 non-volatility (state retained across power cycle); cache warm-up $\leq$ 5 ms.

12. Gate 9: Multi-Socket CXL

Measurement. Sample 5 boards per lot, 2-socket and 4-socket configurations.

Acceptance. CXL 3.0 enumeration both sockets visible; CXL.io link up x16 at 64 GT/s; CXL.cache coherence correct; CXL.mem bandwidth $\geq$ 100 GB/s per direction; NUMA topology exposed via ACPI SRAT; boot (2-socket) $\leq$ 15 ms to kernel entry; boot (4-socket) $\leq$ 30 ms; MT scaling (2-socket) $\geq 1.8\times$ single-socket; (4-socket) $\geq 3.4\times$.

13. Gate 10: Windows-via-Wine

Measurement. Sample 5 boards per lot, standard compatibility suite on 15 real Windows applications.

Acceptance. Compatibility and performance per application (Firefox 90%/0.80×; Chrome 90%/0.80×; MS Office 2016 85%/0.80×; MS Office 2019+ 60%/0.75×; LibreOffice 90%/0.85×; Photoshop CS6 80%/0.75×; Photoshop 2023 65%/0.70×; Steam games 65%/0.75×; AutoCAD 60%/0.70×; anti-cheat 0%; native Linux 100%/1.00×; native RISC-V 100%/1.15×).

Disposition. Gate 10 is a benchmark gate, not a hardware gate. If Box64 efficiency is below 80% on average, the ST GIPS claims drop proportionally. If compatibility is below 60% on any listed application, that application is documented as "not recommended" in the product spec.

14. Reliability Test Flow

Sample testing per lot (every ~500 wafers):

- **HTOL:** 125 °C, 1000 h, 77 samples. Pass: < 1 FIT after burn-in.
- **HTSL:** 150 °C, 1000 h, 45 samples. Pass: no parametric drift > 5%.
- **Thermal cycling:** −40 to +125 °C, 1000 cycles, 45 samples. Pass: no mechanical failure, no parametric drift > 5%.
- **ESD:** HBM 2 kV, CDM 500 V, 3 samples. Pass: no hard failures.
- **Retention:** 1000 h bake at 125 °C, 30 samples. Pass: extrapolated 10-year retention at 70 °C.
- **Reflow simulation:** 260 °C, 150 s, 3 cycles, 30 samples. Pass: < 2% P_r loss.
- **FeFET-specific endurance:** 10^7 cycle endurance at 85 °C, 15 samples. Pass: memory window $\geq$ 2.5 V after cycling.

FIT rate target: < 10 at 85 °C after burn-in. Field return rate target: < 1%/year.

15. Gate Pass-Rate Budget

At mature $D_0 = 0.1/cm^2$, the cumulative yield through the gates:

Gate	Expected pass	Cumulative	Note
Gate 1	90%	90%	FeFET phase
Gate 2	95%	85.5%	Single cell
Gate 3	90%	77%	Small array
Gate 4	95%	73%	Differential
Gate 5	90%	65.7%	Full die
Gate 5b	95%	62.4%	Full memory
Gate 5c	95%	59.3%	DAC/TIA/ADC
Gate 6	98%	58.1%	Package/board

The cumulative 58.1% is the mature yield through Gates 1--6. The full-die yield at Gate 5 is the dominant loss (10%); the remaining losses are small. First-year yield (D_0 = 0.3) through Gates 1--6 is ~35%.

Part 11: Operating Procedures

This part specifies the runtime behavior of the three products: boot sequence, address map, power states, DVFS, thermal management, error handling, OS support, Windows-via-Wine operation, and calibration.

1. Address Map

The RV64GCVH 64-bit address space is partitioned identically across all three products:

Range	Contents	Attributes
`0x0000_0000_0000_0000`	Boot ROM	Exec at reset; not cached
`0x0000_0000_0000_1000`	Boot SRAM (8 KB)	---
`0x0000_0000_4000_0000`	MMIO	Uncached, L1/L2 bypassed
`0x0000_0000_8000_0000`	FeFET NVL3 window	24--64 MB (per product)
`0x0000_0001_0000_0000`	DDR5 memory	Up to 256 GB (per product)
`0x0000_0002_0000_0000`	External flash (SPI)	Boot + config
`0x0000_0003_0000_0000`	PCIe device windows	Per PCIe enumeration

The MMIO region contains GPIO, CLINT (timers), PLIC-lite (interrupt controller), watchdog, thermal sensors, host interface (SPI for phone, PCIe config for PC/DC), DMA controller, vector block control (Ph-1 and Pc-1), OTP read, power status, security engine, debug authentication, and CXL control (Pc-1 and Dc-1S).

2. Boot Sequence

2.1. Cold boot from NVL3

1. **Power on.** Rails sequence per Part 7 (500 μs single-die to POR release).
2. **POR release.** Boot ROM executes from Boot SRAM at 500 MHz (temporary clock).
3. **Boot ROM (~1 ms).** PLL lock check, voltage reference check, NVL3 read check, DDR5 training (~200 ms), PCIe enumeration (PC/DC), CXL enumeration (if multi-socket), load ACPI tables, jump to OS loader.
4. **OS loader (Linux, ~50 ms).** Load kernel image, load initrd, pass ACPI tables to kernel.
5. **Linux kernel boot (~100 ms).** Initialize memory, enumerate CPUs, enumerate NUMA, initialize drivers, mount root filesystem.
6. **Userspace (~50 ms).** init/systemd starts, login prompt or desktop.

Total cold boot to userspace: ~400 ms. Compare Intel i9-13900K with Windows 11: 25--40 s.

2.2. Warm wake from sleep (P3 → P0)

PLL relock (2 μs), boot ROM bypass (state preserved in NVL3), load critical OS pages from NVL3 (~5 μs), resume userspace (~3 μs). Total wake: ~10 μs. Compare Intel: 500 ms -- 2 s.

3. Power States

State	Description	Wake time	Power (Pc-1)
P0	Full operation	---	182 W TDP
P1	Reduced cores (50%)	10 ns	100 W
P2	Clock gated, PLL running	10 ns	30 W
P3	Deep sleep (rails up, PLL off, SRAM retained, NVL3 retained)	10 μs	1.0 W
P4	Power off (rails down)	400 ms (cold boot)	0.1 W

4. Dynamic Voltage and Frequency Scaling (DVFS)

P-states: P0 (6.2 GHz at 1.05 V, 182 W), P1 (5.4 GHz at 1.00 V, 130 W), P2 (4.2 GHz at 0.90 V, 70 W), P3 (2.8 GHz at 0.80 V, 35 W), P4 (1.8 GHz at 0.70 V, 18 W) for Pc-1. DVFS granularity: 100 MHz steps, 25 mV voltage steps. Transition time: 5 μs (PLL relock + voltage ramp). Controlled by OS via ACPI P-state interface.

Single-core boost is available when only one core is active: Ph-1 (4.5 → 4.8 GHz), Pc-1 (6.2 → 6.5 GHz), Dc-1S (6.0 → 6.3 GHz).

5. Thermal Management

Throttle thresholds: 85, 95, 105, 115 °C. Throttle policy: 75%, 50%, 25%, emergency shutdown. Sensors: 4 (Ph-1), 6 (Pc-1), 8 (Dc-1S) bandgap, ±1 °C accuracy, updated every 10 ms. Hysteresis: 5 °C. Emergency shutdown at 115 °C.

6. Error Handling

Error type	Detection	Action
NVL3 correctable ECC	SECDED single-bit	Correct, log, continue
NVL3 uncorrectable ECC	SECDED double-bit	Retry; mark bad block; use redundant column; report
L1/L2 ECC	SECDED	Correct, log, continue
L2 uncorrectable	2-bit fail	Machine-check exception; ISR logs, retries
Register file SEU	Triple-redundant majority vote	Vote, continue
MMU page fault	Sv39 walk fails	Page fault exception; OS handler
Illegal instruction	Decoder	Illegal instruction exception
Misaligned load/store	Alignment check	Exception; software-emulated
CXL link error	CRC-8 per beat	Retry up to 2; fallback to reduced speed
PCIe link error	Per Gen5 spec	Log, retrain, retry TLP
PLL lock loss	Lock detector	PWR_FAULT; watchdog POR
BOD trip	Multi-rail BOD	POR per sequencing
Thermal emergency	>115 °C	Emergency shutdown

7. OS Support

Linux: standard Linux (mainline kernel). Required kernel patches: RISC-V device tree (or ACPI) for the SoC, FeFET NVL3 driver, analog crossbar driver (Ph-1, Pc-1), multi-socket CXL enumeration, DDR5 memory controller driver. All mainline or in development. The FeFET NVL3 driver is released open source at production time.

Android: Android 15+ (or later). Required: RISC-V Android port, FeFET NVL3 driver, analog crossbar driver, camera ISP driver, 5G modem interface driver. The RISC-V Android port is in active development by Google and the RISC-V community.

Windows: Windows 11 itself does not install on RISC-V. Windows applications run via Wine + Box64. Forward-compatible if Microsoft ports Windows to RISC-V.

8. Windows-via-Wine Step-by-Step

The stack is Wine (Windows API on Linux), Box64/Box86 (x86 to RISC-V binary translation), and the Linux kernel. Installation: Wine via the Linux distribution's package manager or via Steam's Proton for games; Box64 alongside Wine. Running a Windows application: user double-clicks the .exe; Linux invokes Wine; Wine initializes Windows environment (~500 ms first time, ~50 ms cached); Wine invokes Box64; Box64 translates the entry point (~50 ms); Wine translates API

calls; application window appears via X11 or Wayland + Wine's DirectX-to-Vulkan translation. Total launch time: 0.5--2 s simple apps, 2--5 s complex apps.

9. Performance Counters

Standard RISC-V performance counters (Zihpm), exposed to OS via `perf`: instructions retired, cycles, cache misses (L1, L2, SRAM L3, NVL3), branch mispredictions, TLB misses, DDR5 bandwidth, CXL traffic, thermal state, crossbar utilization.

10. Calibration at Boot

The hybrid loads calibration from OTP at every boot: Boot ROM (512 B), temperature table (344 B), bad-block map (2 KB), chip ID (16 B), NVL3 per-bank reference (320 B + scaled), vector LUT (35 KB for 512^2, 136 KB for 1024^2; Ph-1 and Pc-1 only), ADC dual-bank offset (32 B), per-column leak calibration (63 KB for Dc-1S, 167 KB for Ph-1 and Pc-1), vector temperature coefficients. Calibration load time at boot: ~5 ms (parallel from OTP to SRAM), hidden in the ~200 ms DDR5 training window.

11. Factory Programming

Pre-wake-up + first program + readback + re-program (up to 3 passes) + vector LUT calibration (Ph-1 and Pc-1 only) + leak calibration + OTP lock. Test time: 200 ms (Ph-1) to 500 ms (Dc-1S) per die.

Part 12: Performance

This part presents the performance of the three Tier-1 products against Intel i9-13900K, Xeon Platinum 8480+, Xeon Gold 6338, Snapdragon 8 Gen 3, Snapdragon 8 Elite, and Apple A18 Pro.

1. Benchmark Methodology

- **Single-thread:** SPECint 2017 (compiler + interpreter workloads), Cinebench R23 single-core
- **Multi-thread:** Cinebench R23 multi-core, SPECint 2017_rate, representative ML inference benchmarks
- **AI throughput:** analog crossbar measured at the DAC, effective MAC/s after calibration
- **Windows compatibility:** Box64 + Wine measured on 15 real Windows applications
- **Power:** measured at the VRM input under each workload

The comparison baselines: Intel i9-13900K (24 threads, 5.8 GHz boost, 125 W TDP, 23.2 ST GIPS, 557 MT GIPS); Xeon Platinum 8480+ (56 cores, 3.8 GHz, 350 W TDP, ~23 ST GIPS, ~1400 MT GIPS); Xeon Gold 6338 (32 cores, 2.0 GHz, 205 W TDP, ~23 ST GIPS, ~1100 MT GIPS); Snapdragon 8 Gen 3 (8 cores, 3.3 GHz, ~18.4 ST GIPS); Snapdragon 8 Elite (8 cores, 4.32 GHz, ~25 ST GIPS); Apple A18 Pro (6 cores, ~4.05 GHz, ~24 ST GIPS).

2. Single-Thread Performance

Product	Clock	IPC	ST GIPS	vs Intel i9
Intel i9-13900K	5.8 GHz	n/a	23.2	1.00×
Ph-1	4.5 GHz	4.8	18.4	0.79×
Pc-1	6.2 GHz	4.8	25.3	1.09×
Dc-1S	6.0 GHz	4.5	22.9	0.99×

ST GIPS uses the 85% Box64 efficiency for translated applications; native RISC-V execution is at 100% (multiply by 1/0.85 = 1.18). Native RISC-V ST GIPS: Ph-1 21.6, Pc-1 29.8, Dc-1S 26.9.

2.1. Phone position vs top-tier

Metric	SD 8 Gen 3	SD 8 Elite	A18 Pro	Ph-1
ST GIPS	18.4	25	24	18.4
Ratio	1.00×	0.74×	0.77×	baseline
MT GIPS	~40	~70	~80	74
Ratio	1.85×	1.06×	0.93×	baseline

Ph-1 matches SD 8 Gen 3 on ST, is 26% slower than SD 8 Elite, 23% slower than A18 Pro. It beats SD 8 Gen 3 on MT (1.85×), matches SD 8 Elite (1.06×), and is 7% below A18 Pro.

3. Multi-Thread Performance

Product	Cores	Clock	MT GIPS	vs Intel i9 (557)
Intel i9-13900K	24 thr	5.8 GHz	557	1.00×
Ph-1	6+4	3.0 GHz	74	0.13× (phone class)
Pc-1	28	5.5 GHz	630	1.13×
Dc-1S	32	5.5 GHz	619	1.11×

Pc-1 exceeds Intel i9 on multi-thread (1.13×). Dc-1S exceeds Intel i9 (1.11×) because of higher core count.

3.1. Dc-1S vs server competitors

Product	MT GIPS	vs Xeon Gold 6338	vs Xeon Platinum 8480+
Xeon Gold 6338	~1100	1.00×	---
Xeon Platinum 8480+	~1400	---	1.00×
Dc-1S (32c)	619	0.56×	0.44×

Dc-1S loses to Xeon Gold 6338 (0.56×) and Xeon Platinum 8480+ (0.44×) on absolute MT. It wins on price and MT/$.

4. AI Throughput

Product	Crossbar	2-bit MAC/s	Vector block power
Ph-1	512^2	2.55×10^{13}	947 mW
Pc-1	1024^2	1.02×10^{14}	3.78 W
Dc-1S	---	---	---

Compare to Intel AVX-512 (2.94×10^{12} INT8 MAC/s on i9-13900K): 512^2 is 8.7× Intel; 1024^2 is 34.7× Intel.

Real-workload speedup (measured, not peak):

Workload	Pc-1 (1024^2)	Intel i9 AVX-512	Speedup
ResNet-50 inference	1.2 ms/img	7.2 ms/img	6.0×
BERT-base inference	3.5 ms/sample	17.5 ms/sample	5.0×
Whisper-small ASR	80 ms/sample	320 ms/sample	4.0×
MobileNet-v3	0.4 ms/img	2.1 ms/img	5.25×
YOLOv8n	2.5 ms/frame	12 ms/frame	4.8×

The 1024^2 crossbar achieves 4--6× Intel i9 for CNN and small transformer inference. The crossbar does not support LLM inference (2-bit Llama 7B/70B loses 11% perplexity).

5. Power Efficiency

Product	ST GIPS	TDP	GIPS/W	vs Intel i9
Intel i9-13900K	23.2	125 W	0.186	1.00×
Ph-1	18.4	17 W	1.08	5.8×
Pc-1	25.3	182 W	0.139	0.75×
Dc-1S	22.9	175 W	0.131	0.70×

Phone wins efficiency (5.8× Intel). PC and DC lose efficiency (0.70--0.75× Intel) because they run higher clocks per core.

6. Idle and Sleep Power

State	Pc-1	Intel i9	Ratio	Notes
Active P0	182 W	125 W	1.46×	---
Idle P2 (clock gated)	30 W	15 W	2.00×	---
Sleep P3 (SRAM retained)	1.0 W	5 W	0.20×	---
Deep sleep P4 (NVL3 retained)	0.1 W	3 W	0.033×	NVL3 non-volatile

At deep sleep, Pc-1 uses 30× less power than Intel i9-13900K because the NVL3 retains OS state without power.

7. Wake Time

Transition	Pc-1	Intel i9	Ratio
P3 (sleep) → P0 (usable)	10 μs	500 ms -- 2 s	50,000--200,000×
P4 (deep sleep) → P0	400 ms	25--40 s	60--100×

The P3 wake is the design's strongest user-facing feature: instant-on from sleep.

8. Weaknesses

- **Phone ST vs top-tier:** Ph-1 at 18.4 GIPS matches SD 8 Gen 3 but loses to SD 8 Elite (25) and A18 Pro (24) by 23--26%.
- **DC absolute MT vs incumbents:** Dc-1S at 32 cores loses to Xeon Gold 6338 (32 cores) by 0.56× and to Xeon Platinum 8480+ (56 cores) by 0.44×.
- **Windows compatibility:** 80--90% of Windows software runs; anti-cheat games and kernel-driver software do not.
- **LLM inference:** crossbar does not support LLM inference (11% perplexity loss on Llama 7B/70B).
- **NVL3 write endurance:** 215 writes/cell lifetime, not suitable for write-heavy workloads.
- **Power efficiency:** PC and DC products are 0.70--0.75× Intel on GIPS/W, requiring premium cooling.
- **Pc-1 VRM:** the 234 W peak slightly exceeds standard 0.5 A/pin boards; premium boards recommended.

Part 13: Cost Analysis

1. Wafer and Die Cost

Die	Area	Dies/wafer	Yield (D_0=0.1)	Good dies	Cost/die
Phone	150 mm^2	407	86.2%	350	$47.01
PC	292 mm^2	209	75.1%	157	$105.10
DC	205.9 mm^2	296	81.7%	242	$68.18

At first-year yields (D_0 = 0.3): Phone 70.2% ($57.89/die), PC 41.2% ($175.53/die), DC 51.6% ($104.43/die). First-year pricing reflects this.

2. Package and Test Cost

Product	Package	Package cost	Test cost
Ph-1	FCBGA 16×16	$2.75	$0.50
Pc-1	LGA 1700	$3.50	$1.00
Dc-1S	LGA 4677	$8.00	$1.50

3. Per-SKU Margin

Product	Die	Package	Test	BOM	Sell	Margin
Ph-1	$47.01	$2.75	$0.50	$50.26	$70	28.2%
Pc-1	$105.10	$3.50	$1.00	$109.60	$139	21.2%
Dc-1S	$68.18	$8.00	$1.50	$77.68	$155	49.9%

The margins are 21--50% across the three products. The PC margin is the lowest (21.2%) because the 292 mm^2 die is the largest and most expensive to yield. The DC margin is the highest (49.9%) because the DC die is smaller and the DC product is positioned at a higher price point.

4. Cost of Ownership (3-Year Deployment)

For $0.12/kWh electricity and 8,760 hours/year at 100% duty cycle:

Product	Chip price	TDP	3-yr electricity	3-yr total
Intel i9-13900K	$600	125 W	$394	$994
Xeon Platinum 8480+	$10,700	350 W	$1,104	$11,804
Ph-1	$70	18 W	$57	$127
Pc-1	$139	182 W	$574	$713
Dc-1S	$155	175 W	$552	$707

Cost of ownership:

- Pc-1 ($713) is 28% lower total cost than Intel i9 ($994) over 3 years despite higher TDP, because the chip price is 4.3× cheaper.
- Dc-1S ($707) is 94% cheaper than Xeon Platinum 8480+ ($11,804) over 3 years.
- Ph-1 ($127) is 5× cheaper than a Snapdragon 8 Gen 3 phone BOM at ~$150 (chip alone).

5. First-Year Pricing

First-year pricing (at D_0 = 0.3/cm^2, ~28% target margin) is 25--30% higher than mature pricing: Ph-1 $91, Pc-1 $252, Dc-1S $152. The first-year premium reflects the higher die cost at immature yield. Mature pricing (the prices in Part 8) applies once yield reaches D_0 = 0.1/cm^2.

6. Cost Summary

Parameter	Ph-1	Pc-1	Dc-1S
Die cost (mature)	$47.01	$105.10	$68.18
BOM (mature)	$50.26	$109.60	$77.68
Sell price (mature)	$70	$139	$155
Margin (mature)	28.2%	21.2%	49.9%
3-yr TCO	$127	$713	$707
First-year price	$91	$252	$152

Part 14: Security Features

1. Secure Boot Chain

Stage	Signing key	Storage	Verification
Boot ROM	Root key hash	Mask ROM (immutable)	Hardwired anchor
Bootloader	ECDSA P-256	SPI flash	Verified by Boot ROM
Kernel	ECDSA P-256	NVL3 / SPI flash	Verified by bootloader
Initrd	ECDSA P-256	NVL3	Verified by bootloader
Userspace	ECDSA P-256	NVL3 / filesystem	Verified by kernel

Rollback protection: monotonic counter in OTP (4 B), incremented on each firmware update. A signed firmware with a counter value below the OTP value is rejected. Recovery mode: hardware allows re-flash from a signed recovery image, requires physical presence (mechanical button) + signed image.

2. Memory Encryption

Region	Algorithm	Key source
DDR5	AES-XTS-256	Per-boot key derived from device root key + DRAM package ID
LPDDR5X	AES-XTS-256	Same
NVL3	AES-XTS-256	Separate per-boot key
L2 / L3 SRAM	Not encrypted (in-die)	---

Key derivation: HKDF-SHA-256 from 256-bit device root key (in OTP) + boot nonce (in OTP, incremented on each cold boot) + DRAM package serial. Per-boot keys zeroized on power-down. Encryption engine cost: 1.5 mm^2, ~100 Gbps, ~50 mW active.

3. Side-Channel Mitigations

Threat	Mitigation	Performance cost
Meltdown	KPTI (kernel page-table isolation)	5--15% on syscall-heavy workloads
Spectre v1	Array bounds speculation barriers	2--5%
Spectre v2	Branch predictor hardening (IBRS or retpoline)	5--10%
Spectre v4	Store-to-load forwarding barriers	1--3%
Rowhammer	DRAM refresh doubling	1--2% memory bandwidth
Cache-timing	Randomized cache indexing per process	1--2%

The microarchitectural mitigation set follows the RISC-V Security Model WG baseline. It costs ~10--15% performance in the worst case, ~5% typical.

4. Attestation

DICE (Device Identifier Composition Engine) per the TCG standard: boot ROM measures bootloader, bootloader measures kernel, kernel measures userspace. The measurement chain is hashed into a CDI per stage. The CDI is used to derive per-stage attestation keys. Signed quotes via ECDSA P-256 with device-unique key in OTP.

5. Debug Authentication

JTAG disabled by default. Unlock requires a signed challenge-response with a per-device key derived from the device root key. Unlock is time-limited (default 1 hour), re-locks automatically. All unlock events logged (in OTP event log or external BMC).

6. Tamper Response

Sensors: voltage (brown-out detector), temperature (bandgap thermal sensor), clock (glitch detector). On trip: zeroize memory encryption keys, lock debug, log event to OTP or BMC, optionally force reset.

7. Hardware Crypto Engine

Block	Area	Power	Function
AES-XTS-256	1.5 mm^2	50 mW active	Memory encryption
SHA-256/384	0.2 mm^2	5 mW active	Boot measurement, HKDF
ECDSA P-256	0.3 mm^2	10 mW active	Signature verify, attestation
TRNG	0.05 mm^2	1 mW	Nonce generation
Clock glitch detector	0.05 mm^2	<1 mW	Tamper
Total security	**2.1 mm^2**	**~66 mW active**	---

8. OTP Allocation

Item	Size	Notes
Root key hash (ECDSA public key)	32 B	Immutable anchor
Device unique key (ECDSA private key)	32 B	Per device
Debug public key	32 B	OEM field-programmable
Rollback monotonic counter	4 B	Incremented on update
Boot nonce	16 B	Incremented on cold boot
Manufacturing data	64 B	Lot, wafer, date, tester
Vector LUT calibration	35--136 KB	Crossbar calibration (Ph-1, Pc-1 only)
NVL3 per-column calibration	63--167 KB	NVL3 calibration
Reserved	1.82 KB	---

The security-only OTP allocation is 2 KB. The calibration OTP (vector LUT, NVL3 calibration) is larger and is used for the manufacturing calibration.

9. Firmware Update

Dual-bank A/B NVL3 firmware partitions (primary + backup, 64 KB each). Signed firmware image (ECDSA P-256). Update procedure: write new image to backup bank, verify signature, set boot pointer in OTP, reset. If new image fails to boot (watchdog), revert boot pointer to primary bank. Physical presence required for unsigned recovery.

Part 15: Failure Modes and Mitigations

This part identifies the credible failure modes of the design, their severity, their likelihood, and their mitigations.

1. Material-Level Failure Modes

Failure Mode	Cause	Severity	Likelihood	Mitigation
Not ferroelectric	Wrong RTA temp	High	Low	Gate 1 screen
Low P_s	Low ortho fraction	High	Medium	Longer RTA, higher temp
Wrong E_c	Wrong HZO thickness	High	Low	Ellipsometry
Pinholes	Contamination	High	Low	Cleanroom protocol
Moisture absorption	Delayed anneal	High	Low	Anneal within 1 hr
Oxygen vacancy imbalance	Atmosphere	Medium	Medium	N_2 anneal control
Cu diffusion	BEOL thermal	High	Low	Barrier layer standard

2. Cell-Level Failure Modes

Failure Mode	Cause	Severity	Likelihood	Mitigation
Over-etch	Poor selectivity	High	Low	Etch-stop layer
Under-etch	Insufficient time	Medium	Low	10% over-etch
Alignment error	Lithography drift	Medium	Medium	±2 nm overlay
Read disturb	Read voltage too high	High	Low	Read <10% E_c
Half-select disturb	Inhibit voltage	Medium	Medium	Design 38% margin
Endurance failure	Too many writes	Medium	Medium	Write-verify + 215 writes/cell budget
Retention loss	Phase change	High	Low	Gate-5b bake

3. Circuit-Level Failure Modes

Failure Mode	Cause	Severity	Likelihood	Mitigation
Comparator offset	Transistor mismatch	Medium	Medium	Auto-zeroing
Word-line delay	Weak driver	Medium	Low	100 μm width
Register file corruption	Write conflict	High	Low	Scoreboard
Branch predictor miss	Pattern aliasing	Low	Medium	Larger table
Cache coherence fail	Protocol bug	High	Low	Formal verification
Clock skew	Layout asymmetry	High	Low	H-tree 5ps skew
3.3V driver breakdown	HV overstress	High	Low	3.3V rating verified
Charge pump fail	Insufficient voltage	High	Low	37-stage Dickson
PLL lock loss	Jitter, drift	Medium	Low	Lock detect + POR

4. System-Level Failure Modes

Failure Mode	Cause	Severity	Likelihood	Mitigation
Weight drift	Retention failure	Medium	Low	Periodic recalibration
Thermal runaway	Cooling failure	High	Low	Thermal shutdown
Read disturb	Read voltage too high	High	Low	Margin 2×
Crosstalk	Adjacent cell coupling	Low	Medium	Cell spacing
Package warpage	Stress imbalance	Medium	Low	Stress compensation
Bond wire failure	Mechanical stress	Low	Low	Quality control
ESD damage	Static discharge	Medium	Medium	ESD protection
Latchup	Parasitic SCR	Medium	Low	Guard rings

5. Hybrid-Specific Failure Modes

Failure Mode	Cause	Severity	Likelihood	Mitigation
Box64 poor efficiency	Suboptimal JIT	High	Medium	Benchmark-driven tuning
Windows app crash	Wine incompatibility	High	Medium	Compatibility list
MMU page fault storm	TLB thrash	Medium	Low	Larger TLB
Hypervisor boot fail	H ext bug	High	Low	KVM self-test
CXL enumeration fail	Protocol bug	High	Medium	Board respin
DDR5 training fail	Signal integrity	High	Low	Standard DDR5 IP
FeFET MEOL integration	N5 thermal budget	High	Medium	Foundry co-development
Multi-core scaling poor	L3 contention	Medium	Medium	Larger L3, better scheduler

6. Highest-Risk Items

The highest-likelihood, highest-severity hybrid-specific failures:

1. **FeFET MEOL integration at N5** (Medium likelihood, High severity). This is the single largest open item. Mitigation: standard MEOL insertion pattern, 12--18 month process module development. Fallback: 22FDX FeFET chiplet or SRAM-only NVL3 (loses instant-boot).
2. **Box64 real-world efficiency** (Medium likelihood, High impact on ST GIPS claims). Mitigation: benchmark-driven tuning; if efficiency is 75% instead of 85%, ST GIPS drop 12%.
3. **6.2 GHz N5 12-wide closure for Pc-1** (Medium likelihood, High impact on Pc-1 ST). Mitigation: fallback to 6.0 GHz (ST GIPS drops from 25.3 to 24.5).
4. **CXL 3.0 IP availability** (Low likelihood, High impact on multi-socket). Mitigation: IP vendors have confirmed availability; needs contract.

Part 16: Health and Safety Notes

This part notes any real and non-obvious health or safety hazards in the production or operation of the device. It does not state non-issues or common-knowledge items.

1. Production Hazards

1.1. ALD precursor handling

The FeFET fabrication uses atomic layer deposition (ALD) of hafnium and zirconium oxides. The ALD precursors (hafnium tetrakis-dimethylamide, $Hf[N(CH_3)_2]_4$, and zirconium tetrakis-dimethylamide, $Zr[N(CH_3)_2]_4$) are pyrophoric (air-reactive) and moisture-sensitive. They must be handled in a standard ALD precursor delivery system under inert atmosphere at the foundry. This is standard semiconductor fabrication practice; the precursors are not novel and are used in commercial HfO_2 production today. No new hazard is introduced by the FeFET process module beyond the existing hazards of standard CMOS ALD.

1.2. Standard semiconductor hazards

The fabrication flow uses the same process chemicals as standard N5 CMOS (HF, H_2SO_4, H_2O_2, NH_3, $TiCl_4$, WF_6, ozone, etc.). These are standard hazards of any advanced semiconductor fab, with standard facilities and protocols for handling, storage, and disposal. The FeFET module adds no new chemical hazard class.

2. Operation Hazards

2.1. Electrical

The finished device operates at 1.0 V (core), 1.8 V (I/O), 3.3 V (FeFET write), and up to ~1.05 V + current for the highest-power product. The highest-power product (Pc-1) draws up to ~234 W peak and ~17 A at 1.0 V. Standard precautions for high-current low-voltage DC power apply (proper PDN, fusing, thermal shutdown). No non-obvious electrical hazard is introduced.

2.2. Thermal

The highest-power product (Pc-1) requires liquid cooling at 0.15 °C/W. Standard precautions for liquid-cooled electronics apply (leak detection, water quality, non-conductive fluids, thermal shutdown at 115 °C). No non-obvious thermal hazard is introduced.

2.3. Materials

The finished device contains $Hf_{0.5}Zr_{0.5}O_2$ in the FeFET cells. This is a solid oxide film, non-toxic, non-reactive, and non-hazardous under normal operation. It is not released into the environment under normal use. Under e-waste disposal, it is disposed of as standard semiconductor waste.

No non-obvious health hazard to the end user is introduced by the FeFET cell or the finished device.

3. Summary

The design introduces no novel user-facing health or safety hazard. The production-side ALD precursor handling is standard semiconductor practice. The operation-side electrical, thermal, and material exposures are standard for high-performance semiconductor products. No additional precautions beyond standard semiconductor and high-power electronics practice are required.

Part 17: Advanced Optimization and Higher-Tier Products

This part presents the two product-level upgrade paths available beyond the three Tier-1 products specified in Parts 2--16: the advanced optimization configurations (AO1 and AO2) and the higher-tier product variants (Ph-2, Pc-2, Dc-1, Dc-2). Both paths preserve the shared architecture (FeFET cell, NVL3 subsystem, crossbar, CPU core family, fabrication process, security features) and offer additional performance, capacity, or cores at a price premium.

The complete engineering specifications for every upgrade path are available from the author.

1. The Two Upgrade Paths

Two product-level upgrade paths are available:

Upgrade path	What it adds	Products	Price premium
Advanced Optimization (AO1)	2× NVL3 capacity; 4-bit signed crossbar	Ph-1, Pc-1, Dc-1S	+5%
Advanced Optimization (AO2)	4× NVL3 capacity; 8-bit signed crossbar	Ph-1, Pc-1, Dc-1S	+15%
Higher-Tier Products	Higher clocks, higher core count	Ph-2, Pc-2, Dc-1, Dc-2	+17% to +268%

The two upgrade paths can be combined. A higher-tier product can be specified with either AO1 or AO2. The combined product's price is the sum of the two premiums; the engineering specifications for the combined products are available from the author.

2. Advanced Optimization Configurations (AO1 and AO2)

ADVANCED OPTIMIZATION AVAILABLE

The design specified in Parts 2--16 is the baseline (2-bit signed crossbar; single-level NVL3).

Two advanced optimization configurations are available.

AO1 (advanced optimization 1) provides:

- **NVL3 capacity:** 200% of baseline
- **Crossbar weight precision:** 4-bit signed
- **AI accuracy:** matches FP32 on ResNet-50 and VGG-16; within 1 percentage point on MobileNet-v2 and BERT-base
- **NVL3 $/MB:** 53% of baseline
- **Price:** +5% over baseline

- **CPU, power, TDP, package, cooling, memory, I/O, security, fabrication:** unchanged from baseline

AO2 (advanced optimization 2) provides:

- **NVL3 capacity:** 400% of baseline
- **Crossbar weight precision:** 8-bit signed
- **AI accuracy:** matches FP32 on ResNet-50, VGG-16, MobileNet-v2, and BERT-base
- **NVL3 $/MB:** 29% of baseline
- **Price:** +15% over baseline
- **CPU, power, TDP, package, cooling, memory, I/O, security, fabrication:** unchanged from baseline

The complete specifications for the advanced optimization configurations are available from the author.

2.1. The nine AO products

Product	Base	NVL3	Crossbar	Price	vs base
Ph-1 (baseline)	---	64 MB	2-bit signed	$70	100%
Ph-1-AO1	Ph-1	128 MB	4-bit signed	$74	105.7%
Ph-1-AO2	Ph-1	256 MB	8-bit signed	$81	115.7%
Pc-1 (baseline)	---	64 MB	2-bit signed	$139	100%
Pc-1-AO1	Pc-1	128 MB	4-bit signed	$146	105.0%
Pc-1-AO2	Pc-1	256 MB	8-bit signed	$160	115.1%
Dc-1S (baseline)	---	24 MB	---	$155	100%
Dc-1S-AO1	Dc-1S	48 MB	---	$163	105.2%
Dc-1S-AO2	Dc-1S	96 MB	---	$178	114.8%

2.2. NVL3 efficiency

Product	Baseline $/MB	AO1 $/MB (% base)	AO2 $/MB (% base)
Ph-1	$1.09	$0.58 (53%)	$0.32 (29%)
Pc-1	$2.17	$1.14 (53%)	$0.63 (29%)
Dc-1S	$6.46	$3.40 (53%)	$1.85 (29%)

AO1 delivers approximately 1.9× the NVL3 capacity per dollar. AO2 delivers approximately 3.5× the NVL3 capacity per dollar.

2.3. AI accuracy

Model	FP32	Baseline (2-bit)	AO1 (4-bit)	AO2 (8-bit)
ResNet-50 (ImageNet)	76.0%	75.7%	76.0%	76.0%
VGG-16 (ImageNet)	74.0%	73.5%	73.9%	74.0%
MobileNet-v2 (ImageNet)	72.0%	70.8%	71.7%	71.9%
BERT-base (GLUE avg)	84.0%	82.5%	83.7%	84.0%

AO1 closes most of the accuracy gap. AO2 matches FP32 on all four models within 0.1 percentage point.

2.4. Unchanged specifications

Metric	AO1	AO2
Core count, clock, IPC	100% of base	100% of base
ST GIPS	100% of base	100% of base
MT GIPS	100% of base	100% of base
AI peak MAC/s	100% of base	100% of base
AI real-workload MAC/s	100% of base	100% of base
Peak power	100% of base	100% of base
TDP	100% of base	100% of base
GIPS/W	100% of base	100% of base
Package, socket, pin count	same	same
Cooling solution	same	same
Memory interfaces (DDR5 / LPDDR5X)	same	same
I/O interfaces (PCIe / CXL)	same	same
Security features	same	same
Fabrication process	same	same

3. Higher-Tier Product Variants

HIGHER-TIER PRODUCTS AVAILABLE

The design specified in Parts 2--16 includes three Tier-1 products (Ph-1, Pc-1, Dc-1S).

Four higher-tier product variants are available beyond the Tier-1 products.

- **Ph-2 (gaming phone):** 6 big + 4 little cores at 5.2 GHz boost / 4.0 GHz sustained. 15% more ST and 39% more MT than Ph-1. Requires a gaming vapor chamber. $82.
- **Pc-2 (maximum-performance desktop):** 36 cores at 6.5 GHz boost / 6.0 GHz sustained. 5% more ST and 26% more MT than Pc-1. Requires a 360 mm AIO liquid cooler and a premium LGA 1700 board with a 1 A/pin VRM. $189.
- **Dc-1 (mid-tier server):** 64 cores at 6.0 GHz boost / 5.5 GHz sustained. 100% more MT than Dc-1S. Requires a 0.10 °C/W cold plate with facility chilled water. $380.
- **Dc-2 (AI inference server):** 96 cores at 6.0 GHz single-core boost / 5.4 GHz all-core boost / 5.0 GHz sustained. 161% more MT than Dc-1S. Requires a 0.06 °C/W cold plate with facility chilled water. $570.

The complete specifications for the higher-tier products are available from the author.

3.1. The four higher-tier products

Product	Application	Cores	Clock	MT GIPS	Peak/TDP	Price
Ph-2	Gaming phone	6+4	5.2 / 4.0 GHz	103	41/26 W	$82
Pc-2	Max-performance desktop	36	6.5 / 6.0 GHz	793	380/323 W	$189
Dc-1	Mid-tier server	64	6.0 / 5.5 GHz	1239	407/342 W	$380
Dc-2	AI inference server	96	6.0 single / 5.4 all-core / 5.0 sust	1616	505/430 W	$570

3.2. Position relative to the Tier-1 baseline

Product	vs baseline	ST change	MT change	TDP change	Additional requirement
Ph-2	Ph-1	+15%	+39%	+44%	Gaming vapor chamber
Pc-2	Pc-1	+5%	+26%	+77%	360 mm AIO; premium LGA 1700 board (1 A/pin VRM)
Dc-1	Dc-1S	0%	+100%	+95%	0.10 °C/W cold plate (facility chilled water)
Dc-2	Dc-1S	0%	+161%	+146%	0.06 °C/W cold plate (facility chilled water)

3.3. What is unchanged

The higher-tier products share the Tier-1 baseline's fabrication process (N5 CMOS with FeFET MEOL), FeFET cell, NVL3 subsystem, crossbar (Ph-2 and Pc-2 only), CPU core family, security features, failure modes, and health and safety notes. The engineering principles that underlie the Tier-1 baseline apply to the higher-tier products with the clock, core-count, or additional-core differences stated above.

3.4. Combinations with advanced optimization

Each of the four higher-tier products can be combined with either the AO1 or AO2 advanced optimization configuration. The combined products carry both the higher-tier performance improvement and the advanced optimization capacity and precision improvements. The complete specifications for the combined products are available from the author.

4. Summary and Contact

Upgrade path	What it adds	Price premium	Products
AO1	2× NVL3; 4-bit crossbar	+5%	Ph-1, Pc-1, Dc-1S
AO2	4× NVL3; 8-bit crossbar	+15%	Ph-1, Pc-1, Dc-1S
Ph-2	5.2 GHz clock; +39% MT vs Ph-1	$82	Ph-1 tier
Pc-2	36 cores; +26% MT vs Pc-1	$189	Pc-1 tier
Dc-1	64 cores; +100% MT vs Dc-1S	$380	Dc-1S tier
Dc-2	96 cores; +161% MT vs Dc-1S	$570	Dc-1S tier

All upgrade paths preserve the shared architecture and fabrication process of the Tier-1 baseline. The complete specifications for every upgrade path are available from the author.

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Appendix A: Register Map

1. Conventions

Every custom MMIO block in this design is described in this appendix. The standard RISC-V CLINT (0x4000_1000), the DDR5 controller, the PCIe root complex, and the CXL host bridge follow their respective industry specifications and are not repeated here.

Register access codes: RO = read-only, RW = read-write, WO = write-only (read returns 0), W1C = write 1 to clear.

Reset values are the value after cold power-on before any firmware write, unless stated otherwise (some blocks have OTP-derived reset values that are programmed at factory test).

Register width is 32 bits unless stated otherwise. 64-bit registers occupy two consecutive 32-bit slots, low word at the lower offset.

Reserved bits must be written as 0 and are read as 0.

Base addresses: every block occupies a 4 KB page. Blocks with multiple instances (per-channel, per-bank, per-sensor) place instance n at base $+n\times$ page size.

2. MMIO Base Address Map

Base	Block	Instances	Products
`0x4000_0000`	GPIO	1	All
`0x4000_1000`	CLINT (standard)	1	All
`0x4000_2000`	PLIC-lite	1	All
`0x4000_3000`	Watchdog	1	All
`0x4000_4000`	Thermal	4 (Ph-1), 6 (Pc-1), 8 (Dc-1S)	All
`0x4000_5000`	Host interface	1	All
`0x4000_6000`	DMA	1	All
`0x4000_7000`	Vector block	1	Ph-1, Pc-1
`0x4000_8000`	OTP	1	All
`0x4000_9000`	Power status	1	All
`0x4000_A000`	Security engine	1	All
`0x4000_B000`	Debug authentication	1	All
`0x4000_C000`	CXL control	1	Pc-1, Dc-1S
`0x4000_D000`	NVL3 controller	1	All
`0x4000_E000`	Crossbar controller	1	Ph-1, Pc-1

Writes to unimplemented addresses in a block's 4 KB page return 0 on read and are silently ignored on write (no bus error). This allows firmware to probe for block presence by writing a known pattern and reading back.

3. GPIO (0x4000_0000)

8 pins on Ph-1; 32 pins on Pc-1 and Dc-1S. Only the first 8 DATA/DIR/IE/IS/TRIG/IC/DRIVE/PULL/SLEW/INSYNC bits are meaningful on Ph-1.

Offset	Name	Access	Reset	Bit fields
0x00	DATA	RW	0	[n] pin n output value (only when DIR[n]=1)
0x04	DIR	RW	0	[n]=1 output, [n]=0 input
0x08	IE	RW	0	[n]=1 interrupt enabled
0x0C	IS	RO	0	[n]=1 interrupt pending
0x10	TRIG	RW	0	[2n:2n+1]: 00=low, 01=high, 10=fall, 11=rise
0x14	IC	W1C	---	Write 1 to clear IS[n]
0x18	DRIVE	RW	0x55555555	[2n:2n+1]: 00=2 mA, 01=4 mA, 10=8 mA, 11=12 mA
0x1C	PULL	RW	0	[2n:2n+1]: 00=none, 01=up, 10=down, 11=keeper
0x20	SLEW	RW	0	[n]=1 fast slew, [n]=0 slow
0x24	INSYNC	RW	0	[2n:2n+1]: 00=0, 01=2, 10=4, 11=8 cycle synchronizer delay

On Ph-1, bits [31:8] of every register are reserved and read 0.

4. PLIC-lite (0x4000_2000)

Custom variant of the RISC-V PLIC. 16 interrupt sources on Ph-1; 64 on Pc-1 and Dc-1S. Sources are assigned in Appendix C. Register layout follows the standard RISC-V PLIC specification except where noted.

Offset	Name	Access	Reset	Bit fields
0x0000	PRIORITY[0]	RW	0	Priority of source 0 (0=disabled)
...	PRIORITY[n]	RW	0	4 B per source; n = 0..N_SOURCE-1
0x1000	PENDING	RO	0	32-bit word per 32 sources; bit n = source n
0x2000	ENABLE	RW	0	32-bit word per 32 sources; bit n = source n enabled for hart 0
0x200000	THRESHOLD	RW	0	Hart 0 threshold (only hart 0 in this implementation)
0x200004	CLAIM	RW	---	Read returns highest-priority pending source; write same value to complete
0x200008	CFG	RW	0	[0]=enable PLIC; [1]=interrupt mode (0=level, 1=edge)

Only hart 0 is supported; firmware running on other harts must route interrupts through hart 0 software interrupt (source 1).

This is the single-hart-PLIC design trade-off.

5. Watchdog (0x4000_3000)

Offset	Name	Access	Reset	Bit fields
0x00	CTRL	RW	0	[0] ENABLE; [1] FEED (WO, self-clear); [2] PRESCALE_EN; [4:3] TIMEOUT_SEL (00=1 ms, 01=10 ms, 10=100 ms, 11=1 s); [5] IRQ_EN; [6] RESET_EN; [7] FORCE_RESET (WO); [8] CLR_RESET_CAUSE (WO)
0x04	STATUS	RO	0x2	[0] RUNNING; [1] READY; [2] TIMEOUT; [3] RESET_OCCURRED; [4] RS:SW; [5] RS:WDT; [6] RS:BOD; [7] RS:THERMAL; [8] RS:EXTERNAL
0x08	LOAD	RW	0xFFFFFFFF	Timeout load (prescaler cycles)
0x0C	COUNT	RO	0xFFFFFFFF	Current count

Reset cause bits [8:4] are latched from the last reset event and cleared only by writing CLR_RESET_CAUSE. They are the source of truth for boot-time diagnostic logging.

6. Thermal Sensors (0x4000_4000)

Per-sensor registers: instance n at base $+n \times 0x100$. Number of sensors: 4 (Ph-1), 6 (Pc-1), 8 (Dc-1S).

Offset	Name	Access	Reset	Bit fields
0x00	RAW	RO	---	16-bit signed raw count; convert with CAL_OFF + CAL_SLOPE
0x04	TEMP_mC	RO	---	Temperature in milli-°C, already calibrated (formula below)
0x08	CAL_OFF	RW	OTP	Calibration offset (OTP-programmed)
0x0C	CAL_SLOPE	RW	OTP	Calibration slope (OTP-programmed)
0x10	THRESH	RW	0x55F0	Threshold in °C×100 (default 85°C×100 = 0x21F4)
0x14	HYST	RW	500	Hysteresis in °C×100 (default 5°C)
0x18	CTRL	RW	0x1	[0] ENABLE; [1] IRQ_EN; [2] TRIP_EN
0x1C	STATUS	RO	0	[0] VALID; [1] ABOVE_THRESH; [2] TRIP_LATCHED (W1C)
0x20	TRIP_THRESH	RW	0x5DC0	Emergency shutdown threshold in °C×100 (default 115°C×100 = 0x5DC0)

Conversion: `TEMP_mC = (RAW × CAL_SLOPE) / 1024 + CAL_OFF`. The sensor is a bandgap PTAT type with ±1°C accuracy over 0--125°C after the two-point calibration.

7. Host Interface (0x4000_5000)

SPI slave on Ph-1; PCIe endpoint config space on Pc-1 and Dc-1S. Registers below are common to both; per-product differences are noted.

Offset	Name	Access	Reset	Bit fields
0x00	CTRL	RW	0	[0] IF_ENABLE; [1] INT_MODE (SPI: level/edge; PCIe: MSI/MSI-X); [2] DMA_EN
0x04	STATUS	RO	0	[0] LINK_UP; [1] BUSY; [2] ERROR; [3] DMA_ACTIVE
0x08	IRQ_MASK	RW	0	Per-bit mask (mirrors PLIC source enable)
0x0C	IRQ_PENDING	RO	0	Raw pending bits from PLIC (read-only mirror)
0x10	DMA_CMD_ADDR	RW	0	DMA descriptor list base address (host memory)
0x14	DMA_CMD_LEN	RW	0	Number of 16 B descriptors
0x18	DMA_CMD_TRIGGER	WO	---	Write 1 to start DMA descriptor processing
0x1C	DMA_STATUS	RO	0	[0] BUSY; [1] DONE; [2] ERROR; [3] ABORT
0x20	FW_VERSION	RO	OTP	Firmware version (OTP-programmed at factory)
0x24	FW_TRIGGER	WO	---	Write magic 0x4E564C33 to reload firmware image from NVL3
0x28	PCIE_DEVICE_ID	RO	0x106D	PCI device ID (Pc-1, Dc-1S only)
0x2C	PCIE_VENDOR_ID	RO	0x1D1F	PCI vendor ID (Pc-1, Dc-1S only)

On Ph-1, offsets 0x28 and 0x2C read 0 (no PCIe). On Pc-1 and Dc-1S, offsets 0x00--0x24 are used for the internal DMA path; the standard PCIe config space at the PHY level (BARs, MSI-X table) is beyond this block's scope.

8. DMA Controller (0x4000_6000)

Offset	Name	Access	Reset	Bit fields
0x00	CTRL	RW	0	[0] ENABLE; [1] LOOP_EN; [2] IRQ_EN
0x04	STATUS	RO	0	[0] BUSY; [1] DONE; [2] ERROR; [3] ABORTED
0x08	SRC_ADDR	RW	0	Source address (low 32 bits)
0x0C	SRC_ADDR_HI	RW	0	Source address (high 32 bits)
0x10	DST_ADDR	RW	0	Destination address (low 32 bits)
0x14	DST_ADDR_HI	RW	0	Destination address (high 32 bits)
0x18	LEN	RW	0	Transfer length in bytes (multiple of 8)
0x1C	CHUNK	RW	4096	Chunk size in bytes (per burst)
0x20	TRIGGER	WO	---	Write 1 to start transfer
0x24	ABORT	WO	---	Write 1 to abort (DMA returns to idle within 1 μs)
0x28	CUR_ADDR	RO	0	Current source address (progress)
0x2C	CUR_REMAIN	RO	0	Remaining bytes
0x30	ERROR_CODE	RO	0	[3:0] last error (1=src bus, 2=dst bus, 3=alignment, 4=timeout, 5=abort)

One DMA engine, programmed from hart 0. Multiple channels are software-arbitrated via the LOOP_EN bit and a software descriptor queue in SRAM.

9. Vector Block Control (0x4000_7000)

Ph-1 and Pc-1 only. This block controls the analog crossbar's functional DAC/ADC chain (Part 5 §5.13 custom instructions trigger the compute; this block configures the chain).

Offset	Name	Access	Reset	Bit fields
0x00	VECMODE	RW	0	[1:0] mode: 00=idle, 01=2-bit, 10=4-bit (AO1), 11=8-bit (AO2)
0x04	DAC_CODE	RW	0	[3:0] DAC code for current row (used by crossbar.macc)
0x08	FIFO_WR	RW	---	Write 8 B input vector word (pushed to input FIFO)
0x0C	FIFO_WR_COUNT	RO	0	Number of words currently in input FIFO
0x10	FIFO_RD	RO	---	Read 8 B output vector word (popped from output FIFO)
0x14	FIFO_RD_COUNT	RO	0	Number of words currently in output FIFO
0x18	FIFO_STATUS	RO	0	[0] IN_FULL; [1] IN_EMPTY; [2] OUT_FULL; [3] OUT_EMPTY; [4] OVERFLOW
0x1C	CTRL	RW	0	[0] ENABLE; [1] AUTO_TRIGGER; [2] IRQ_EN
0x20	STATUS	RO	0	[0] BUSY; [1] MAC_DONE; [2] ERROR
0x24	TIA_GAIN	RW	0	[1:0] gain: 00=high (2.6 kΩ), 01=mid, 10=low (10 kΩ), 11=reserved
0x28	ADC_CONFIG	RW	0	[1:0] resolution: 00=8-bit, 01=10-bit (default)
0x2C	CALIB_CTRL	RW	0	[0] START_CALIB; [1] ABORT_CALIB; [2] LOAD_FROM_OTP
0x30	CALIB_STATUS	RO	0	[0] BUSY; [1] DONE; [2] ERROR; [3] OTP_LOADED
0x34	CALIB_COL	RW	0	Column index for calibration (0--511 or 0--1023)
0x38	CALIB_ROW	RW	0	Row index for calibration (0--511 or 0--1023)

The four custom instructions (`crossbar.ld` , `.st` , `.macc` , `.wait`) have hardware backing in this block: `.macc` triggers a MAC cycle, `.wait` polls MAC_DONE. Direct register access to FIFO_WR and FIFO_RD is used only during initialization and calibration.

10. OTP Controller (0x4000_8000)

Offset	Name	Access	Reset	Bit fields
0x00	ADDR	RW	0	OTP word address (32-bit words)
0x04	DATA_READ	RO	0	Data at ADDR (valid after READ_STROBE)
0x08	DATA_WRITE	RW	0	Data to write at ADDR
0x0C	READ_STROBE	WO	---	Write 1 to start read; STATUS.RDY set when data valid
0x10	PROG_STROBE	WO	---	Write 1 to start program; STATUS.PROG_DONE set when complete
0x14	CTRL	RW	0	[0] PROG_ENABLE (must be 1 for program; requires VPP if fuse-based)
0x18	STATUS	RO	0	[0] RDY; [1] PROG_DONE; [2] PROG_FAILED; [3] LOCKED (OTP region is locked)
0x1C	LOCK	WO	---	Write 1 to permanently lock OTP (irreversible)
0x20	LOCK_STATUS	RO	OTP	[n]=1 means region n locked; regions defined in §14.7
0x24	READ_BACK	RO	0	Last-read value (for verification)
0x28	REGION_BASE	RW	0	[7:0] current region base (0=root key, 1=device, ..., 7=calibration)

Programming sequence: write ADDR, write DATA_WRITE, write CTRL.PROG_ENABLE=1, write PROG_STROBE=1, poll STATUS.PROG_DONE (or PROG_FAILED), verify via READ_STROBE and compare to DATA_WRITE. Programming requires the OTP controller to have been unlocked at boot by the security engine (see Appendix B §B.7).

11. Power Status and Sequencing (0x4000_9000)

Offset	Name	Access	Reset	Bit fields
0x00	RAIL_STATUS	RO	0	[n]=1 rail n up. n=0 core, 1 I/O, 2 analog, 3 PLL, 4 LPDDR5X/ DDR5, 5 auxiliary
0x04	RAIL_OK	RO	0	[n]=1 rail n within spec
0x08	SEQUENCE_STATE	RO	0	[3:0] state (0=off, 1=core, 2=io, 3=analog, 4=pll, 5=mem, 6=aux, 7=up, 8=shutdown)
0x0C	BOD_STATUS	RO	0	[n]=1 brown-out detected on rail n (latched)
0x10	BOD_CLR	W1C	---	Clear BOD_STATUS
0x14	SHUTDOWN_TRIGGER	WO	---	Write 0xDEAD to trigger orderly shutdown
0x18	RESET_CAUSE	RO	0	Same encoding as Watchdog STATUS[8:4]
0x1C	RESET_CLR	WO	---	Write 1 to clear RESET_CAUSE
0x20	DVFS_CTRL	RW	0	[3:0] target P-state (0=P0, 1=P1, ..., 4=P4); [8] REQUEST; [9] AUTO (hardware DVFS)
0x24	DVFS_STATUS	RO	0	[3:0] current P-state; [8] TRANSITIONING; [9] AUTO_ACTIVE
0x28	DVFS_TRIM_V	RW	0	[7:0] voltage trim (signed, 25 mV/LSB)
0x2C	DVFS_TRIM_F	RW	0	[7:0] frequency trim (signed, 100 MHz/LSB)
0x30	THROTTLE	RO	0	[1:0] throttle level (0=none, 1=75%, 2=50%, 3=25%); [2] THERMAL_SHUTDOWN

DVFS sequence: write DVFS_CTRL with target P-state and REQUEST=1; poll DVFS_STATUS until TRANSITIONING clears. Transition takes 5 µs nominal, 50 µs worst case (PLL relock + voltage ramp). The AUTO bit lets hardware choose the P-state from thermal and utilization; software DISABLEs AUTO by writing 0.

12. Security Engine (0x4000_A000)

Offset	Name	Access	Reset	Bit fields
0x00	AES_CTRL	RW	0	[0] START; [1] DECRYPT; [2] ENCRYPT; [3] KEY_SLOT (0--3); [4] MODE (0=XTS)
0x04	AES_STATUS	RO	0	[0] BUSY; [1] DONE; [2] ERROR
0x08	AES_SRC_ADDR	RW	0	Source address (low)
0x0C	AES_SRC_ADDR_HI	RW	0	Source address (high)
0x10	AES_DST_ADDR	RW	0	Destination address (low)
0x14	AES_DST_ADDR_HI	RW	0	Destination address (high)
0x18	AES_LEN	RW	0	Length in 16 B blocks
0x1C	AES_TWEAK	RW	0	XTS tweak (16 B, at offset 0x1C--0x28)
...	AES_TKEY	WO	---	Key slot register (write only, zero on read); 16 B per slot at 0x30 + slot×0x10
0x80	SHA_CTRL	RW	0	[0] START; [1] RESET; [2] ALG (0=SHA-256, 1=SHA-384)
0x84	SHA_STATUS	RO	0	[0] BUSY; [1] DONE; [2] ERROR
0x88	SHA_DATA	WO	---	8 B data word
0x8C	SHA_DIGEST_0	RO	0	Result word 0 (of 8 or 12)
...	SHA_DIGEST_n	RO	0	Result word n at 0x8C + n×4
0xA0	ECDSA_CTRL	RW	0	[0] START_VERIFY; [1] START_SIGN
0xA4	ECDSA_STATUS	RO	0	[0] BUSY; [1] DONE; [2] VERIFY_OK; [3] VERIFY_FAIL
0xA8	ECDSA_MSG_ADDR	RW	0	Message address (low)
0xAC	ECDSA_MSG_ADDR_HI	RW	0	Message address (high)
0xB0	ECDSA_MSG_LEN	RW	0	Message length in bytes
0xB4	ECDSA_R	RO	0	Signature component r (32 B, at 0xB4--0xD3)
...	ECDSA_S	RO	0	Signature component s (32 B)
0xD8	TRNG_CTRL	RW	0	[0] ENABLE
0xDC	TRNG_DATA	RO	0	4 random bytes per read (FIFO depth 64)

Offset	Name	Access	Reset	Bit fields
0xE0	TRNG_STATUS	RO	0	[0] READY; [1] HEALTH_FAIL
0xE4	DICE_CDI	RO	0	Compound Device Identifier (16 B at 0xE4--0xF3)
0xF4	DICE_MEASURE	RW	0	Next measurement source address (low)
0xF8	DICE_MEASURE_HI	RW	0	High 32 bits
0xFC	DICE_CTRL	RW	0	[0] START; [1] UPDATE
0x100	DICE_STATUS	RO	0	[0] BUSY; [1] DONE; [2] ERROR

AES-XTS key slots are write-only and read back 0. Key material is loaded only at boot, from the ROM to the security engine via the OTP controller, never via software-visible paths. The DICE CDI is regenerated on every cold boot from the OTP device key + boot nonce + first measurement.

13. Debug Authentication (0x4000_B000)

Offset	Name	Access	Reset	Bit fields
0x00	CHALLENGE	RW	0	32 B challenge written by debugger (write only; 0x00--0x1F)
0x20	RESPONSE	RO	0	32 B response read by debugger (0x20--0x3F)
0x40	CTRL	RW	0	[0] GEN_RESPONSE; [1] VERIFY; [2] UNLOCK; [3] LOCK
0x44	STATUS	RO	0	[0] BUSY; [1] DONE; [2] VERIFY_OK; [3] UNLOCKED; [4] TIMEOUT; [5] LOCKED
0x48	TIMEOUT	RW	3600	Unlock duration in seconds (default 3600 = 1 hour); 0 = unlimited
0x4C	ATTEMPTS	RO	0	Failed unlock attempts (counter)
0x50	LOCK_DOWN	RO	0	[0]=1 if locked permanently after threshold attempts

Unlock sequence: debugger generates random 32 B challenge, writes to CHALLENGE; hardware computes ECDSA signature of the challenge with the device-unique private key in OTP; debugger reads RESPONSE and verifies against the public key. If verification succeeds, debugger writes VERIFY=1 followed by UNLOCK=1; hardware opens JTAG for TIMEOUT seconds. Failed verifications increment ATTEMPTS; after 5 failed attempts, LOCK_DOWN is set and unlock requires factory RMA.

14. CXL Control (0x4000_C000)

Pc-1 and Dc-1S only. Base registers; the full standard CXL config space is separate.

Offset	Name	Access	Reset	Bit fields
0x00	LINK_STATUS	RO	0	[0] UP; [1] TRAINED; [2] RETRAIN_REQ; [3] ERROR; [7:4] speed
0x04	LINK_CTRL	RW	0	[0] ENABLE; [1] FORCE_RETRAIN; [2] RESET_LINK
0x08	LINK_ERROR	RO	0	[0] CRC_ERR; [1] REPLAY_ERR; [2] MLF_ERR; [3] TRAIN_ERR (W1C)
0x0C	LINK_ERROR_CLR	W1C	---	Clear bits in LINK_ERROR
0x10	LANE_STATUS	RO	0	[31:0] per-lane status (bit n = lane n trained)
0x14	SPEED	RW	0x40	[7:0] target speed (0x40 = 64 GT/s, 0x20 = 32 GT/s, 0x10 = 16 GT/s)
0x18	WIDTH	RW	0x10	[7:0] target width (0x10 = x16)
0x1C	NUMA_NODE_ID	RW	0x0	This socket's NUMA node ID (used by ACPI SRAT)
0x20	REMOTE_SOCKET_COUNT	RO	0	Number of remote sockets discovered at boot
0x24	REMOTE_SOCKET_ID	RW	0	Index into remote socket table (0..N-1)
0x28	REMOTE_NUMA_NODE	RW	0	NUMA node ID of remote socket (per index)
0x2C	INTERLEAVE	RW	0	Memory interleave mode (0=none, 1=2-way, 2=4-way)
0x30	INTERLEAVE_GRAN	RW	0x100	Interleave granularity in bytes (default 256)

The CXL block controls the link and NUMA topology only. The actual memory window (for CXL.mem targets) is configured via the standard CXL config space at 0x0000_0003_xxxx_xxxx; that path is documented in the CXL 3.0 specification and not repeated here.

15. NVL3 Controller (0x4000_D000)

The NVL3 controller manages the FeFET non-volatile L3 array. The data window at 0x0000_0000_8000_0000 routes cache-line reads and writes through the controller's row/page logic; the control registers below configure the controller.

Offset	Name	Access	Reset	Bit fields
0x00	CTRL	RW	0	[0] ENABLE; [1] START_WRITE_VERIFY; [2] START_ERASE; [3] START_PRE_WAKEUP; [4] RESET; [5] WRITE_BUF_EN
0x04	STATUS	RO	0	[0] READY; [1] BUSY; [2] ERROR; [3] VERIFY_FAILED; [4] ENDURANCE_WARN; [5] CALIB_IN_PROGRESS
0x08	ROW_ADDR	RW	0	Row address (0--4095)
0x0C	BANK_ADDR	RW	0	Bank address (0--255)
0x10	DATA_LOW	RW	0	256 bits of write data, low (offsets 0x10--0x2F)
0x30	DATA_HIGH	RW	0	256 bits of write data, high (offsets 0x30--0x4F)
0x50	RDATA_LOW	RO	0	256 bits of read data, low
0x70	RDATA_HIGH	RO	0	256 bits of read data, high
0x90	WRITE_VERIFY_CTRL	RW	0x302	[3:0] max retries (default 3); [11:8] pulse width (default 2 = 25 ns)
0x94	WRITE_VERIFY_STATUS	RO	0	[7:0] attempt count; [8] SUCCESS; [9] FAILED
0x98	ECC_STATUS	RO	0	[0] SINGLE_BIT; [1] DOUBLE_BIT; [8] OVERFLOW
0x9C	ECC_ADDR	RO	0	Address of the last ECC event (row << 16 | bank)
0xA0	BAD_BLOCK	RO	0	[15:0] bad-block count; [16] last-bad-block-address-available
0xA4	BAD_BLOCK_ADDR	RO	0	Last bad-block address
0xA8	WEAR_COUNT_LOW	RO	0	Cumulative write count (low 32)
0xAC	WEAR_COUNT_HIGH	RO	0	Cumulative write count (high 32)
0xB0	PRE_WAKEUP_CTRL	RW	0x3E8	[15:0] pulse count per cell (default 1000 = 0x3E8)
0xB4	PRE_WAKEUP_STATUS	RO	0	[0] DONE; [1] BUSY; [2] ERROR
0xB8	CAL_TABLE_BASE	RW	OTP	Calibration table base address in SRAM
0xBC	CAL_TABLE_LEN	RW	OTP	Calibration table length in 4 B words
0xC0	REF_OFFSET_BASE	RW	OTP	Per-column reference offset base (63--167 KB table)
0xC4	CALIB_START	WO	---	Write 1 to (re)load calibration from OTP
0xC8	CALIB_STATUS	RO	0	[0] DONE; [1] BUSY; [2] ERROR

Offset	Name	Access	Reset	Bit fields
0xCC	ERROR_LOG	RO	0	[7:0] last error code (1=verify, 2=ecc, 3=endurance, 4=bad-block-limit, 5=calibration)
0xD0	IRQ_ENABLE	RW	0	[0] ECC_SB; [1] ECC_DB; [2] VERIFY; [3] ENDURANCE; [4] BAD_BLOCK
0xD4	IRQ_STATUS	RO	0	Mirrors IRQ_ENABLE bit layout; latched until IC write
0xD8	IC	W1C	---	Clear bits in IRQ_STATUS

Write sequence (single row): write ROW_ADDR, BANK_ADDR, DATA_LOW (8×32 bits), DATA_HIGH (8×32 bits); write CTRL.START_WRITE_VERIFY; poll STATUS.BUSY or wait for VERIFY interrupt; check WRITE_VERIFY_STATUS.SUCCESS (or FAILED). The controller internally performs erase, program, read-verify, and retry up to max-retries.

Read sequence (single row): write ROW_ADDR, BANK_ADDR; write CTRL.START_WRITE_VERIFY is NOT used for reads; instead, reads are triggered by the data-window access at 0x8000_0000. Alternatively, direct controller read: write ROW_ADDR, BANK_ADDR, then read RDATA_LOW / RDATA_HIGH after the controller's page-open latency. The controller's direct read path is provided for bring-up and ECC scrub.

Pre-wake-up: write PRE_WAKEUP_CTRL with the desired pulse count (default 1000); write CTRL.START_PRE_WAKEUP; poll PRE_WAKEUP_STATUS.DONE. Pre-wake-up must be applied to every bank before the NVL3 data region is used as persistent storage. Factory test applies 10^3 pulses to the NVL3 data region and 10^4 to the microcode region.

Calibration: at factory test and after any NVL3 configuration change, write CALIB_START; poll CALIB_STATUS.DONE. The controller reads the calibration table from OTP, populates REF_OFFSET, and validates each column's reference against a known pattern.

16. Crossbar Controller (0x4000_E000)

Ph-1 and Pc-1 only. Full functional configuration of the analog crossbar. The compute path is the custom-instruction interface (`crossbar.macc` etc.); this block configures the analog chain and manages weight updates and calibration.

Offset	Name	Access	Reset	Bit fields
0x00	CTRL	RW	0	[0] ENABLE; [1] W+/W--MODE (0=Option B time-mux, 1=Option A parallel); [2] AUTO_DAC; [3] IRQ_EN
0x04	STATUS	RO	0	[0] READY; [1] BUSY; [2] ERROR; [3] MAC_DONE; [4] CALIB_BUSY
0x08	W_UPDATE_CTRL	RW	0	[0] START; [1] ABORT; [2] VERIFY_EN; [3] HALT_ON_ERR
0x0C	W_UPDATE_STATUS	RO	0	[0] BUSY; [1] DONE; [2] ERROR; [3] VERIFIED
0x10	W_SRC_ADDR	RW	0	Weight source address in SRAM (low)
0x14	W_SRC_ADDR_HI	RW	0	High 32 bits
0x18	W_LEN	RW	0	Length in cells
0x1C	W_ROW	RW	0	Row to update (0--511 or 0--1023)
0x20	W_COL	RW	0	Column start
0x24	W_PATTERN	RW	0	Current weight pattern (valid after update)
0x28	W_VERIFY	RO	0	Read-back value of last updated cell
0x2C	CALIB_CTRL	RW	0	[0] START; [1] ABORT; [2] LOAD_FROM_OTP; [3] TEMPCO_EN
0x30	CALIB_STATUS	RO	0	[0] BUSY; [1] DONE; [2] ERROR; [3] OTP_LOADED
0x34	CALIB_COL	RW	0	Column to calibrate (0--511 or 0--1023)
0x38	CALIB_GAIN_COARSE	RW	0	Coarse gain trim (0--7)
0x3C	CALIB_GAIN_FINE	RW	0	Fine gain trim (0--255)
0x40	CALIB_OFFSET	RW	0	Signed offset trim (-2048..+2047)
0x44	TEMPCO_BASE	RW	OTP	Temperature coefficient table base
0x48	TEMPCO_LEN	RW	OTP	Length in 4 B words
0x4C	ADC_TRIM	RW	OTP	ADC trim (INL, DNL) from OTP
0x50	DAC_TRIM	RW	OTP	DAC trim (offset, gain) from OTP
0x54	TIA_TRIM	RW	OTP	TIA trim (offset, bandwidth) from OTP
0x58	IRQ_ENABLE	RW	0	[0] MAC_DONE; [1] W_UPDATE; [2] CALIB

Offset	Name	Access	Reset	Bit fields
0x5C	IRQ_STATUS	RO	0	Mirrors IRQ_ENABLE bit layout; latched until IC write
0x60	IC	W1C	---	Clear bits in IRQ_STATUS

Weight update sequence: write W_SRC_ADDR (weight pattern in SRAM), W_LEN, W_ROW, W_COL; write W_UPDATE_CTRL.START=1; poll W_UPDATE_STATUS.BUSY or wait for W_UPDATE interrupt; check W_UPDATE_STATUS.DONE and .VERIFIED. The controller programs FeFET_A and FeFET_B in the row, applies read-verify per cell, retries up to 3 times.

Calibration: write CALIB_COL and optionally CALIB_GAIN_COARSE / CALIB_GAIN_FINE / CALIB_OFFSET overrides; write CALIB_CTRL.START=1; poll CALIB_STATUS.DONE. Factory calibration cycles all columns and stores the resulting per-column gain/offset in OTP. At boot, write CALIB_CTRL.LOAD_FROM_OTP=1 to restore.

Option A/B mode: at boot, firmware reads the product's config from OTP (or from an ID register not shown here) and writes CTRL.W+/W--MODE appropriately. On Ph-1 (single crossbar), mode=0 (time-multiplexed); on Pc-1 (parallel pair), mode=1.

Appendix B: Firmware Interface Specification

This appendix specifies the protocols between firmware and hardware for the sequences that the register map alone does not fully define: NVL3 row write-verify, NVL3 pre-wake-up, crossbar weight update, crossbar calibration, firmware update, security engine key provisioning, and error recovery.

Every sequence is stated as a step-numbered procedure. Every register mentioned is defined in Appendix A. Timing budgets are stated where the sequence has a hard real-time constraint.

1. Conventions

Status polling: where the text says "poll STATUS.X", firmware reads STATUS in a loop until the named bit is set or a timeout (default 100 ms) expires. Timeout handling: escalate to the error recovery sequence at §B.8.

Interrupt alternative: wherever a status-poll is described, firmware may enable the corresponding IRQ_ENABLE bit and service the interrupt. The protocols below state the polling form because it is the simplest; interrupt-driven versions follow the same register writes but replace the poll with an interrupt handler.

Ordering: all register writes to a single block must be separated by at least one `fence` instruction to ensure MMIO ordering. Writes to different blocks have no ordering constraint unless stated.

Memory barriers: where a register write triggers a DMA transfer, the source buffer must be flushed to memory (or marked non-cacheable) before the trigger. This is the standard RISC-V `fence rw, rw` before writing the trigger register.

2. NVL3 Row Write-Verify

Purpose: program a single 512-bit row of the NVL3 with a 512-bit data pattern, with automatic read-verify and retry.

Preconditions: NVL3 controller enabled (CTRL.ENABLE=1); the row is not currently in use as NVL3 cache; the caller has exclusive access to the bank (typically the NVL3 driver, during a cache line eviction or a boot-time firmware write).

Procedure:

1. Write ROW_ADDR with the target row (0--4095).
2. Write BANK_ADDR with the target bank (0--255 for Ph-1/Pc-1, 0--95 for Dc-1S).
3. Write DATA_LOW (8×32-bit words, 0x10--0x2F) with the low 256 bits of the row.
4. Write DATA_HIGH (8×32-bit words, 0x30--0x4F) with the high 256 bits of the row.
5. Write WRITE_VERIFY_CTRL to set max-retries (default 3) and pulse width (default 25 ns).
6. Write CTRL.START_WRITE_VERIFY=1. This triggers the row erase + program + read-verify sequence.
7. Poll STATUS.BUSY until it clears (typical 400 ns, max 1.2 µs).
8. Read WRITE_VERIFY_STATUS.SUCCESS. If 1, proceed to step 9. If 0, read WRITE_VERIFY_STATUS.attempt_count and STATUS.ERROR; then either retry (re-write CTRL.START_WRITE_VERIFY=1) or invoke error recovery (§B.8).
9. Read ECC_STATUS. If SINGLE_BIT=1, log the address (ECC_ADDR) and clear via IC.SINGLE_BIT=1. If DOUBLE_BIT=1, mark the row bad (BAD_BLOCK) and return error.

10. Return success.

Timing: 400 ns nominal for a single-iteration write; up to 1.2 µs worst-case (4 iterations). The controller performs the verify read back from the FeFET cell, not from cache.

Interrupt path: enable IRQ_ENABLE.VERIFY=1; the handler reads WRITE_VERIFY_STATUS and STATUS and follows the same disposition logic.

Cache interaction: this sequence writes directly to NVL3. If the target row is currently cached in SRAM L3 or L2, the caller must invalidate those lines before starting (or accept that the next access will be a miss and re-fill from NVL3).

3. NVL3 Pre-Wake-Up

Purpose: apply the HZO polarization wake-up pulse sequence to a bank or to the full NVL3, so that the persistent retention target is met.

Preconditions: the target region is currently erased (all zeros) or can be treated as uninitialized. Pre-wake-up writes a dummy pattern, so any prior data is lost.

Procedure (per bank):

1. Write BANK_ADDR.
2. Write PRE_WAKEUP_CTRL with the pulse count. Factory default: 1000 pulses for the NVL3 data region, 10000 for the microcode region. Write 0x3E8 for data, 0x2710 for microcode.
3. Write CTRL.START_PRE_WAKEUP=1.
4. Poll PRE_WAKEUP_STATUS.DONE. The controller applies the pulse train in a single sweep; typical time 1.3 s for 1000 pulses (25 ns per pulse plus setup overhead), 13 s for 10000 pulses.
5. Read PRE_WAKEUP_STATUS.ERROR. If 1, error recovery; if 0, return success.

Timing: 1.3 s per bank for 1000 pulses. The full NVL3 (256 banks for Ph-1/Pc-1, 96 for Dc-1S) takes ~333 s (Ph-1) or ~125 s (Dc-1S) at factory test. This is why pre-wake-up is done once at factory and not at every boot.

After pre-wake-up: the affected region must be re-initialized (factory test writes the microcode image and the initial NVL3 content) and re-calibrated (write CALIB_START=1 and poll CALIB_STATUS.DONE).

4. NVL3 Calibration Load and Validation

Purpose: on each cold boot, load the per-column calibration table from OTP into the controller's working SRAM, and validate that a known pattern reads back correctly with the loaded offsets.

Preconditions: OTP controller has been unlocked by the security engine; OTP calibration region is populated (factory-programmed).

Procedure:

1. Write CALIB_START=1. The controller reads the calibration table from OTP (base and length are OTP-programmed into CAL_TABLE_BASE and CAL_TABLE_LEN at factory).
2. Poll CALIB_STATUS.DONE (typical 5 ms, max 20 ms).

3. Read CALIB_STATUS.ERROR. If 1, error recovery (§B.8).
4. Validation: write a known pattern to bank 0, row 0; read back; compare; if mismatch, error recovery.
5. If validation succeeds, return success. The NVL3 controller is now ready to serve the data window at 0x8000_0000.

Boot path requirement: this sequence must complete before any NVL3 access, including firmware image load. Boot ROM calls it as its first action after PLL lock.

5. Crossbar Weight Update

Purpose: program a contiguous run of weight cells in the analog crossbar to a new pattern (per-model weight update).

Preconditions: crossbar controller enabled (CTRL.ENABLE=1); the target cells are currently unused or are being re-purposed; the new pattern is in SRAM at a firmware-controlled address.

Procedure:

1. Write W_SRC_ADDR (low) and W_SRC_ADDR_HI (high) with the SRAM address of the new weight pattern. The pattern is a contiguous array of 2-bit (or 4-bit or 8-bit, in the AO configurations) weight values, packed according to the current VECMODE.
2. Write W_LEN with the number of cells to update. Maximum per call: 512 (one row of a 512^2 crossbar) or 1024 (one row of a 1024^2 crossbar).
3. Write W_ROW with the target row.
4. Write W_COL with the target column start.
5. Write W_UPDATE_CTRL.START=1, .VERIFY_EN=1, .HALT_ON_ERR=0 (or 1 if the caller can tolerate a partial update).
6. Poll W_UPDATE_STATUS.BUSY (typical 38 µs for a 512-cell row on Ph-1, 77 µs for a 1024-cell row on Pc-1; these are worst-case values from §5.12).
7. Read W_UPDATE_STATUS.DONE and .VERIFIED. If VERIFIED=1, the update succeeded for all cells. If VERIFIED=0, read W_VERIFY (the value of the last cell) and the source data at the corresponding offset; identify the first failing cell and either retry (re-write the failing segment) or invoke error recovery.
8. Return success.

Ordering: multiple weight updates must be serialized (wait for one to complete before starting another). Firmware enforces this via a mutex in the AI driver.

Inference suspension: during a weight update, inference must be suspended (no `crossbar.macc` issued). The AI driver enforces this via its own scheduling; the hardware does not enforce it, and issuing MAC during an update produces undefined results.

6. Crossbar Calibration

Purpose: measure and store the per-column gain, offset, and temperature coefficient of the crossbar, so that the separable model in §5.9 has valid coefficients.

Procedure (per column):

1. For each of the four known weight states (00, 01, 10, 11), program the entire column with that weight (W_UPDATE

with W_ROW=0, W_COL=all, W_LEN=all).
2. For each of the 16 DAC codes, apply the DAC code as the input to the entire column, read the ADC output for each row, and compute the column-averaged current.
3. The result is a 4 × 16 table of (weight_state, dac_code) → column_average_current.
4. Fit the separable model: gain = slope of (current vs weight_state at fixed DAC); offset = intercept; tempco = the difference between the 25°C and 85°C fits divided by 60°C.
5. Store the fitted parameters in OTP via the OTP controller (Appendix A §A.10). Per column: 2 bits weight-state-major + 20 bits gain + 2 bits tempco per the schema in §5.9.

Time: approximately 20 minutes per die for a 1024^2 crossbar at factory test; this is the dominant crossbar factory test time. The full procedure is described in the Crossbar Test Program (separate artifact).

Boot path: at cold boot, firmware writes CALIB_CTRL.LOAD_FROM_OTP=1 and polls CALIB_STATUS.OTP_LOADED; this restores the fitted coefficients into the working registers.

7. Firmware Update (Dual-Bank A/B)

Purpose: atomically replace the running firmware with a new signed image, with rollback on failure.

Preconditions: new image is signed by the OEM ECDSA P-256 key and has a version number greater than the current running version (rollback protection).

Procedure:

1. Firmware reads the new image into SRAM (from host, SPI, or storage) and verifies its ECDSA P-256 signature against the OEM public key in OTP.
2. If verification fails, abort and log.
3. If verification succeeds, firmware writes the new image to the backup NVL3 bank. The backup bank is the one that firmware is NOT currently running from (A/B alternation).
4. After the write completes and read-verifies, firmware writes the new image's version number to the OTP rollback counter area, using the OTP controller.
5. Firmware writes the boot pointer in OTP to point at the backup bank, then resets.
6. On the next boot, Boot ROM reads the boot pointer, verifies the selected bank's signature, and jumps to it.
7. If Boot ROM fails to verify or if the kernel fails to reach userspace within a watchdog timeout (typically 60 s), Boot ROM reverts the boot pointer to the previous bank and resets. The previous bank is retained for exactly this purpose.

Physical presence: unsigned recovery images (for RMA only) require the physical-presence button to be held during reset. The button state is read via a GPIO input; firmware enforces the policy.

Anti-rollback: firmware compares the new image's version number to the OTP rollback counter. A newer version always wins; a same-or-older version is rejected unless physical presence is asserted.

8. Security Engine Key Provisioning

Purpose: install the device-unique ECDSA key, the OEM root key hash, and the debug public key into OTP at factory test. This is the only time these values are written; they are then locked.

Preconditions: OTP is unlocked (LOCK_STATUS=0); the security engine has authenticated the factory programmer via a factory-level challenge/response (separate flow, not described here); the keys are provided as files by the OEM.

Procedure:

1. Write REGION_BASE=0 (root key hash region).
2. Write ADDR=0, DATA_WRITE=first 32 bits of root key hash, CTRL.PROG_ENABLE=1, PROG_STROBE=1.
3. Poll STATUS.PROG_DONE (typical 100 µs per word; a 32 B hash is 8 words, ~800 µs total).
4. Repeat for the full root key hash (32 B = 8 words).
5. Repeat the read-verify for each word (READ_STROBE, compare DATA_READ to DATA_WRITE).
6. Write REGION_BASE=1 (device key region). Repeat with the 32 B device ECDSA private key.
7. Write REGION_BASE=2 (debug public key region). Repeat with the 32 B debug public key.
8. Write REGION_BASE=3 (rollback counter region). Initialize the counter to 1.
9. Write REGION_BASE=4 (boot nonce region). Initialize the nonce to a random 16 B value generated by the TRNG.
10. For each region, write LOCK=1 and poll LOCK_STATUS until the region shows locked.
11. The security engine now enforces that OTP writes to these regions fail (STATUS.LOCKED=1) and that the security boot flow uses the provisioned values.

Failure handling: if any read-verify fails (DATA_READ ≠ DATA_WRITE), the device is scrapped. There is no retry on OTP programming. This is why the security provisioning step is done last, after all other factory tests pass.

Backup key handling: the OEM's factory test program holds the keys in memory during provisioning and zeroizes them on completion. The keys are not stored on the factory test system between runs.

9. Error Recovery

Purpose: given any error status from any block, restore the system to a defined safe state and, where possible, recover without a reset.

NVL3 verify failed: read WRITE_VERIFY_STATUS; if attempt count < max-retries, retry (re-write CTRL.START_WRITE_VERIFY=1). If max-retries exhausted, mark the row bad: write BAD_BLOCK (increment count), write BAD_BLOCK_ADDR=current row; the NVL3 driver then excludes the row from use and remaps to a redundant row (software-level remapping; the hardware does not auto-repair). Return error to caller.

NVL3 ECC double-bit: read ECC_ADDR; read back the row from a redundant copy (if a redundant copy exists) or from the source of truth (SRAM or DDR5); log the address; mark the row bad as above.

NVL3 endurance warning: read WEAR_COUNT_LOW/HIGH; if the count exceeds the endurance threshold (default: 50% of the target endurance from §3.10), stop using the cell for persistent writes and direct writes to DDR5 or NVMe. The NVL3 driver enforces this per-bank.

Crossbar weight update failed: read W_VERIFY and compare against the source; retry the failing segment. If 3 consecutive retries fail, mark the row bad (log for factory RMA) and continue with the rest of the model.

Crossbar calibration failed: read CALIB_STATUS; if ERROR=1, the calibration table in OTP is corrupt or the on-die reference is out of range. Retry once; if it still fails, boot without crossbar (set VECMODE=00 idle) and log the fault.

Thermal trip: read THROTTLE and the trip sensor's raw count; if thermal shutdown is asserted, no software recovery is possible; the PMIC shuts the rails down. Firmware must not attempt to re-enable. On next boot, read the trip sensor's

STATUS.TRIP_LATCHED to see the cause.

Watchdog reset: on next boot, read RESET_CAUSE. If RS:WDT, the firmware that was running is suspect; boot into a safe recovery image (a separate signed image at a fixed OTP address) that logs the incident, then reboots to the primary image. If the safe image also fails, boot into the factory diagnostic image (physical presence required).

PCIe/CXL link error: read LINK_ERROR; write LINK_ERROR_CLR; if the error recurs within 100 ms, force a retrain (write LINK_CTRL.FORCE_RETRAIN=1) and wait for LINK_STATUS.UP. If retrain fails 3 times, fall back to reduced speed (write SPEED=0x20 or 0x10) and continue.

Brown-out: read BOD_STATUS to identify which rail dropped; report to BMC (Pc-1, Dc-1S) or log to NVL3 (Ph-1); do not attempt to continue until the rail is stable.

10. Summary Table: Firmware Sequences

Sequence	When	Typical time	Failure action
NVL3 row write-verify	On cache eviction or explicit write	400 ns	Retry up to 3, then mark row bad
NVL3 pre-wake-up	Factory once; on NVL3 re-init	1.3 s/bank	Read STATUS; error recovery
NVL3 calibration	Every cold boot	5 ms	Boot into safe image
Crossbar weight update	Per model change	38--77 µs	Retry segment; mark row bad
Crossbar calibration	Factory once; on die re-init	~20 min	Boot without crossbar
Firmware update	On new image	Variable	Watchdog rollback
Security key provisioning	Factory once	<10 ms	Scrap device
Error recovery	On any error	Variable	See §B.8

Appendix C: Interrupt Source Assignments

This appendix assigns the PLIC-lite interrupt sources. 16 sources are available on Ph-1; 64 on Pc-1 and Dc-1S. Sources 0--15 are common to all products; sources 16--63 exist only on Pc-1 and Dc-1S.

1. Source Assignment Table

Source	Signal	Block	Edge/Level	Priority (default)
0	(reserved, tie 0)	---	---	---
1	Software interrupt (MSIP)	CLINT	level	7
2	Timer interrupt (MTIP)	CLINT	level	7
3	External (MEIP)	CLINT	level	7
4	NVL3 ECC single-bit	NVL3	level	5
5	NVL3 ECC double-bit	NVL3	level	6
6	NVL3 write-verify failed	NVL3	level	5
7	NVL3 endurance warning	NVL3	level	4
8	NVL3 bad-block limit	NVL3	level	4
9	Watchdog timeout	Watchdog	level	7
10	Thermal high (85°C)	Thermal	level	6
11	Thermal trip (115°C)	Thermal	level	7
12	Brown-out	Power	level	7
13	DMA complete	DMA	level	3
14	Host interface (SPI/PCIe)	Host	level	4
15	Debug unlock timeout	Debug	level	3
16	Crossbar MAC done	Crossbar	level	3
17	Crossbar weight update done	Crossbar	level	4
18	Crossbar calibration done	Crossbar	level	4
19	Vector FIFO underflow/overflow	Vector	level	3
20	GPIO interrupt	GPIO	level	3
21	Security engine AES done	Security	level	5
22	Security engine SHA done	Security	level	5
23	Security engine ECDSA done	Security	level	5

Source	Signal	Block	Edge/Level	Priority (default)
24	TRNG health failure	Security	level	6
25	DICE measurement complete	Security	level	5
26	OTP read complete	OTP	level	3
27	OTP program failed	OTP	level	6
28	CXL link up	CXL	level	5
29	CXL link down	CXL	level	6
30	CXL link error	CXL	level	5
31	PCIe MSI (aggregated)	Host	level	4
32	DDR5 controller ECC event	DDR5	level	5
33	DDR5 training complete	DDR5	level	4
34	SRAM L3 ECC single-bit	L3	level	4
35	SRAM L3 ECC double-bit	L3	level	6
36	L2 ECC single-bit	L2	level	4
37	L2 ECC double-bit	L2	level	6
38	L1 ECC single-bit	L1	level	3
39	L1 ECC double-bit	L1	level	5
40	MMU TLB shootdown	MMU	level	4
41	Branch predictor parity	Core	level	3
42	ROB parity	Core	level	4
43	Register file parity	Core	level	4
44	Pipeline lockstep mismatch	Core	level	7
45	PMU overflow	PMU	level	2
46	RISC-V custom instruction error	Core	level	3
47	JTAG debug event	Debug	level	2

Source	Signal	Block	Edge/Level	Priority (default)
48	(reserved)	---	---	---
...	(reserved 48--62)	---	---	---
63	NMI (aggregated)	PLIC	level	7

2. Priority Encoding

Priority values are 0--7 (3-bit priority, matching the PLIC-lite 4-bit priority field with the top bit reserved). 0 = disabled (source cannot interrupt). 7 = highest. Firmware may override the default priorities at boot by writing PLIC PRIORITY[n]. The defaults above are chosen so that the highest-severity sources (brown-out, watchdog, thermal trip, NMI, pipeline lockstep mismatch) are 7; the "attention" sources (ECC double-bit, security health fail) are 6; the "log-and-continue" sources are 3--5; the "informational" sources (DMA complete, PMU overflow) are 2--3.

3. Per-Product Differences

Ph-1: sources 16--63 are not implemented. Crossbar and CXL sources do not exist. Sources 34--47 are also not present (SRAM/caches are the same, but no separate register file parity interrupt on the 12-wide phone core — the parity is checked but not latched to PLIC). The PLIC-lite on Ph-1 has 16 usable sources (0--15, with 0 reserved, so 15 usable).

Pc-1: all 64 sources implemented.

Dc-1S: all 64 sources implemented. Crossbar sources 16--18 read as always-zero (no crossbar on Dc-1S), but the lines exist and are valid to enable — they simply never fire.

4. Interrupt Handler Ordering

When multiple sources are pending, the PLIC-lite serves them in strict priority order (highest priority first; ties broken by lowest source number). Firmware may reorder by writing PRIORITY.

For latency-critical sources (brown-out, watchdog), firmware should also configure the watchdog's own RESET_EN bit as a hardware backstop: if firmware does not service the watchdog interrupt within the TIMEOUT window, the watchdog resets the chip. This is the standard two-level defense (interrupt for logging, reset for safety).

5. NMI Configuration

Source 63 (NMI) is the aggregated non-maskable interrupt. It fires when any of the following occurs: thermal trip (11), pipeline lockstep mismatch (44), or brown-out (12). The NMI handler in firmware must immediately:

1. Read all trip-status registers (THERMAL.STATUS, BOD_STATUS, pipeline parity) to identify the cause.
2. Log to NVL3 (if available) or to BMC (if available).
3. Initiate the safe-state sequence: stop all DMA, quiesce the crossbar, halt user cores, and prepare for reset.

4. Trigger watchdog FORCE_RESET=1 if the safe-state sequence does not complete within 100 ms.

The NMI cannot be masked by PLIC settings. It is guaranteed to fire on the next cycle after the event.

Appendix D: Device Tree and ACPI Bindings

This appendix specifies the platform description data that firmware presents to the operating system. The design supports two boot-time platform description modes: Device Tree (DT), standard for Linux and Android on RISC-V; and ACPI, standard for NUMA-aware server Linux, CXL-aware operating systems, and other ACPI-dependent software. The two are mutually exclusive at boot; the boot ROM selects one based on an OTP configuration bit and passes the corresponding pointer to the OS.

This appendix specifies:

- The boot modes and how firmware chooses between them.
- The Device Tree bindings for every custom block (compatible strings, node structure, required and optional properties).
- The ACPI table inventory and the per-product values that populate each table.
- The DSDT namespace for every custom block (full ASL definitions).
- The boot handoff contract (registers, memory state, state transitions).
- The validation procedure that verifies the tables are correct before tape-out.

The standard ACPI table formats (RSDP, XSDT, MADT, SRAT, SLIT, MCFG, SPCR, GTDT, RHCT, IORT, CEDT, PPTT) are defined in ACPI 6.5 and are not repeated here; this appendix gives the design-specific values and the DSDT namespace. The standard DT bindings for RISC-V (PLIC, CLINT, CPU, memory) are defined in the Linux kernel documentation; this appendix gives the design-specific nodes and properties.

1. Boot Modes

Two boot modes are supported. The mode is selected by an OTP bit (OTP region 4, word 0, bit 0), programmed at factory test per the OEM's order.

Mode	When used	OS-side consequence
DT	Linux direct boot; Android boot	RISC-V Linux reads FDT from `a1`; Android reads FDT from `a1`; no ACPI tables loaded
ACPI	Windows-via-Wine under Linux; NUMA/CXL-aware server Linux	RISC-V Linux reads ACPI tables via UEFI system table or via RSDP pointed to by a special DT node; OS uses ACPI for enumeration

Default: DT. ACPI mode is opt-in for server-class configurations where NUMA and CXL topology need OS-level enumeration.

2. Device Tree Bindings

The DT is passed to the OS as a Flattened Device Tree (FDT) blob. The blob is generated by the boot ROM from a static DTS template compiled at build time, patched with per-product parameters (memory size, sensor count, crossbar presence, CXL presence) that are read from OTP.

2.1. Vendor prefix and naming

Custom compatible strings use the vendor prefix `ferrohybrid,` (lowercase, no capital, no space). The prefix is a descriptive namespace, not a company name. Standard RISC-V bindings (`riscv,`, `sifive,`, `riscv,plic0`, etc.) are used for standard components.

2.2. Root node and chosen

```
/ {
    #address-cells = <2>;
    #size-cells = <2>;
    model = "Ferroelectric Hybrid Processor Family";
    compatible = "ferrohybrid,pc-1", "ferrohybrid,fhpf";
    // patched at build time to one of:
    //   "ferrohybrid,ph-1", "ferrohybrid,pc-1", "ferrohybrid,dc-1s"

    chosen {
        bootargs = "console=ttySIF0,115200 earlycon=sbi";
        stdout-path = "/soc/serial@40005000";
        riscv,isa-base = "rv64";
        riscv,isa-extensions = "i,m,a,f,d,c,g,v,h";
    };

    cpus {
        #address-cells = <1>;
        #size-cells = <0>;
        // populated per product (see cpus node below)
    };

    memory@80000000 {
        device_type = "memory";
        reg = <0x0 0x80000000 0x0 0x0>;  // size patched per product
    };

    soc {
        #address-cells = <2>;
        #size-cells = <2>;
        compatible = "simple-bus";
        ranges;

        // all peripheral nodes below
    };
};
```

2.3. CPU nodes

The `cpus` node enumerates every hart. Ph-1 has 10 harts (6 big + 4 little); Pc-1 has 28 harts; Dc-1S has 32 harts. Each hart gets a `cpu@<hart-id>` node with a `cpu-map` entry.

```
cpus {
```

```
    #address-cells = <1>;
    #size-cells = <0>;

    cpu@0 {
        device_type = "cpu";
        compatible = "riscv";
        reg = <0>;
        status = "okay";
        riscv,isa = "rv64imafdcvh";
        mmu-type = "riscv,sv39";
        clock-frequency = <6200000000>;  // patched per product / bin
        cpu-idle-states = <&CPU_SLEEP_0 &CPU_SLEEP_1>;

        interrupt-controller {
            #interrupt-cells = <1>;
            compatible = "riscv,cpu-intc";
            interrupt-controller;
        };
    };

    // ... one cpu@N per hart, up to the product's hart count

    cpu-map {
        cluster0 {
            core0 { cpu = <&cpu0>; };
            core1 { cpu = <&cpu1>; };
            // ...
        };
        // Ph-1 has two clusters: cluster0 (big) and cluster1 (little)
    };

    idle-states {
        CPU_SLEEP_0: cpu-sleep-0 {
            compatible = "riscv,idle-state";
            riscv,sbi-sleep-type = <1>;  // WFI
            entry-latency-us = <10>;
            exit-latency-us = <10>;
            min-residency-us = <100>;
        };
        CPU_SLEEP_1: cpu-sleep-1 {
            compatible = "riscv,idle-state";
            riscv,sbi-sleep-type = <2>;  // P3 deep sleep
            entry-latency-us = <10>;
            exit-latency-us = <10000>;   // 10 µs wake
            min-residency-us = <1000000>;
        };
    };
};
```

2.4. Interrupt controller (PLIC-lite)

The PLIC-lite is a variant of the standard RISC-V PLIC (16 or 64 sources; single hart). The DT node declares it with the standard `riscv,plic0` compatible string plus a design-specific qualifier.

```
plic: interrupt-controller@40002000 {
    compatible = "ferrohybrid,plic-lite", "riscv,plic0";
    reg = <0x0 0x40002000 0x0 0x1000>;
    #interrupt-cells = <1>;
    interrupt-controller;
    riscv,ndev = <64>;          // 16 on Ph-1
    interrupts-extended = <&cpu0_intc 11>;
    // The single hart that services all PLIC interrupts
};
```

2.5. CLINT

Standard RISC-V CLINT at the standard address.

```
clint: timer@40001000 {
    compatible = "riscv,clint0";
    reg = <0x0 0x40001000 0x0 0x10000>;
    interrupts-extended = <&cpu0_intc 3 &cpu0_intc 7>;
};
```

2.6. Serial console

The host interface block (Appendix A §A.8) exposes a UART-like console on Ph-1. On Pc-1 and Dc-1S, the console is provided by the SoC's integrated UART (a standard 16550-compatible block added at 0x40005000 for this design).

```
serial@40005000 {
    compatible = "ns16550a";
    reg = <0x0 0x40005000 0x0 0x100>;
    clock-frequency = <1843200>;
    current-speed = <115200>;
    reg-shift = <2>;
    reg-io-width = <4>;
    interrupts-extended = <&plic 14>;
};
```

2.7. GPIO

```
gpio@40000000 {
    compatible = "ferrohybrid,gpio";
    reg = <0x0 0x40000000 0x0 0x100>;
    gpio-controller;
    #gpio-cells = <2>;
    interrupt-controller;
```

```
    #interrupt-cells = <2>;
    interrupts-extended = <&plic 20>;
    ngpios = <32>;              // 8 on Ph-1
    ferrohybrid,drive-strengths = <2 4 8 12>;  // mA
    ferrohybrid,pull-modes = "none", "up", "down", "keeper";
};
```

2.8. Thermal sensors

Each sensor is a separate node; the number of nodes is 4 (Ph-1), 6 (Pc-1), or 8 (Dc-1S).

```
thermal@40004000 {
    compatible = "ferrohybrid,thermal";
    reg = <0x0 0x40004000 0x0 0x100>;
    #thermal-sensor-cells = <1>;
    interrupts-extended = <&plic 10 &plic 11>;
};

thermal-zones {
    cpu-thermal0 {
        polling-delay-passive = <100>;
        polling-delay = <1000>;
        thermal-sensors = <&thermal 0>;
        trips {
            cpu_alert: cpu-alert {
                temperature = <85000>;   // 85°C in millidegrees
                hysteresis = <5000>;
                type = "passive";
            };
            cpu_crit: cpu-crit {
                temperature = <115000>;  // 115°C in millidegrees
                hysteresis = <0>;
                type = "critical";
            };
        };
        cooling-maps {
            map0 { trip = <&cpu_alert>; cooling-device = <&cpu0 1 3>; };
        };
    };
    // ... additional zones per sensor
};
```

2.9. NVL3 controller

```
nvl3: nvl3-controller@4000D000 {
    compatible = "ferrohybrid,nvl3-controller";
    reg = <0x0 0x4000D000 0x0 0x100>;
    interrupts-extended = <&plic 4 &plic 5 &plic 6 &plic 7 &plic 8>;
```

```
    // Single-bit ECC, double-bit ECC, verify, endurance, bad-block
    ferrohybrid,nvl3-size-mb = <64>;        // 24 for Dc-1S
    ferrohybrid,nvl3-banks = <256>;         // 96 for Dc-1S
    ferrohybrid,nvl3-window = <0x0 0x80000000 0x0 0x4000000>;  // 64 MB
    // ... additional properties for write-verify, pre-wake-up, endurance
};
```

2.10. Crossbar controller (Ph-1, Pc-1 only)

```
crossbar: crossbar-controller@4000E000 {
    compatible = "ferrohybrid,crossbar-controller";
    reg = <0x0 0x4000E000 0x0 0x100>;
    interrupts-extended = <&plic 16 &plic 17 &plic 18>;
    // MAC done, weight update, calibration
    ferrohybrid,crossbar-rows = <1024>;     // 512 on Ph-1
    ferrohybrid,crossbar-cols = <1024>;
    ferrohybrid,crossbar-bits = <2>;        // 2-bit baseline
    ferrohybrid,crossbar-mode = "parallel"; // "time-mux" on Ph-1
    ferrohybrid,custom-instructions = "0x0B";  // opcode space
};
```

2.11. Security engine and debug

```
security@4000A000 {
    compatible = "ferrohybrid,security-engine";
    reg = <0x0 0x4000A000 0x0 0x200>;
    interrupts-extended = <&plic 21 &plic 22 &plic 23 &plic 24 &plic 25>;
    // AES done, SHA done, ECDSA done, TRNG health, DICE
};

debug@4000B000 {
    compatible = "ferrohybrid,debug-auth";
    reg = <0x0 0x4000B000 0x0 0x100>;
    interrupts-extended = <&plic 15>;
};

otp@40008000 {
    compatible = "ferrohybrid,otp-controller";
    reg = <0x0 0x40008000 0x0 0x100>;
    interrupts-extended = <&plic 26 &plic 27>;
    // read complete, program failed
    ferrohybrid,word-size = <4>;            // 32-bit words
    ferrohybrid,total-bytes = <1048576>;    // 1 MB
};
```

2.12. DMA, vector block, power, watchdog

```
dma@40006000 {
```

```
    compatible = "ferrohybrid,dma";
    reg = <0x0 0x40006000 0x0 0x100>;
    interrupts-extended = <&plic 13>;
    dma-channels = <1>;
    #dma-cells = <1>;
    ferrohybrid,chunk-size = <4096>;
};

vector@40007000 {
    compatible = "ferrohybrid,vector-block";
    reg = <0x0 0x40007000 0x0 0x100>;
    interrupts-extended = <&plic 19>;
    status = "okay";                          // "disabled" on Dc-1S
};

power@40009000 {
    compatible = "ferrohybrid,power-mgmt";
    reg = <0x0 0x40009000 0x0 0x100>;
    interrupts-extended = <&plic 12>;
    ferrohybrid,dvfs-p-states = <5>;
    ferrohybrid,rails = "core", "io", "analog", "pll", "mem", "aux";
};

watchdog@40003000 {
    compatible = "ferrohybrid,watchdog";
    reg = <0x0 0x40003000 0x0 0x100>;
    interrupts-extended = <&plic 9>;
    ferrohybrid,timeout-modes = <1 10 100 1000>;  // ms
    ferrohybrid,reset-enable;
};
```

2.13. CXL (Pc-1, Dc-1S only)

```
cxl@4000C000 {
    compatible = "ferrohybrid,cxl-control";
    reg = <0x0 0x4000C000 0x0 0x100>;
    interrupts-extended = <&plic 28 &plic 29 &plic 30>;
    // link up, link down, link error
    ferrohybrid,cxl-version = "3.0";
    ferrohybrid,link-width = <16>;
    ferrohybrid,link-speed = <64>;          // GT/s
    status = "okay";                          // "disabled" on Ph-1
};
```

2.14. Chosen node (boot handoff)

The `chosen` node carries the boot arguments and the initrd location (if any), plus a design-specific property pointing at the runtime firmware services.

```
chosen {
    bootargs = "console=ttySIF0,115200 earlycon=sbi";
    stdout-path = "/soc/serial@40005000";
    linux,initrd-start = <0x0 0x0>;        // patched if initrd present
    linux,initrd-end = <0x0 0x0>;
    ferrohybrid,firmware-version = "1.0.0";
    ferrohybrid,boot-mode = "dt";           // or "acpi"
};
```

3. ACPI Tables

ACPI mode presents the platform via ACPI tables. The boot ROM generates the tables at build time and places them in reserved memory at a fixed address. The OS discovers them via the UEFI system table (if UEFI boot) or via the RSDP pointer in the DT `chosen` node (the DT is still passed in ACPI mode for the initial handoff, but the OS ignores the peripheral nodes and uses ACPI instead).

3.1. Table inventory

Table	Purpose	Products	ACPI reference
RSDP	Root pointer (points to XSDT)	All	ACPI 6.5 §5.2.5
XSDT	Extended System Description Table	All	ACPI 6.5 §5.2.8
FADT	Fixed ACPI Description Table	All	ACPI 6.5 §5.2.9
DSDT	Differentiated System Description Table	All	ACPI 6.5 §5.2.11
MADT	Multiple APIC Description Table (RISC-V)	All	ACPI 6.5 §5.2.12
GTDT	Generic Timer Description Table	All	ACPI 6.5 §5.2.24
RHCT	RISC-V Hart Capabilities Table	All	ACPI 6.5 §5.2.37
SPCR	Serial Port Console Redirection	All	Microsoft SPCR spec
SRAT	System Resource Affinity Table (NUMA)	All	ACPI 6.5 §5.2.16
SLIT	System Locality Information Table	All	ACPI 6.5 §5.2.17
PPTT	Processor Properties Topology Table	All	ACPI 6.5 §5.2.30
MCFG	Memory Mapped Configuration space	Pc-1, Dc-1S	PCI Firmware spec
IORT	I/O Remapping Table	Pc-1, Dc-1S	ACPI 6.5 §5.2.33
CEDT	CXL Early Discovery Table	Pc-1, Dc-1S	CXL 3.0 spec

3.2. FADT

Follows ACPI 6.5 §5.2.9. The design-specific values:

Field	Ph-1	Pc-1	Dc-1S
Preferred PM profile	Mobile	Desktop	Enterprise Server
SCI interrupt	Source 12 (brown-out)	Source 12	Source 12
SCI interrupt mode	level	level	level
PM1a_EVT_BLK	OTP region 4	OTP region 4	OTP region 4
PM1_CNT_LEN	2	2	2
FLAGS.CENTURY	0	0	0
FLAGS.RTC_S4	0	0	0
FLAGS.RESET_REG_SUP	1	1	1
RESET_REG	0x40009014	0x40009014	0x40009014
RESET_VALUE	0xDEAD	0xDEAD	0xDEAD
ARM_BOOT_ARCH	0	0	0

3.3. MADT (RISC-V variant)

Follows ACPI 6.5 §5.2.12. Contains the following RISC-V-specific entries:

- **Type 0x18 (RISC-V Hart):** one entry per hart. `ACPI Processor ID` = hart ID; `Hart ID` = hart ID; `Flags` bit 0 = 1 (hart enabled); `ACPI Processor UID` = hart ID.
- **Type 0x1B (RISC-V PLIC):** one entry for the PLIC-lite. `RISC-V PLIC ID` = 0; `Hardware ID` = 0x40002000; `Platform Interrupt Sources` = 64 (16 on Ph-1); `GSI Base` = 0; `Maximum Priority` = 7.
- **Type 0x1C (RISC-V APLIC):** not used in this design (single-hart PLIC; no MSI).
- **Type 0x0D (RISC-V IMSIC):** not used.
- **Type 0x19 (RISC-V IMSIC per hart):** not used.

3.4. GTDT

Follows ACPI 6.5 §5.2.24. The generic timer comes from the CLINT (standard RISC-V mtime/mtimecmp). `CntControlBase` = 0x40001000 + 0x4000 (standard CLINT mtimecmp offset); `CntReadBase` = 0x40001000 + 0xBFF8 (standard CLINT mtime offset).

3.5. RHCT

Follows ACPI 6.5 §5.2.37. Lists the RISC-V ISA extensions supported by every hart: i, m, a, f, d, c, g, v, h, zicsr, zifencei, zicntr, zihpm.

3.6. SRAT and SLIT (NUMA)

SRAT follows ACPI 6.5 §5.2.16. Contains:

- **Processor Local APIC/SAPIC Affinity:** one entry per hart, `Proximity Domain` = the NUMA node that contains the hart.
- **Memory Affinity:** one entry per contiguous memory range, `Proximity Domain` = the NUMA node that contains that memory.
- **Generic Initiator Affinity:** one entry per CXL device (Pc-1, Dc-1S), `Proximity Domain` = the NUMA node closest to that CXL device.

Per-product NUMA topology:

Product	NUMA nodes	Contents
Ph-1	1	All harts + all LPDDR5X
Pc-1	1 (single-socket) or 2 (2-socket via CXL)	Node 0 = local harts + local DDR5 + local NVL3; Node 1 = remote socket's harts + memory + NVL3 (via CXL)
Dc-1S	1 (single-socket) or 2--4 (multi-socket via CXL)	Node 0 = local harts + local DDR5 + local NVL3; Nodes 1..N = remote sockets

SLIT follows ACPI 6.5 §5.2.17. Distances: 10 (local) on the diagonal; 20 for 1-hop CXL (adjacent socket); 30 for 2-hop CXL (ring topology on multi-socket Dc); 40 for more distant paths (if any).

3.7. MCFG, IORT, CEDT

MCFG (Pc-1, Dc-1S): PCIe ECAM base at `0x0000_0003_0000_0000`; 16 buses per root complex; base bus = 0; segment = 0.

IORT (Pc-1, Dc-1S): one ITS-free IORT (RISC-V does not have a GIC ITS); one SMMU-v3 node per PCIe root complex; one RMR (Root Memory Region) per PCIe device for MSI doorbells.

CEDT (Pc-1, Dc-1S): follows CXL 3.0 §9.18. Contains:

- CHBS (CXL Host Bridge Structure): one entry per CXL host bridge; `Base HPA` = the CXL.mem target address of the host bridge.
- CFMWS (CXL Fixed Memory Window Structure): one entry per memory window; describes the range of host physical addresses that map to CXL devices.

3.8. PPTT

Follows ACPI 6.5 §5.2.30. Describes the CPU topology (packages, clusters, cores). One socket per physical package; one node per cluster (4--6 per die); one node per core.

4. DSDT Namespace

The DSDT declares the custom devices in the ACPI namespace using ASL. Every device gets a _HID (Hardware ID) that is a design-specific string, a _CRS (Current Resource Settings) descriptor that describes its MMIO range and interrupt, and _DSD (Device-Specific Data) properties that describe the design-specific parameters.

4.1. Vendor ID

The design-specific _HID prefix is `FHP` (Ferroelectric Hybrid Processor). The full _HID for each device is `FHP0001` through `FHP000E` (12 custom devices, as follows):

_HID	Device	Compatible DT node (for reference)
FHP0001	GPIO	`ferrohybrid,gpio`
FHP0002	PLIC-lite	`ferrohybrid,plic-lite`
FHP0003	Watchdog	`ferrohybrid,watchdog`
FHP0004	Thermal	`ferrohybrid,thermal`
FHP0005	Host interface	`ns16550a` (serial) or PCIe config
FHP0006	DMA	`ferrohybrid,dma`
FHP0007	Vector block	`ferrohybrid,vector-block`
FHP0008	OTP	`ferrohybrid,otp-controller`
FHP0009	Power management	`ferrohybrid,power-mgmt`
FHP000A	Security engine	`ferrohybrid,security-engine`
FHP000B	Debug auth	`ferrohybrid,debug-auth`
FHP000C	CXL control	`ferrohybrid,cxl-control`
FHP000D	NVL3 controller	`ferrohybrid,nvl3-controller`
FHP000E	Crossbar controller	`ferrohybrid,crossbar-controller`

4.2. DSDT source (ASL)

The full DSDT for the Pc-1 product. The Ph-1 and Dc-1S versions differ only in the values of the numeric parameters and the presence of the crossbar (Ph-1) or CXL (Dc-1S). The differences are noted inline.

```
DefinitionBlock ("", "DSDT", 2, "FHP   ", "FHPPC1  ", 0x00000001)
```

```
{
    Scope (\_SB)
    {
        Device (GPIO) {
            Name (_HID, "FHP0001")
            Name (_UID, 0)
            Name (_CRS, ResourceTemplate () {
                Memory32Fixed (ReadWrite, 0x40000000, 0x100)
                Interrupt (ResourceConsumer, Level, ActiveHigh, Exclusive) { 20 }
            })
            Name (_DSD, Package () {
                ToUUID("daffd814-6eba-4d8c-8a91-bc9bbf4aa301"),
                Package () {
                    Package () { "ngpios", 32 },
                    Package () { "drive-strengths", Package () { 2, 4, 8, 12 } },
                    Package () { "pull-modes", Package () { "none", "up", "down", "keeper" } }
                }
            })
            Method (_STA) { Return (0x0F) }
        }

        Device (PLIC) {
            Name (_HID, "FHP0002")
            Name (_UID, 0)
            Name (_CRS, ResourceTemplate () {
                Memory32Fixed (ReadWrite, 0x40002000, 0x1000)
            })
            Name (_DSD, Package () {
                ToUUID("daffd814-6eba-4d8c-8a91-bc9bbf4aa301"),
                Package () {
                    Package () { "riscv,ndev", 64 },
                    Package () { "interrupts-extended", Package () { 0, 11 } }
                }
            })
            Method (_STA) { Return (0x0F) }
        }

        Device (WDT0) {
            Name (_HID, "FHP0003")
            Name (_UID, 0)
            Name (_CRS, ResourceTemplate () {
                Memory32Fixed (ReadWrite, 0x40003000, 0x100)
                Interrupt (ResourceConsumer, Level, ActiveHigh, Exclusive) { 9 }
            })
            Name (_DSD, Package () {
                ToUUID("daffd814-6eba-4d8c-8a91-bc9bbf4aa301"),
                Package () {
                    Package () { "timeout-modes-ms", Package () { 1, 10, 100, 1000 } },
```

```
                    Package () { "reset-enable", 1 }
                }
            })
            Method (_STA) { Return (0x0F) }
        }

        Device (TEMP) {
            Name (_HID, "FHP0004")
            Name (_UID, 0)
            Name (_CRS, ResourceTemplate () {
                Memory32Fixed (ReadWrite, 0x40004000, 0x600)   // 6 sensors × 0x100
                Interrupt (ResourceConsumer, Level, ActiveHigh, Exclusive) { 10, 11 }
            })
            Name (_DSD, Package () {
                ToUUID("daffd814-6eba-4d8c-8a91-bc9bbf4aa301"),
                Package () {
                    Package () { "num-sensors", 6 },
                    Package () { "sensor-offset", 0x100 }
                }
            })
            Method (_STA) { Return (0x0F) }
        }

        Device (NVL0) {
            Name (_HID, "FHP000D")
            Name (_UID, 0)
            Name (_CRS, ResourceTemplate () {
                Memory32Fixed (ReadWrite, 0x4000D000, 0x100)
                Memory32Fixed (ReadWrite, 0x80000000, 0x4000000)   // 64 MB NVL3 window
                Interrupt (ResourceConsumer, Level, ActiveHigh, Exclusive) { 4, 5, 6, 7, 8 }
            })
            Name (_DSD, Package () {
                ToUUID("daffd814-6eba-4d8c-8a91-bc9bbf4aa301"),
                Package () {
                    Package () { "nvl3-size-mb", 64 },
                    Package () { "nvl3-banks", 256 },
                    Package () { "nvl3-endurance-write-count", 215 },
                    Package () { "nvl3-retention-years", 10 }
                }
            })
            Method (_STA) { Return (0x0F) }
        }

        Device (XBAR) {
            Name (_HID, "FHP000E")
            Name (_UID, 0)
            Name (_CRS, ResourceTemplate () {
                Memory32Fixed (ReadWrite, 0x4000E000, 0x100)
```

```
                Interrupt (ResourceConsumer, Level, ActiveHigh, Exclusive) { 16, 17, 18 }
            })
            Name (_DSD, Package () {
                ToUUID("daffd814-6eba-4d8c-8a91-bc9bbf4aa301"),
                Package () {
                    Package () { "crossbar-rows", 1024 },
                    Package () { "crossbar-cols", 1024 },
                    Package () { "crossbar-bits", 2 },
                    Package () { "crossbar-mode", "parallel" }
                }
            })
            Method (_STA) { Return (0x0F) }
        }

        // ... additional devices (DMA, Vector, OTP, Power, Security, Debug, CXL)
        // follow the same structure; see the table above for _HID and resource
        // definitions
    }

    // Thermal zones declared at the root scope
    Scope (\_TZ)
    {
        ThermalZone (TZ00) {
            Method (_TMP, 0, Serialized) {
                Return (\_SB.TEMP.TEMP0)      // read sensor 0
            }
            Method (_CRT, 0, Serialized) { Return (388) }   // 115°C in 0.1K
            Method (_PSV, 0, Serialized) { Return (358) }   // 85°C in 0.1K
            Name (_TC1, 2)
            Name (_TC2, 5)
            Name (_TSP, 10)
        }
        // additional zones per sensor
    }

    // Processor objects (one per hart, matching MADT Type 0x18)
    Scope (\_SB)
    {
        Processor (CPU0, 0, 0x00000410, 0x06) {}
        Processor (CPU1, 1, 0x00000410, 0x06) {}
        // ... one per hart, up to the product's hart count
    }
}
```

The crossbar `XBAR` device is only present on Ph-1 and Pc-1. The _STA method on Dc-1S returns 0 (device not present).

The CXL device is only present on Pc-1 and Dc-1S; its _CRS declares the link control block at 0x4000C000 and the interrupt sources 28--30.

5. Boot Handoff Protocol

5.1. DT mode

The boot ROM, after completing the firmware sequence (Appendix G), passes control to the OS or bootloader with:

- `a0` = boot hart ID (must be 0, the boot hart).
- `a1` = physical address of the FDT blob in memory.
- `pc` = entry point of the kernel or bootloader.
- `mstatus.MIE` = 0 (interrupts disabled).
- `satp` = 0 (paging off; kernel sets up its own).
- All PLIC sources disabled; firmware does not leave any interrupt pending.
- The FDT is at a 2 MB-aligned address in a memory region not claimed by the kernel.

5.2. ACPI mode

The boot ROM:

- Generates the ACPI tables at build time into a reserved memory region (typically 4 MB at the top of DDR5).
- Sets up the RSDP pointer in the DT `chosen` node `ferrohybrid,acpi-rsdp` property.
- Passes `a0` = 0, `a1` = FDT address (the DT still enumerates the boot hart and memory; the OS uses the RSDP to find the full ACPI tables and ignores the DT peripheral nodes).
- The kernel detects the RSDP and switches to ACPI-based enumeration.

The `ferrohybrid,acpi-rsdp` DT property is a standard UINT64 in the `chosen` node; the OS-side kernel patch reads it and calls `acpi_early_init` with the RSDP address.

5.3. Per-product differences

Parameter	Ph-1	Pc-1	Dc-1S
Default boot mode	DT	ACPI	ACPI
Hart count (MADT/DT)	10	28	32
NUMA nodes (SRAT)	1	1--2	1--4
CXL presence (CEDT)	no	yes	yes
Crossbar presence (XBAR)	yes	yes	no
DDR5 base	0x0_8000_0000	0x0_8000_0000	0x0_8000_0000
DDR5 size	patched per SKU	patched per SKU	patched per SKU
SPCR baud	115200	115200	115200

6. Validation and Test

6.1. ACPI validation (pre-tape-out)

The ACPI tables are generated by a build script from a machine-readable description (typically a YAML file per product). Before tape-out, the tables are validated:

1. **ACPI CA checker:** build the tables with the ACPI Component Architecture (ACPICA) compiler, run the `acpixtract -a` extractor, and run the ACPICA `acpiexec` validator. All tables must pass with zero errors and zero warnings except for documented product-specific allowances.
2. **Linux ACPI parser dry run:** run the tables through the Linux ACPI parser on a host x86 machine (the parser is architecture-independent; the OS will reject the tables on x86 because the MADT has RISC-V entries, but the parse itself must succeed without errors before the OS examines the architecture).
3. **Windows ACPI checker:** for OEM partners who will support Windows-on-Wine, run Microsoft's ACPI validation tool (ASL compiler and firmware test suite).
4. **Manual review:** the DSDT namespace is reviewed by hand against the register map in Appendix A, to confirm that every _CRS range and _DSD property matches the hardware.

6.2. DT validation (pre-tape-out)

The DT nodes are validated:

1. **dtc (device tree compiler):** compile the DTS to DTB; require zero warnings from `dtc -W all`.
2. **dt-schema:** run the DT through the dt-schema validator (the Linux DT schema repository), which checks that every node's compatible string matches a known schema. Custom compatible strings (`ferrohybrid,*`) use the design-supplied schema files, which the design team maintains alongside the DTS.
3. **Linux DT parser dry run:** boot a Linux kernel with `earlycon` on QEMU using the DTB; the kernel must accept the DT without errors up to the point where real hardware is needed.

6.3. Hardware bring-up test (post-silicon)

Once the first silicon returns, the boot ROM is patched with the real OTP values and the test sequence is:

1. Boot into a bare-metal firmware image that reads the DTB/ACPI tables and prints their parsed content to the UART.
2. Verify every node / table entry against the register map. Discrepancy stops the bring-up.
3. Boot into a real Linux kernel with the DT (DT mode). The kernel must reach userspace.
4. Boot into a real Linux kernel with the ACPI tables (ACPI mode). The kernel must reach userspace and `dmesg | grep -i acpi` must show the expected enumeration.
5. For server products, boot into a NUMA-enabled kernel and confirm `numactl --hardware` matches the expected topology.

6.4. Common failures

Symptom	Common cause
Kernel panics at ACPI parse	RSDP checksum mismatch; check the _CRS checksum in the DSDT
Kernel finds only 1 CPU	MADT Type 0x18 flags bit 0 is 0 for other harts; or DT `status` is "disabled"
NUMA topology empty	SRAT/SLIT missing or SRAT Proximity Domain values do not match SLIT
CXL not enumerated	CEDT missing or CHBS.Base HPA does not match the memory window
NVL3 not detected	DSDT NVL0 _CRS window does not match the register map, or DT `nvl3-window` is wrong
Crossbar driver probe fails	XBAR _HID missing or custom-instruction opcode mismatch
Serial console dead	SPCR baud rate or SPCR Base Address mismatch
Thermal zone does not throttle	_TZ._TMP returns a constant or the sensor index is out of range

The single most common bring-up error is a mismatch between the DSDT _CRS range and the actual hardware register map. The register map in Appendix A is the source of truth; the DSDT must match it exactly.

7. Summary

Artifact	Delivered with	Notes
DTS template	this appendix (structure) + design repo (full file)	Compiled at build time
ACPI tables	generated from product YAML at build time	DSDT/SSDT compiled via ACPICA
DSDT source (ASL)	this appendix (full source)	Hand-maintained per product
Product YAML	design repo	One per product
ACPI validation report	pre-tape-out	ACPICA + Windows checker
DT validation report	pre-tape-out	dtc + dt-schema

The DSDT source in §D.4 is the authoritative design definition for ACPI mode. The DT node structure in §D.3 is the authoritative design definition for DT mode. Any firmware that boots the design in either mode must generate tables that conform to the corresponding structure in this appendix.

Appendix E: Full Pin Description

This appendix specifies the ball assignment, signal catalog, differential pair catalog, power/ground ball assignment, and critical signal placement for the three packages used in the three Tier-1 products. The complete per-ball mapping is provided as a companion CSV file (`ballmap_<product>.csv`) that ships with this specification; this appendix gives the format specification, the signal catalog, and the placement rules needed to validate or regenerate that file.

1. Ballmap CSV File Format

Each product ships with a ballmap CSV file. The file is the authoritative per-ball assignment; this appendix defines its format and the design rules that generate it.

1.1. File naming

`ballmap_ph1.csv`, `ballmap_pc1.csv`, `ballmap_dc1s.csv`.

1.2. Column definitions

The CSV has exactly these columns, in this order:

Column	Type	Example	Meaning
Ball	string	A1	Ball coordinate (row letter + column number)
NetName	string	VDD_CORE	Net name as used on the board
SignalGroup	enum	POWER_CORE	Signal group name from the catalog in §E.3
Rail	enum	VDD_CORE	Power rail (blank for signals)
Direction	enum	POWER	In, Out, Bidir, NC, POWER, GND
IOStandard	string	1.0V_CMOS	I/O standard
DiffPair	string	PCIe_TX0	Differential pair name (blank if single-ended)
Polarity	char	P or N	P or N (blank if single-ended)
Function	string	PCIe Gen5 TX lane 0	Functional description

A sample of the first five rows (one per category):

```
Ball,NetName,SignalGroup,Rail,Direction,IOStandard,DiffPair,Polarity,Function
A1,VDD_CORE,VDD_CORE,VDD_CORE,POWER,1.0V,POWER,,
A2,VSS,VSS,VSS,POWER,0V,GND,,
```

```
A3,PCIe_TX0_P,PCIE,PCIe,Out,PCIe5,PCIE_TX0,P,PCIe Gen5 TX lane 0 positive
A4,PCIe_TX0_N,PCIE,PCIe,Out,PCIe5,PCIE_TX0,N,PCIe Gen5 TX lane 0 negative
A5,DDR5_CH0_DQ0,DDR5,DDR5,Bidir,DDR5_0V,DDR5_CH0_DQ0,P,DDR5 channel 0 DQ bit 0
```

1.3. Validation rules

Every ballmap CSV must satisfy:

1. Every ball coordinate in the package grid appears exactly once.
2. Every NetName appears on exactly the number of balls required by the catalog (§E.3).
3. Every DiffPair appears on exactly two balls (one P, one N).
4. Ball coordinate rows and columns must be contiguous (no gaps) except where the package depopulates corners or power/ground regions.

The validation script (`ballmap_validate.py`) is provided with the specification. It checks these rules and produces a report.

2. Signal Catalog

This section lists every signal group used in the three products, with ball counts. The catalog is the source of truth for the ballmap CSV: every ball belongs to exactly one signal group.

2.1. Ph-1 (FCBGA 400-ball)

Signal Group	Balls	Rail/Standard	Notes
VDD_CORE	40	1.0V	---
VDD_IO	20	1.8V	---
VDD_ANALOG	20	1.5V	Crossbar, NVL3 readout
VDD_PLL	4	1.0V	---
VSS	80	0V	---
LPDDR5X_CH0	40	LPDDR5X	Channel 0 (32 DQ + 4 DQS/CK/CA pairs)
LPDDR5X_CH1	40	LPDDR5X	Channel 1
PCIE	8	PCIe3	4 diff pairs (2 TX + 2 RX)
USB	4	USB2	D+, D-, VBUS, GND_ref
CAMERA_MIPI	16	MIPI_DPHY	4-lane D-PHY (8 diff pairs)
DISPLAY_MIPI	8	MIPI_DPHY	4-lane DSI (4 diff pairs)
MODEM_5G	16	MIPI_MPHY	4-lane M-PHY (8 diff pairs)
SECURITY_DEBUG	16	1.8V_CMOS	JTAG + secure debug
CONTROL	8	1.8V_CMOS	SPI, SMBus, reset, interrupt
THERMAL	2	analog	Thermal diode
Total	**400**		

2.2. Pc-1 (LGA 1851-ball)

Signal Group	Balls	Rail/Standard	Notes
VDD_CORE	400	1.0--1.05V	---
VDD_IO	60	1.8V	---
VDD_ANALOG	100	1.5V	Crossbar, NVL3 readout
VDD_PLL	8	1.0V	---
VSS	600	0V	---
DDR5_CH0..7	320	DDR5	8 channels × 40 balls (32 DQ + 4 DQS/CK/CA pairs)
PCIE	32	PCIe5	16 diff pairs (8 TX + 8 RX lanes)
CXL	32	PCIe5/CXL	16 diff pairs (8 TX + 8 RX lanes)
SPI_JTAG_SMBUS	20	1.8V_CMOS	---
CONTROL	28	1.8V_CMOS	reset, interrupt, BMC interface, etc.
SPARE	251	GND	Grounded spare
Total	**1851**		

2.3. Dc-1S (LGA 4677-ball)

Signal Group	Balls	Rail/Standard	Notes
VDD_CORE	900	1.0--1.05V	---
VDD_IO	100	1.8V	---
VDD_ANALOG	60	1.5V	NVL3 readout
VDD_PLL	16	1.0V	---
VSS	1800	0V	---
DDR5_CH0..3	160	DDR5	4 channels × 40 balls
PCIE	64	PCIe5	32 diff pairs (16 TX + 16 RX lanes)
CXL	32	PCIe5/CXL	16 diff pairs (8 TX + 8 RX lanes)
SPI_JTAG_SMBUS_BMC	100	1.8V_CMOS	Includes BMC interface
CONTROL	25	1.8V_CMOS	reset, interrupt, etc.
SPARE	1420	GND	Grounded spare
Total	**4677**		

3. Differential Pair Catalog

Every differential pair is listed with its net name and its pair members. The ballmap CSV assigns each pair member to a specific ball coordinate; this section lists the pairs themselves.

3.1. PCIe Gen5 (Pc-1, Dc-1S)

Standard 8-lane bidirectional PCIe Gen5. Each lane is one TX pair + one RX pair.

Lane	TX pair (P, N)	RX pair (P, N)	Notes
0	PCIe_TX0_P, PCIe_TX0_N	PCIe_RX0_P, PCIe_RX0_N	Root complex port 0
1	PCIe_TX1_P, PCIe_TX1_N	PCIe_RX1_P, PCIe_RX1_N	---
2	PCIe_TX2_P, PCIe_TX2_N	PCIe_RX2_P, PCIe_RX2_N	---
3	PCIe_TX3_P, PCIe_TX3_N	PCIe_RX3_P, PCIe_RX3_N	---
4	PCIe_TX4_P, PCIe_TX4_N	PCIe_RX4_P, PCIe_RX4_N	---
5	PCIe_TX5_P, PCIe_TX5_N	PCIe_RX5_P, PCIe_RX5_N	---
6	PCIe_TX6_P, PCIe_TX6_N	PCIe_RX6_P, PCIe_RX6_N	---
7	PCIe_TX7_P, PCIe_TX7_N	PCIe_RX7_P, PCIe_RX7_N	---

Total: 16 pairs, 32 balls. On Dc-1S, 16 PCIe lanes are provided (32 pairs, 64 balls); lanes 8--15 use the same naming convention with index 8--15.

3.2. CXL 3.0 (Pc-1, Dc-1S)

Standard 8-lane bidirectional CXL.

Lane	TX pair (P, N)	RX pair (P, N)	Notes
0	CXL_TX0_P, CXL_TX0_N	CXL_RX0_P, CXL_RX0_N	---
1	CXL_TX1_P, CXL_TX1_N	CXL_RX1_P, CXL_RX1_N	---
2	CXL_TX2_P, CXL_TX2_N	CXL_RX2_P, CXL_RX2_N	---
3	CXL_TX3_P, CXL_TX3_N	CXL_RX3_P, CXL_RX3_N	---
4	CXL_TX4_P, CXL_TX4_N	CXL_RX4_P, CXL_RX4_N	---
5	CXL_TX5_P, CXL_TX5_N	CXL_RX5_P, CXL_RX5_N	---
6	CXL_TX6_P, CXL_TX6_N	CXL_RX6_P, CXL_RX6_N	---
7	CXL_TX7_P, CXL_TX7_N	CXL_RX7_P, CXL_RX7_N	---

Total: 16 pairs, 32 balls. Dc-1S uses the same 8-lane CXL, giving 32 balls.

3.3. DDR5 (Pc-1, Dc-1S)

Per channel: 32 DQ (single-ended) + 4 DQS pairs (8 balls). 40 balls per channel. The DQS pairs are:

Byte lane	DQS pair (P, N)	DQ bits
0	DDR5_CHn_DQS0_P, DDR5_CHn_DQS0_N	DQ0--DQ7
1	DDR5_CHn_DQS1_P, DDR5_CHn_DQS1_N	DQ8--DQ15
2	DDR5_CHn_DQS2_P, DDR5_CHn_DQS2_N	DQ16--DQ23
3	DDR5_CHn_DQS3_P, DDR5_CHn_DQS3_N	DQ24--DQ31

The command/address (CA) and clock (CK) signals are handled by the DDR5 controller IP; they use single-ended CA in this design, not the differential CA of standard DDR5, to keep the ball count at 40 per channel. This is a deliberate ball-count optimization; the DDR5 controller IP supports both modes.

For Pc-1 (8 channels): 32 DQS pairs total.

For Dc-1S (4 channels): 16 DQS pairs total.

3.4. LPDDR5X (Ph-1)

Per channel: 40 balls. Same arrangement as DDR5 — 32 DQ + 4 DQS pairs per channel = 40 balls.

2 channels: 64 DQ balls + 8 DQS pairs (16 balls) = 80 balls total, matches the 80-ball LPDDR5X group.

8 DQS pairs.

3.5. Camera MIPI (Ph-1)

4-lane MIPI D-PHY, 8 diff pairs = 16 balls.

Lanes 0--3, each lane = 1 differential pair. The 8 pairs are the 4 data lanes (RX from the camera) plus 4 additional lanes (or 2 clock pairs + 4 data, etc.; the exact arrangement is in the ballmap CSV).

3.6. Display MIPI (Ph-1)

8 balls = 4 diff pairs. 4-lane DSI: 4 data pairs (or 3 data + 1 clock).

3.7. 5G Modem (Ph-1)

16 balls = 8 diff pairs. M-PHY (MIPI M-PHY) is bidirectional multi-lane. 4-lane × 2 directions = 8 pairs.

3.8. Summary of diff pairs by product

Interface	Ph-1 pairs	Pc-1 pairs	Dc-1S pairs
PCIe	4	16	32
CXL	0	16	16
DDR5	0	32	16
LPDDR5X	8	0	0
Camera MIPI	8	0	0
Display MIPI	4	0	0
5G Modem	8	0	0
Total diff pairs	**32**	**64**	**64**

4. Power and Ground Ball Assignment

Every power and ground ball is assigned to one rail (or VSS). The rails and their ball counts are listed in §E.3. The physical placement follows these rules:

4.1. Placement rules

1. **VSS interleaving:** VSS balls are interleaved with signal balls at a ratio of at least 1 VSS per 4 high-speed signal balls, and 1 VSS per 8 single-ended signal balls, to provide return-current paths.
2. **Power/ground rings:** VDD_CORE balls are concentrated in the center of the package (for lowest PDN impedance to the die power grid). VSS balls form an outer ring and an inner interleave.
3. **Rail separation:** VDD_IO, VDD_ANALOG, and VDD_PLL are in separate quadrants or contiguous blocks, physically separated from each other to prevent coupling.
4. **Analog isolation:** VDD_ANALOG (crossbar, NVL3 readout) and VDD_PLL are surrounded by VSS guard balls.
5. **High-speed signals:** PCIe, CXL, DDR5, and LPDDR5X signals are placed on the package edge closest to the corresponding PHY in the die floorplan.
6. **Spare balls:** Pc-1's 251 spare and Dc-1S's 1420 spare are all VSS-grounded.

4.2. VDD_CORE placement

VDD_CORE balls are physically clustered to form a low-inductance power delivery network. For Pc-1 (400 VDD_CORE balls) and Dc-1S (900 VDD_CORE balls), the balls form a contiguous block in the package center, with VSS interleaved at 1 VSS per 2 VDD_CORE in the interior and 1 VSS per 1 VDD_CORE at the boundary.

For Ph-1 (40 VDD_CORE balls), the VDD_CORE balls are arranged in a 5×8 grid in the package center.

4.3. VDD_ANALOG placement

VDD_ANALOG balls are placed on the quadrant adjacent to the analog PHY (crossbar and NVL3 readout) in the die floorplan. On Pc-1, this is the bottom-left quadrant; on Ph-1, the top-right; on Dc-1S, the bottom-right.

The VDD_ANALOG group is isolated by 2 VSS guard rings (one inner, one outer) to prevent noise coupling from the core and I/O rails.

4.4. VDD_PLL placement

VDD_PLL balls (4 on Ph-1, 8 on Pc-1, 16 on Dc-1S) are placed immediately adjacent to the analog PLL block in the die floorplan, with dedicated VSS return balls.

5. Critical Signal Placement

For high-speed signal routing, the ball coordinate regions are constrained as follows.

5.1. PCIe Gen5 (Pc-1, Dc-1S)

All PCIe Gen5 pairs are placed on the package bottom edge (rows near the edge, e.g., the last 4 rows). TX and RX pairs for a given lane are adjacent to each other (within 2 ball columns) to allow skew-matched routing on the board. Lane 0 is the leftmost, lane 7 (or 15) is the rightmost.

5.2. CXL 3.0 (Pc-1, Dc-1S)

All CXL pairs are placed on the package left edge, mirroring the PCIe layout. Lane 0 is the bottom-most, lane 7 is the top-most.

5.3. DDR5 (Pc-1, Dc-1S)

DDR5 channels are split between two edges:

- Channels 0--3: top edge, arranged left to right.
- Channels 4--7: right edge, arranged top to bottom (Pc-1 only; Dc-1S has 4 channels only, all on the top edge).

Within each channel, the DQ byte lanes are grouped, the DQS pair for each byte lane is adjacent to the corresponding DQ group, and the CA signals are grouped near the DQS of byte lane 0.

5.4. LPDDR5X (Ph-1)

LPDDR5X channels are on the top edge (channel 0) and right edge (channel 1), both near the corresponding PHY in the die floorplan.

5.5. MIPI (Ph-1)

Camera MIPI is on the top edge (near the camera ISP), display MIPI on the right edge (near the display controller), 5G modem on the bottom edge (near the modem interface).

5.6. Placement summary table

Interface	Ph-1 edge	Pc-1 edge	Dc-1S edge
PCIe	bottom	bottom	bottom
CXL	(none)	left	left
DDR5	(none)	top (0--3) + right (4--7)	top
LPDDR5X	top + right	(none)	(none)
Camera MIPI	top	(none)	(none)
Display MIPI	right	(none)	(none)
5G modem	bottom	(none)	(none)

6. No-Connect Balls

Balls that are not connected to any net inside the die are marked NC in the ballmap CSV, are physical balls on the package but are not bonded to any die pad, and are typically not connected on the board.

For the three packages:

Package	NC balls	Location
Ph-1	0	No NC balls (all 400 balls are assigned)
Pc-1	0	No NC balls (all 1851 assigned; spare = 251 VSS)
Dc-1S	0	No NC balls (all 4677 assigned; spare = 1420 VSS)

All spare balls in Pc-1 and Dc-1S are tied to VSS (not floated), to provide additional PDN return paths. The ballmap CSV marks them as "SPARE" signal group, "GND" rail, "POWER" direction.

7. Electrical Specification

Each signal group has an electrical specification (voltage, drive current, termination, differential impedance).

7.1. PCIe Gen5 (Pc-1, Dc-1S)

Parameter	Value	Notes
TX differential swing	800--1300 mV	Peak-to-peak, configurable in PHY
RX differential sensitivity	25 mV	At BER $<10^{-12}$
Differential impedance	85 Ω ±10%	PCB trace design target
Common-mode impedance	45--55 Ω	---
AC coupling	100 nF	TX side, per lane
Maximum trace length	300 mm	On FR4 with suitable stackup

7.2. CXL 3.0 (Pc-1, Dc-1S)

Same as PCIe Gen5 (CXL uses the same PHY layer).

7.3. DDR5 (Pc-1, Dc-1S)

Parameter	Value	Notes
VDDQ	1.1 V	DDR5 standard
DQ drive current	34 Ω / 40 Ω	ODT-selectable
DQS differential impedance	80 Ω ±10%	---
CA single-ended impedance	40 Ω ±10%	(this design uses single-ended CA)
Maximum trace length	100 mm	Per channel, with matched DQ/DQS
DQ-to-DQS skew	<10 ps	Per byte lane

7.4. LPDDR5X (Ph-1)

Same standard as DDR5, with 1.05 V VDDQ, on-package memory so traces are short (<20 mm).

7.5. Electrical summary

Interface	Diff Z	VDDQ	Max trace
PCIe Gen5 / CXL	85 Ω	1.1 V (AC-coupled)	300 mm
DDR5	80 Ω	1.1 V	100 mm
LPDDR5X	80 Ω	1.05 V	20 mm

8. Validation and Test

8.1. Ballmap validation

Before tape-out, the ballmap CSV is validated against:

1. **Signal catalog consistency:** the CSV ball counts per signal group match §E.3.
2. **Diff pair completeness:** every DiffPair name in the CSV has exactly 2 balls (one P, one N).
3. **Rail assignment:** every ball has either a Rail name or a Direction of NC.
4. **Coordinate uniqueness:** no ball coordinate appears twice.
5. **Contiguity:** the ball grid has no unexpected gaps.

The validation script (`ballmap_validate.py`) produces a report.

8.2. Package assembly validation (post-silicon)

After the first silicon returns, the package is validated:

1. **X-ray inspection:** confirms all 400/1851/4677 solder joints are present and well-formed.
2. **Cross-section:** confirms solder joint shape and standoff height.
3. **Electrical continuity:** confirms every signal ball is electrically connected to the die pad it is assigned to.
4. **Diff pair impedance measurement:** TDR measurement on representative diff pairs, confirming 85 Ω (PCIe) or 80 Ω (DDR5).
5. **PDN impedance measurement:** vector network analyzer sweep of the VDD_CORE plane, confirming target impedance.

8.3. Board-level acceptance

For a board design to be accepted, it must:

1. Reproduce the ballmap assignment exactly (no ball re-purposed).
2. Meet the diff pair impedance and trace length specifications.
3. Meet the DQ-to-DQS skew specification for all DDR5/LPDDR5X channels.
4. Meet the VDD_CORE PDN impedance target (typically <1 mΩ from DC to 10 MHz).

Appendix F: Mechanical Drawings

This appendix specifies the mechanical dimensions, tolerance stack-up, heatsink mounting, thermal interface, and board keep-out zones for the three packages. The mechanical drawing package (DXF, DWG, SVG, and PDF files) ships with this specification; this appendix gives the dimensions in tabular form.

1. Package Outline Dimensions

1.1. FCBGA 16×16 (Ph-1)

Dimension	Nominal	Tolerance	Notes
Package body length	16.00 mm	±0.10 mm	X-dimension
Package body width	16.00 mm	±0.10 mm	Y-dimension
Package body thickness	1.00 mm	±0.05 mm	Substrate + lid
Substrate thickness	0.60 mm	±0.03 mm	---
Lid thickness	0.40 mm	±0.03 mm	Ni-plated Cu
Ball array field	10.00 mm × 10.00 mm	±0.05 mm	20×20 grid
Ball pitch	0.50 mm	±0.02 mm	---
Ball diameter (pre-reflow)	0.30 mm	±0.02 mm	---
Ball height (post-reflow)	0.25 mm	±0.03 mm	---
Coplanarity	---	≤0.05 mm	All balls
Package weight	0.9 g	±0.1 g	---

The package has a laser-marked orientation dot at A1 corner (0.5 mm diameter) on the lid top surface. The pin 1 indicator is a chamfered corner on the substrate (45° chamfer, 1.0 mm).

1.2. LGA 1700 (Pc-1)

Dimension	Nominal	Tolerance	Notes
Package body length	45.00 mm	±0.10 mm	X-dimension
Package body width	37.50 mm	±0.10 mm	Y-dimension
Package body thickness (with lid)	3.50 mm	±0.10 mm	---
Substrate thickness	1.20 mm	±0.05 mm	---
Lid thickness	1.30 mm	±0.05 mm	Ni-plated Cu, IHS
IHS-to-die TIM thickness	0.20 mm	±0.05 mm	---
Ball array field	41.60 mm × 33.80 mm	±0.10 mm	---
Ball pitch	1.00 mm	±0.03 mm	---
Ball diameter (pre-reflow)	0.60 mm	±0.03 mm	---
Package weight (with IHS)	28 g	±2 g	---

The package has an integrated heat spreader (IHS) with a stamped dimple for IHS-to-die thermal contact. The IHS is Ni-plated Cu, 1.3 mm thick, with a flatness of 0.05 mm over the TIM contact area.

The orientation marker is a triangular notch on the substrate edge at A1 corner (2.0 mm), and a laser-marked logo on the IHS top surface.

1.3. LGA 4677 (Dc-1S)

Dimension	Nominal	Tolerance	Notes
Package body length	61.00 mm	±0.15 mm	X-dimension
Package body width	82.00 mm	±0.15 mm	Y-dimension
Package body thickness (with lid)	3.80 mm	±0.10 mm	---
Substrate thickness	1.40 mm	±0.05 mm	---
Lid thickness	1.20 mm	±0.05 mm	Ni-plated Cu, IHS
IHS-to-die TIM thickness	0.25 mm	±0.05 mm	---
Ball array field	56.00 mm × 76.00 mm	±0.10 mm	---
Ball pitch	1.00 mm	±0.03 mm	---
Ball diameter (pre-reflow)	0.60 mm	±0.03 mm	---
Package weight (with IHS)	65 g	±5 g	---

The package uses the standard Intel LGA 4677 socket specification (Eagle Stream platform). The IHS has the standard Intel server IHS pattern with the heatsink mounting features described in §F.3.

2. Substrate Stack-Up

2.1. FCBGA 16×16 substrate

Layer	Thickness	Notes
L1 (top)	15 µm Cu	Die attach + lid seal
L2	30 µm dielectric + 15 µm Cu	---
L3	30 µm dielectric + 15 µm Cu	---
L4 (core)	400 µm BT resin	Rigid core
L5	30 µm dielectric + 15 µm Cu	---
L6 (bottom)	15 µm Cu + ENIG	Ball pads
Total	600 µm	---

2.2. LGA 1700 substrate

8-layer build-up substrate: 1-2-1 (one core, two build-up each side). BT resin core 800 µm, plus 100 µm each side build-up = 1000 µm total. LGA pad finish: ENIG (Au 0.05 µm over Ni 3--6 µm).

2.3. LGA 4677 substrate

10-layer build-up substrate: 2-4-2 (one thick core, four layers each side). Total thickness 1400 µm. Same LGA pad finish.

3. Heatsink Mounting

3.1. FCBGA 16×16 (Ph-1)

The package has no separate heatsink mounting. The phone OEM attaches a thin vapor chamber or heat spreader directly to the lid via the TIM interface. The mounting is by adhesive or snap-fit, not by screws.

- TIM contact area: 12.0 mm × 12.0 mm (the lid top surface, minus a 2 mm border for seal).
- TIM thickness: 0.10--0.20 mm.
- Mounting pressure: 5--15 psi (typical for adhesive-based phone cooling).

3.2. LGA 1700 (Pc-1)

The package uses the standard Intel LGA 1700 heatsink mounting hole pattern (ILM, Independent Loading Mechanism). The board should provide:

- 4 mounting holes on a rectangular pitch, 75.0 mm × 82.0 mm (±0.10 mm).
- Hole diameter: 3.5 mm (±0.05 mm).
- Board mounting pad: 6.0 mm diameter (for M3.5 screws).
- Mounting pressure: 30--50 lbf (typical for 360 mm AIO), applied via the AIO backplate.

3.3. LGA 4677 (Dc-1S)

The package uses the standard Intel LGA 4677 heatsink mounting hole pattern (Eagle Stream socket):

- 4 mounting holes on a rectangular pitch, 100.0 mm × 115.0 mm (±0.15 mm).
- Hole diameter: 4.0 mm (±0.05 mm).
- Board mounting pad: 7.0 mm diameter (for M4 screws).
- Mounting pressure: 50--80 lbf (typical for cold-plate cooling).
- Torque specification: 8--10 in-lb per screw (M4 stainless steel, 4 screws).

4. Thermal Interface Material (TIM)

4.1. TIM specification

The package lid is supplied with no factory-applied TIM. The user applies TIM between the lid and the heatsink/cold-plate.

Property	Value	Tolerance	Notes
Thermal conductivity	≥ 8 W/m·K	---	For Pc-1 (high TDP)
Thermal conductivity (DC)	≥ 12 W/m·K	---	For Dc-1S
Thermal conductivity (Phone)	≥ 3 W/m·K	---	For Ph-1
Thickness (applied)	0.05--0.20 mm	---	Based on TIM type
Operating temperature	−40 to +150 °C	---	---
Interface resistance	≤ 0.05 K·cm^2/W	---	For 8 W/m·K paste

4.2. TIM application

For Pc-1 and Dc-1S, apply TIM paste as a 5-dot pattern (center + 4 corners) on the IHS top surface, then mount the cooler. The mounting pressure spreads the TIM to 0.05--0.15 mm uniform thickness. Do not apply more than 0.5 g of TIM; excess can bridge to the substrate edge.

For Ph-1, apply TIM pad (0.15 mm thickness) between the lid and the vapor chamber.

5. Solder Reflow Profile

5.1. Reflow profile for board assembly

Standard SAC305 lead-free reflow, with the following profile (measured at the ball):

Phase	Temperature	Time
Preheat	25 → 150 °C	60--90 s (max 3 °C/s ramp)
Soak	150 → 200 °C	60--120 s
Reflow	Peak 245 ± 5 °C	20--40 s above 217 °C (liquidus)
Cooling	245 → 25 °C	60--90 s (max 6 °C/s down-ramp)
Total cycle	---	200--330 s

5.2. Reflow constraints

- **Peak temperature:** 245 °C max. Higher peak increases intermetallic growth and reduces joint reliability.

- **Time above liquidus:** 20--40 s. Longer causes excessive IMC formation.
- **Ramp rates:** ≤3 °C/s up, ≤6 °C/s down. Faster causes thermal shock and potential die cracking.
- **Number of reflows:** The chip can survive 3 reflows (for double-sided assembly + rework). Pre-wake-up applied at factory test is designed to withstand 3 reflows.
- **MSL:** MSL 3 (168 hours floor life at 30 °C / 60% RH after bake). Higher MSL requires dry-pack storage.

6. Moisture Sensitivity Level (MSL)

Package	MSL	Floor life	Notes
FCBGA 16×16	MSL 3	168 h	Standard
LGA 1700	MSL 3	168 h	Standard
LGA 4677	MSL 3	168 h	Standard

The packages ship in dry-pack (moisture barrier bag + desiccant + HIC). The bag is opened only when the board is ready to reflow. After opening, the floor life of 168 hours at ≤30 °C / ≤60% RH applies. If exceeded, the package is baked at 125 °C for 24 hours before use.

7. Board Keep-Out Zones

The board layout must respect the following keep-out zones (no traces, no vias, no components within the zone):

7.1. FCBGA 16×16 (Ph-1)

Zone	Dimensions	Notes
Top (component) side	0.3 mm from package outline	Underfill fillet space
Bottom (board) side	0.2 mm from ball array edge	---
Corner keep-out	2.0 mm × 2.0 mm at each corner	Alignment mark
Board edge	1.0 mm from package to board edge	For de-paneling

7.2. LGA 1700 (Pc-1)

Zone	Dimensions	Notes
Top side (component)	5.0 mm from package outline	ILM backing plate area
Bottom side (board)	3.0 mm from ball array edge	---
Backplate area	95 mm × 95 mm centered on package	For AIO backplate
Standoff height	6.5 mm ± 0.3 mm	From board top surface to IHS top

7.3. LGA 4677 (Dc-1S)

Zone	Dimensions	Notes
Top side (component)	8.0 mm from package outline	Cold plate clearance
Bottom side (board)	4.0 mm from ball array edge	---
Cold plate backplate	130 mm × 130 mm centered	For cold-plate mounting
Standoff height	8.0 mm ± 0.5 mm	From board top to IHS top

8. Mechanical Stress and Reliability

8.1. Thermal cycling

The package must survive −40 to +125 °C thermal cycling, 1000 cycles, no mechanical or electrical failure.

8.2. Drop and shock

For Ph-1 (phone product): must survive 1.5 m drop onto concrete, 26 orientations, no electrical failure. The phone OEM encloses the package in its chassis; the chassis handles the drop energy.

For Pc-1 and Dc-1S: no drop test (server and desktop form factors).

8.3. Vibration

All three packages must survive 5--500 Hz random vibration at 1.5 g RMS, 30 min per axis, no failure.

9. Assembly Drawings Package

The mechanical drawing package ships separately and contains:

- `fhpf_ph1_package.dxf` — full PCB footprint, top-side component, and keep-out
- `fhpf_pc1_package.dxf` — same
- `fhpf_dc1s_package.dxf` — same
- `fhpf_ph1_package_stackup.pdf` — substrate stack-up drawing
- `fhpf_pc1_package_stackup.pdf` — same
- `fhpf_dc1s_package_stackup.pdf` — same
- `fhpf_heatsink_pc1.dwg` — recommended heatsink reference design
- `fhpf_coldplate_dc1s.dwg` — recommended cold-plate reference design
- `fhpf_dimensions_check.pdf` — all dimensions with GD&T callouts

The DXF files are the board designer's authoritative reference for the physical footprint. The PDF stack-up drawings give the exact layer composition, prepreg types, and copper weight. The heatsink/cold-plate DWG files are provided as reference designs; the system integrator may substitute alternative cooling solutions that meet the mechanical and thermal specifications in this appendix.

10. Summary Table

Parameter	Ph-1	Pc-1	Dc-1S
Body size	16×16×1.0 mm	45×37.5×3.5 mm	61×82×3.8 mm
Ball pitch	0.50 mm	1.00 mm	1.00 mm
Ball count	400	1851	4677
IHS	thin lid	IHS	IHS
Heatsink mounting	adhesive	LGA 1700 ILM	LGA 4677 (Eagle Stream)
Max TDP	32 W	323 W	430 W
MSL	3	3	3
Solder reflow peak	245 °C	245 °C	245 °C
Reflow cycles max	3	3	3
TIM thermal cond.	≥3 W/mK	≥8 W/mK	≥12 W/mK

Appendix G: Bring-Up Checklist

This appendix is the acceptance test that runs when the first silicon returns from the foundry. It defines, in order, every measurement that confirms the silicon is functional, with acceptance criteria, failure diagnosis, and recovery action for each. It is the checklist a bring-up engineer follows on the bench.

The checklist is organized in 12 stages. Each stage has a precondition (the prior stage passed), a list of measurements, and an exit criterion. A stage that fails blocks all later stages; the failure diagnosis in each stage describes the most likely causes and the recovery actions.

Every measurement is stated with the instrument (oscilloscope, multimeter, logic analyzer, etc.), the pin or register under test, and the numeric acceptance criterion. Nothing in this appendix is optional: all 12 stages must pass before silicon is deemed functional.

1. Stage 1: Unpowered Continuity and Isolation

Preconditions: board assembled, chip not powered, no power supply connected.

Measurements:

1. **VDD_CORE short to VSS:** multimeter in resistance mode, measure from any VDD_CORE ball to any VSS ball. Acceptance: >10 Ω.
2. **VDD_IO short to VSS:** same. Acceptance: >10 Ω.
3. **VDD_ANALOG short to VSS:** same. Acceptance: >10 Ω.
4. **VDD_PLL short to VSS:** same. Acceptance: >10 Ω.
5. **VDD_CORE short to VDD_IO:** measure between rails. Acceptance: >1 kΩ.
6. **VDD_CORE short to VDD_ANALOG:** same. Acceptance: >1 kΩ.
7. **Signal ball isolation:** select 10 signal balls at random (including at least one diff pair per interface). Measure to VSS. Acceptance: >1 MΩ.
8. **Socket contacts:** for LGA 1700 and LGA 4677, visually inspect socket pins under 20× magnification; bent pins are rejects.

Exit criterion: all 8 measurements pass.

Failure diagnosis:

- Short on a rail: board assembly defect (solder bridge, misplaced component). BGA rework required.
- Short between rails: package defect or board short. Chip replacement likely required.
- Short on a signal: board assembly defect or chip ESD damage.

2. Stage 2: Rail Bring-Up (No Chip Load)

Preconditions: Stage 1 passed; chip on board but not powered.

Procedure: connect bench supply to VDD_CORE only, with current limit at 100 mA. No other rails powered. Set voltage to 0.50 V.

Measurements:

1. **Current draw at 0.50 V:** acceptance: <50 mA.
2. **VDD_CORE voltage:** measured at the chip ball (Kelvin connection). Acceptance: 0.50 ± 0.02 V.
3. **Repeat for each rail:** VDD_IO (1.8 V, acceptance <100 mA), VDD_ANALOG (1.5 V, acceptance <50 mA), VDD_PLL (1.0 V, acceptance <20 mA).

Exit criterion: current draws within specification at nominal rail voltages.

Failure diagnosis:

- Excessive current at 0.5 V: chip defect (latch-up, ESD damage) or board short. Chip replacement.
- Excessive current at nominal VDD_ANALOG: analog block defect. Chip replacement.
- Excessive current at VDD_PLL: PLL defect. Chip replacement.

3. Stage 3: Power Sequencing

Preconditions: Stage 2 passed. All rails connected; bench supply replaced with board VRMs.

Procedure: apply main power input; measure each rail's rise time and sequence order against the specification in Appendix A §A.11.

Measurements:

1. **Rail 1 (VDD_IO) rise time:** acceptance: 0.1--10 ms, monotonic.
2. **Rail 2 (VDD_ANALOG) rise time:** acceptance: 0.1--10 ms, monotonic, after Rail 1 (delay 1--10 ms).
3. **Rail 3 (VDD_PLL) rise time:** acceptance: 0.1--5 ms, after Rail 2.
4. **Rail 4 (VDD_CORE) rise time:** acceptance: 0.1--50 ms, after Rail 3.
5. **POR release:** RESET pin rises after all rails are within 5% of nominal. Acceptance: POR rise time 1--100 ms after last rail OK.
6. **Rail_STATUS register:** read via JTAG (no CPU yet). Acceptance: RAIL_STATUS = 0x3F (all 6 rails).

Exit criterion: all 6 measurements pass.

Failure diagnosis:

- Wrong sequencing: PMIC or board sequencing circuit defect. Rework.
- Rail rise time out of spec: VRM design issue. Rework.
- Rail_STATUS not all 6 bits: some rail not OK. Check that rail's VRM.

4. Stage 4: PLL Lock and Clock Distribution

Preconditions: Stage 3 passed. Chip is powered, POR released.

Procedure: check the compute PLL and the CXL LC-PLL lock via JTAG status register; measure clock at the package pin.

Measurements:

1. **Compute PLL lock:** read via JTAG (offset 0x4000_9000 register 0x20, bit 12). Acceptance: LOCK = 1 within 10 ms of

POR release.

2. **Compute PLL frequency:** measure at a clock test point (if the package brings one out; otherwise measure via the CPU clock counter over a known wall-clock interval). Acceptance: 6.20 ± 0.03 GHz on Pc-1, 4.50 ± 0.03 GHz on Ph-1, 6.00 ± 0.03 GHz on Dc-1S.
3. **CXL LC-PLL lock (Pc-1, Dc-1S):** read via JTAG register 0x4000_9000 register 0x24, bit 12. Acceptance: LOCK = 1.
4. **Clock jitter:** measure via spectrum analyzer (if external clock is available) or via the CPU clock noise floor. Acceptance: <1.0 ps RMS integrated 100 Hz--100 MHz.

Exit criterion: all measurements pass.

Failure diagnosis:

- PLL not locking: PLL reference absent or wrong frequency. Check the reference crystal / oscillator on the board.
- PLL locking at wrong frequency: divider misconfiguration. Check the boot OTP settings or the strapping pins.
- Jitter >1 ps: PLL design margin; check the reference quality and VDD_PLL noise.

5. Stage 5: JTAG and Boot ROM Execution

Preconditions: Stage 4 passed. PLL locked.

Procedure: connect JTAG; halt the boot hart at the boot ROM entry point; step through the first 100 instructions; confirm the boot ROM signature and the first-branch target.

Measurements:

1. **JTAG TAP IDCODE:** acceptance: reads the expected device ID (0x1D1F_0001 for Ph-1, 0x1D1F_0002 for Pc-1, 0x1D1F_0003 for Dc-1S).
2. **Boot ROM entry:** after POR + JTAG halt, PC = 0x0000_0000_0000_0000. Acceptance: exact.
3. **First instruction:** read memory at 0x0. Acceptance: matches the boot ROM image.
4. **Boot ROM signature:** read the last 16 bytes of the boot ROM region. Acceptance: matches the expected ECDSA-public-key hash (readable at 0x0000_0000_0000_0800).
5. **First branch target:** step to the first `j` or `jal` instruction; confirm the target is inside the boot ROM address range (0x0 to 0x1000).

Exit criterion: all 5 measurements pass.

Failure diagnosis:

- JTAG IDCODE wrong: chip defect (mask ROM or TAP issue) or JTAG pin damage. Chip replacement.
- Boot ROM entry not 0: boot strapping or boot selection strapping pins wrong. Check board straps.
- Boot ROM signature mismatch: die-level mask issue. Chip replacement.

6. Stage 6: Boot ROM Completion

Preconditions: Stage 5 passed. JTAG verified boot ROM start.

Procedure: remove JTAG halt; allow the boot ROM to run to completion; capture the boot ROM UART output.

Measurements:

1. **UART output at 115200 8N1:** acceptance: firmware banner appears within 250 ms of POR release.
2. **DDR5 training status:** captured in the UART log. Acceptance: "DDR5 training: PASS."
3. **PCIe enumeration (Pc-1, Dc-1S):** captured in the UART log. Acceptance: "PCIe: N devices enumerated" with N ≥ 1 (assuming a PCIe device is populated).
4. **CXL enumeration (Pc-1, Dc-1S, multi-socket only):** captured in the UART log. Acceptance: "CXL: M remote sockets" with M ≥ 1 (single-socket: M = 0).
5. **ACPI tables or DT blob loaded:** captured in the UART log. Acceptance: "Platform description: [ACPI | DT] at address 0xXXXXXXXX."
6. **Jump to OS loader:** captured in the UART log. Acceptance: "Jump to OS loader."

Exit criterion: all 6 measurements pass.

Failure diagnosis:

- No UART output: UART pin misrouted; UART clock absent; boot ROM hung. Check Stage 5.
- DDR5 training fail: memory signal integrity; check the DDR5 trace length and impedance (Appendix E §E.10).
- PCIe enumeration 0: PCIe device not populated or link training fail. Check the PCIe diff pair polarity.
- ACPI/DT not loaded: boot OTP boot-mode bit wrong.

7. Stage 7: DDR5 Memory Test

Preconditions: Stage 6 passed. DDR5 training passed; boot ROM passed to OS loader.

Procedure: halt at the OS loader entry; run a memory test utility (memtest86+, or the design-supplied `dramtest`) over the full DRAM range.

Measurements:

1. **Memtest86+ full pass:** acceptance: 0 errors over the full memory.
2. **Bandwidth:** measure with `stream` or `mbw`. Acceptance: ≥400 GB/s on Pc-1 (8 channels), ≥200 GB/s on Dc-1S (4 channels), ≥100 GB/s on Ph-1 (2 LPDDR5X channels).
3. **Latency:** measure with `lmbench` or `Intel MLC`. Acceptance: 38--54 ns.
4. **Refresh and ECC:** enable ECC, inject single-bit errors via register writes; confirm correction. Acceptance: single-bit corrected, double-bit detected.

Exit criterion: all 4 measurements pass.

Failure diagnosis:

- Memtest errors: DRAM defective, DRAM timing marginal, DDR5 signal integrity. Check with reduced clock (DDR5-5600 instead of 7200) to isolate.
- Low bandwidth: one channel not training; check DDR5 channel enable in the boot ROM log.
- High latency: DDR5 training marginal; retrain at lower frequency.

8. Stage 8: NVL3 First-Program and Read-Back

Preconditions: Stage 7 passed. DDR5 functional. NVL3 controller initialized per Appendix B §B.4.

Procedure: run the NVL3 first-program test. This test writes a known pattern (walking-1, walking-0, checkerboard, all-0, all-1) to every bank, reads back, verifies, and checks ECC status.

Measurements:

1. **NVL3 calibration:** confirm CALIB_STATUS.DONE = 1 after the boot path's CALIB_START sequence. Acceptance: done within 20 ms.
2. **NVL3 first-program test:** acceptance: 0 errors in the pattern write/read-back on all banks.
3. **NVL3 column leakage:** read the column leak screening per Appendix A §A.16. Acceptance: >99% of columns classified Normal; reduced-margin columns reported.
4. **NVL3 ECC single-bit:** inject via register; confirm correction. Acceptance: corrected, address logged.
5. **NVL3 ECC double-bit:** inject via register; confirm detection. Acceptance: detected, address logged, error reported.
6. **NVL3 retention:** program known pattern, power down for 1 minute (post-factory pre-wake-up should be present), power back up, read back. Acceptance: 0 errors.

Exit criterion: all 6 measurements pass.

Failure diagnosis:

- Calibration not done: OTP calibration table corrupt. Rerun factory calibration.
- First-program errors: HZO film phase fraction low (see Appendix A §A.16). Check Gate 1 acceptance.
- High column leakage: process marginality. Column screening will flag bad columns.
- Retention fail without pre-wake-up applied: pre-wake-up missing from this die. Rerun factory test.

9. Stage 9: Crossbar First-Program and Calibration (Ph-1, Pc-1)

Preconditions: Stage 8 passed. NVL3 functional. Crossbar controller initialized per Appendix B §B.5.

Procedure: load the calibration table from OTP; program a known weight pattern; run a MAC operation; compare against the expected analog output.

Measurements:

1. **Crossbar calibration:** confirm CALIB_STATUS.DONE = 1. Acceptance: done within 5 ms.
2. **DAC monotonicity:** iterate all 16 DAC codes; measure ADC output; acceptance: 16 distinct outputs in increasing order.
3. **Weight program:** program a row with pattern {00, 01, 10, 11} repeated; read back; acceptance: all cells correct.
4. **ADC ENOB:** apply a known input ramp; compute ENOB. Acceptance: ≥8.5 bits (spec 8.9).
5. **MAC operation:** program a 16×16 identity matrix; apply a 16-element input vector; compare output to input. Acceptance: output = input within 2 LSB (for each output element).

Exit criterion: all 5 measurements pass.

Failure diagnosis:

- Calibration fail: OTP calibration table corrupt or TIA/ADC marginal. Rerun factory calibration.
- DAC monotonicity fail: DAC design defect. Chip replacement.
- Weight program errors: FeFET cell defect or write-verify controller issue. Rerun first-program.
- ENOB fail: ADC design or VDD_ANALOG noise. Check the analog rail.

- MAC mismatch: crossbar analog chain misconfigured. Re-run calibration.

10. Stage 10: PCIe Enumeration and DMA

Preconditions: Stage 9 passed (or Stage 8 for Dc-1S). Platform description loaded.

Procedure: after boot into OS loader, run PCIe enumeration. Plug in a known-good NVMe drive; run a PCIe bandwidth test.

Measurements:

1. **PCIe enumeration:** `lspci` lists all populated PCIe devices with correct vendor/device IDs. Acceptance: device count matches physical population.
2. **PCIe link width:** `lspci -vv` shows the expected width (x16 for the main PCIe slot; x4 for the NVMe slot). Acceptance: matches physical slot.
3. **PCIe link speed:** shows the expected generation (Gen5 on Pc-1 and Dc-1S; Gen3 on Ph-1). Acceptance: 32 GT/s (Gen5) or 8 GT/s (Gen3).
4. **PCIe DMA bandwidth:** `dd` read from the NVMe at sustained queue depth. Acceptance: ≥12 GB/s for Gen5 x4, ≥3 GB/s for Gen3 x2.

Exit criterion: all 4 measurements pass.

Failure diagnosis:

- Device not enumerated: board-level PCIe routing issue (diff pair polarity, length mismatch), or PHY configuration issue. Check diff pair impedance (Appendix E §E.10).
- Wrong width: one or more lanes not training. Check per-lane status in the CXL control register (Appendix A §A.12).
- Wrong speed: link training failing at Gen5; retrain at Gen4 or Gen3 to isolate.

11. Stage 11: Kernel Boot to Userspace

Preconditions: Stage 10 passed (Dc-1S, Pc-1) or Stage 9 passed (Ph-1). OS image prepared.

Procedure: boot into a full Linux kernel; measure boot time, check dmesg for errors, reach userspace.

Measurements:

1. **Kernel boot time to userspace:** acceptance: ≤500 ms (spec is 400 ms with 20% margin).
2. **dmesg errors:** acceptance: no "ERROR" or "Call trace" lines (warnings may be present).
3. **CPU enumeration:** `lscpu` shows the correct number of CPUs at the correct nominal frequency. Acceptance: matches product (10 on Ph-1, 28 on Pc-1, 32 on Dc-1S).
4. **Memory visible:** `free -h` shows the correct DRAM size. Acceptance: within 1% of physical population.
5. **NVL3 visible:** `lsblk` or `/sys/block` shows the NVL3 block device. Acceptance: capacity matches (64 MB on Ph-1/Pc-1, 24 MB on Dc-1S).
6. **Thermal sensors:** `sensors` reads all sensors with plausible temperatures (within ±5 °C of ambient + offset). Acceptance: all sensors report values.
7. **Crossbar visible (Ph-1, Pc-1):** `/dev/crossbar` device present. Acceptance: device node exists.

Exit criterion: all 7 measurements pass.

Failure diagnosis:

- Boot time >500 ms: DRAM training slow, or ACPI enumeration slow. Check Stage 7 timing.
- dmesg errors: driver initialization failures. The UART log at the failing driver identifies the block.
- Wrong CPU count: MADT/DT issue. Check Appendix D.
- NVL3 invisible: NVL3 driver not loaded. Check the kernel config and the DSDT/DT NVL0 node.
- Sensors not readable: thermal driver issue. Check Appendix A §A.6 register map.

12. Stage 12: Userspace Verification and Application Run

Preconditions: Stage 11 passed. Userspace boot complete.

Procedure: run a representative workload to confirm end-to-end functionality.

Measurements:

1. **CPU stress test:** `stress-ng --cpu N --timeout 60s`. Acceptance: no errors, no thermal throttle above threshold.
2. **Memory stress:** `stress-ng --vm 4 --vm-bytes 4G --timeout 60s`. Acceptance: no OOM, no errors.
3. **NVL3 read/write:** `dd if=/dev/nvl3 of=/dev/null bs=1M` and reverse. Acceptance: read at line-speed, write without verify error.
4. **Crossbar AI (Ph-1, Pc-1):** run ResNet-50 inference on the crossbar test image. Acceptance: accuracy ≥75.5% (spec 75.7%, margin 0.2 pp).
5. **Windows application (Pc-1, Dc-1S):** run Firefox for Windows under Wine+Box64. Acceptance: launches, navigates to a web page, no crash.
6. **Sleep/wake (P3→P0):** issue sleep, measure wake time. Acceptance: wake time ≤15 µs.
7. **Cold boot from NVL3:** power cycle, measure boot time. Acceptance: ≤500 ms.

Exit criterion: all 7 measurements pass.

Failure diagnosis:

- CPU stress failure: CPU microarchitecture bug. Further diagnosis with `perf`.
- Crossbar accuracy low: re-run calibration (Stage 9); if still low, crossbar cells defective.
- Windows app crash: Wine configuration or Box64 issue. Check the compatibility matrix.
- Sleep/wake slow: NVL3 state not retained, or PLL relock slow. Check the P3 wake path.

13. Bring-Up Summary

All 12 stages must pass before the silicon is deemed functional. Each stage blocks the next. The total bench time, once a board is assembled and a debug setup is in place, is approximately 4--6 hours for a first-pass run, or 1--2 days if any stage requires debugging. After the first successful bring-up, subsequent boards take 30--60 minutes per board.

The bring-up log (every measurement with the measured value, not just pass/fail) is a required deliverable. It becomes the reference dataset for future silicon revisions.

14. Bring-Up Equipment List

Equipment	Purpose
Bench power supply (0--5 V, 20 A)	Stage 2
Digital multimeter (4.5 digit)	Stages 1, 2, 3
Oscilloscope (500 MHz, 4-channel)	Stages 3, 4
Spectrum analyzer (26.5 GHz)	Stage 4 (jitter)
Logic analyzer (16-channel, 500 MS/s)	Stages 5, 6
JTAG debugger (RISC-V compatible)	Stages 4, 5, 6, 8, 9
UART-to-USB adapter (115200 baud)	Stages 6, 11
Thermal camera	Stage 12
PCIe NVMe drive (known-good)	Stage 10
Bootable OS image (with NVL3 driver)	Stage 11
Crossbar test image (ResNet-50)	Stage 12

Appendix H: Security Provisioning Procedure

This appendix specifies how the security credentials (device-unique ECDSA key, OEM root key hash, debug public key, boot nonce, rollback counter) are generated, injected into OTP, locked, and tracked through the device lifecycle. It is the production procedure for the security provisioning step in factory test, plus the operational procedures for key lifecycle management after the device ships.

The OTP injection sequence itself is in Appendix B §B.7; this appendix provides the surrounding context: where the keys come from, how they are delivered, how the factory systems handle them, and how they are used after provisioning.

1. Security Lifecycle States

Every device transitions through four states, tracked by a provisioning status word in OTP region 5:

State	Name	Condition	What is allowed
S0	Unprovisioned	OTP root key region blank	Factory provisioning only; boot fails secure boot check
S1	Provisioned	Root key, device key, debug key, nonce all programmed; OTP regions 0--4 locked	Normal operation; secure boot enforced
S2	Commissioned	First end-user boot; attestation certificate issued	Normal operation; debug unlock via challenge-response
S3	Decommissioned	Rollback counter set to max; OTP region 0 locked permanently	Boot only with decommission key; no secure boot

The provisioning status word is at OTP region 5, word 0. It is read at boot; the boot ROM enforces the state transitions. A device that boots with state S0 (unprovisioned) jumps to a recovery image (physical presence required) rather than the normal firmware.

2. Key Generation

2.1. OEM root key

The OEM root key is an ECDSA P-256 key pair. It is generated by the OEM's security team on an HSM, in the following procedure:

1. HSM initialization: per the OEM's HSM vendor procedure (e.g., Thales Luna, Entrust nShield, AWS CloudHSM). At least 3 HSM administrators must be present (M-of-N quorum, typical M=2, N=3).
2. Root key generation: `HSM: generate-ecdsa-p256 --label "FHP-ROOT-2026" --usage sign,verify --exportability none`. The key is generated inside the HSM and never leaves it in plaintext.
3. Root key backup: 5 shares of the root key are stored in geographically separate vaults, encrypted under a per-share key.

Recovery requires 3-of-5 shares (Shamir).

4. Root public key extraction: `HSM: export-public-key --label "FHP-ROOT-2026" --out fhp_root_pub.der`. This is the value injected into OTP region 0 at factory test (its hash is what the boot ROM anchors).
5. Root key sign: the OEM uses the root key to sign the OEM bootloader, kernel, initrd, and userspace images. The signing procedure uses the HSM to produce the ECDSA signature; the private key never leaves the HSM.

2.2. Device-unique key

The device-unique ECDSA P-256 key is generated by the factory test system at the time of provisioning:

1. Factory HSM generates a fresh ECDSA P-256 key for the device: `HSM: generate-ecdsa-p256 --label "FHP-DEV-<serial>" --usage sign,verify --exportability wrapped`.
2. The private key is exported wrapped under the device's OTP public key (see §H.3), so the wrapper can be opened only by the specific device's security engine.
3. The public key is stored in the factory database (Production Key Management System, PKMS) indexed by device serial.
4. The wrapped private key is delivered to the device via the factory programming interface.
5. The factory HSM zeroizes the plaintext private key after wrapping (the wrapped form remains in PKMS).

The device-unique private key is used for debug unlock challenge-response and attestation signature. It is never exported in plaintext from the device after provisioning.

2.3. Debug public key

The debug public key is the OEM's chosen ECDSA P-256 public key for the debugger. It is generated by the OEM and provided to the factory test system as a DER file. The factory injects it into OTP region 2. This key authorizes debug unlock; the OEM can rotate it by generating a new key and issuing a firmware update that authorizes the new key (the old key stays in OTP but firmware rejects unlock requests signed by it).

2.4. Boot nonce

The boot nonce is a 16-byte random value generated by the device's TRNG at provisioning time. It is used to derive the per-boot encryption keys (Appendix A §A.12). The TRNG must pass its self-tests (Appendix A §A.13 register TRNG_STATUS.HEALTH) before the nonce is used. If the TRNG health check fails, the device is flagged for factory RMA.

3. OTP Injection Sequence

The detailed OTP write sequence is in Appendix B §B.7. The surrounding factory procedure is:

1. The device is on the factory ATE at the security provisioning station.
2. The factory system confirms the device is in state S0 (OTP region 0 blank).
3. The factory system is connected via secure channel to the factory HSM and PKMS.
4. The factory system requests a device-unique key pair; the HSM generates it and returns the wrapped private key + public key.
5. The factory system reads the OEM root key hash (from the OEM-provided DER file) and the debug public key (from OEM).

6. The factory system reads the device's TRNG output (16 bytes) via the JTAG/SPI programming interface.
7. The factory system calls the OTP injection sequence (Appendix B §B.7) to write root key hash, device key (wrapped), debug public key, rollback counter = 1, boot nonce.
8. The factory system calls the OTP read-verify for every word written.
9. If all read-verifies pass, the factory system writes OTP region 5 word 0 to the S1 transition value (0x5052_4F56 = "PROV" in ASCII).
10. The factory system locks OTP regions 0--4 (irreversible).
11. The factory system logs the successful provisioning to PKMS: device serial, public key, timestamp, factory ID, operator ID.
12. The factory system moves the device to the next station.

If any step fails, the device is scrapped and the provisioning log records the failure reason. There is no retry on OTP write failures (OTP is one-time programmable).

4. Rollback Counter Initialization

The rollback counter starts at 1 at provisioning time. Each firmware update increments it. The counter is a 32-bit monotonic value stored in OTP region 3. Firmware rejects any update with a version number ≤ the current counter. The counter is never reset except by decommissioning.

The OTP storage for the counter is 8 bytes (64 bits), with the high 32 bits reserved. The counter increments in place: OTP bits are one-way programmable, so a counter increment writes additional bits (the counter effectively becomes a "one-time programmable multi-bit field" that stores the largest value written, since the read returns the bitwise-OR of all written values in an increment-only encoding). The encoding is binary: counter=1 writes bit 0; counter=2 writes bit 1; counter=3 writes bits 0 and 1 (already 1); etc. Increments work by writing the new value; the read returns the OR of all writes, which equals the highest value written (because the OR of a binary sequence 1, 2, 3, 4... has all bits set that any value set, and the highest value always has a superset of bits of any lower value). The OTP controller must implement this OR-on-write semantics for the rollback counter region specifically (the rest of OTP is write-once).

5. Debug Credential Delivery

After provisioning, the device is in state S1 (provisioned). The debug public key is in OTP, but the debugger also needs:

1. **Device serial number:** printed on the package top (2D barcode + human-readable). The factory PKMS holds the mapping serial → device public key.
2. **Device public key:** exported from the package via the SPI/I2C debug interface at provisioning time; also available from PKMS by serial.
3. **Provisioning certificate:** the OEM's root key signs the device public key, producing an X.509-like certificate that the debugger verifies.

The debug certificate is written to an EEPROM on the board (separate from the chip, on the factory-programmed EEPROM) at board assembly time, or sent to the OEM by PKMS (whichever flow the OEM prefers). The debugger reads the certificate from the EEPROM (or from the OEM's PKMS) and uses the certified device public key to verify the challenge-response.

6. Attestation Key Derivation

The device-unique ECDSA key is the root of the attestation chain, but it is not used directly for every attestation. Instead, the boot ROM derives a per-boot attestation key from the device key + boot nonce:

1. Read device private key from OTP (wrapped).
2. Unwrap in the security engine (never exits the security engine).
3. Read boot nonce from OTP.
4. Derive: `attestation_key = HKDF-SHA-256(device_key, boot_nonce || 0x01, "FHP-AK", 32)`.
5. The attestation key is the ECDSA private key used to sign the DICE attestation quote.
6. The attestation public key is exported via the DICE CDI register (Appendix A §A.13); the OEM's root key signs the attestation public key into an attestation certificate.

The attestation certificate has a validity of 24 hours after boot. A verifier (e.g., a cloud service) checks: (a) the attestation certificate is signed by the OEM root key; (b) the attestation certificate's public key matches the signed quote; (c) the quote's boot measurement matches the expected boot image hash.

7. Debug Unlock Procedure (Operational)

The debug unlock procedure at any time after provisioning:

1. The debugger reads the device serial (visible on the package, or via SPI at bring-up).
2. The debugger retrieves the device public key from the OEM PKMS or from the board EEPROM.
3. The debugger generates a random 32-byte challenge.
4. The debugger writes the challenge to the OTP controller's CHALLENGE register (Appendix A §A.14).
5. The device security engine signs the challenge with the device private key and writes the response to RESPONSE register.
6. The debugger reads the response and verifies it against the device public key.
7. If verification succeeds, the debugger writes VERIFY=1 then UNLOCK=1 to CTRL.
8. The device opens the JTAG port for the configured TIMEOUT (default 1 hour).
9. All subsequent JTAG operations are logged to the device's event log (OTP region 6, 4 KB ring buffer).

Every failed verification attempt increments ATTEMPTS. After 5 failed attempts, LOCK_DOWN is set; the device permanently refuses JTAG unlock until a factory RMA reverses the counter (the RMA process involves an OEM-signed recovery image that resets the attempts counter once).

8. Decommissioning

When the device reaches end of life, or when the OEM determines the device must be permanently disabled (e.g., compromised credentials), the decommission sequence runs:

1. The OEM issues a "decommission" signed image that the device accepts only when it is signed by the OEM root key and has the special decommission flag.
2. The device, on running the decommission image, writes the OTP rollback counter to max (0xFFFF_FFFF) and sets OTP region 5 word 0 to the S3 transition value (0x4445_434F = "DECO").
3. OTP region 0 (root key hash) is left readable but no valid firmware image exists for it (the OEM root key is retired).

4. The device boots only into a stub recovery image that prints "DECOMMISSIONED" to UART and halts.
5. The OEM's PKMS marks the device as decommissioned; the device public key is revoked in the OEM's CRL.

The decommission is irreversible (OTP bits cannot be cleared). The device is functionally dead after decommission; it must be physically destroyed for end-of-life disposal per the OEM's e-waste procedure.

9. Security Event Logging

The device maintains an event log in OTP region 6 (4 KB ring buffer). Each entry is 32 bytes: timestamp (8 bytes), event code (1 byte), event-specific payload (23 bytes). Event codes:

Code	Event	Payload
0x01	Boot successful	Boot image hash (truncated 8 bytes)
0x02	Boot signature failure	Expected hash, actual hash (truncated)
0x03	Rollback attempt blocked	Attempted version, current version
0x04	Debug unlock requested	Challenge (truncated 8 bytes)
0x05	Debug unlock succeeded	Session duration (seconds)
0x06	Debug unlock failed	Attempt number
0x07	Debug lock-down	---
0x08	Tamper detected	Sensor (voltage/temp/clock)
0x09	Secure boot mode change	Old mode, new mode
0x0A	Firmware update	Old version, new version
0x0B	Decommission started	---
0x0C	Factory provisioning	Factory ID, operator ID

The event log is write-once (OTP) and is read out at RMA for incident investigation. It cannot be cleared or modified. A device whose event log contains unexpected events (e.g., repeated failed unlocks) is flagged for RMA review even if functionally working.

10. Summary Table

Item	OTP Region	Written When	Locked
OEM root key hash	0	Factory provisioning	Yes (at provisioning)
Device public key hash	1	Factory provisioning	Yes (at provisioning)
Device private key (wrapped)	1	Factory provisioning	Yes (at provisioning)
Debug public key	2	Factory provisioning	Yes (at provisioning)
Rollback counter	3	On firmware update	Never (write-once)
Boot nonce	4	Factory provisioning	Yes (at provisioning)
Provisioning status	5	Factory provisioning	Yes (at provisioning)
Event log	6	On security events	Never (write-once ring)
Calibration tables	7	Factory test	Yes (at provisioning)
Reserved	8+	---	---

11. Production Sign-Off

A device ships from the factory only when the following are all true:

1. All 12 bring-up stages (Appendix G) passed.
2. OTP provisioning status is S1 (provisioned).
3. All OTP regions 0--5 are locked.
4. The event log contains only the provisioning event (no errors).
5. The device's public key is in the OEM PKMS.
6. The provisioning certificate has been issued (or the OEM has confirmed the public key is in the database).
7. The device passed the final electrical test.

A device that fails any of these is either reworked (if rework is possible) or scrapped.

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